Batterjee Medical College
For Science and Technology

ECG Interpretation

Name: Muhammad Reihan
Designation: Associate professor
Department: Medicine

Learning Objectives

Knowledge & understanding

- **Describe in sequence the steps in ECG interpretation**
- **Identify the rhythm, site of origin, possible causes, and clinical significance of ECG changes.**

Skill

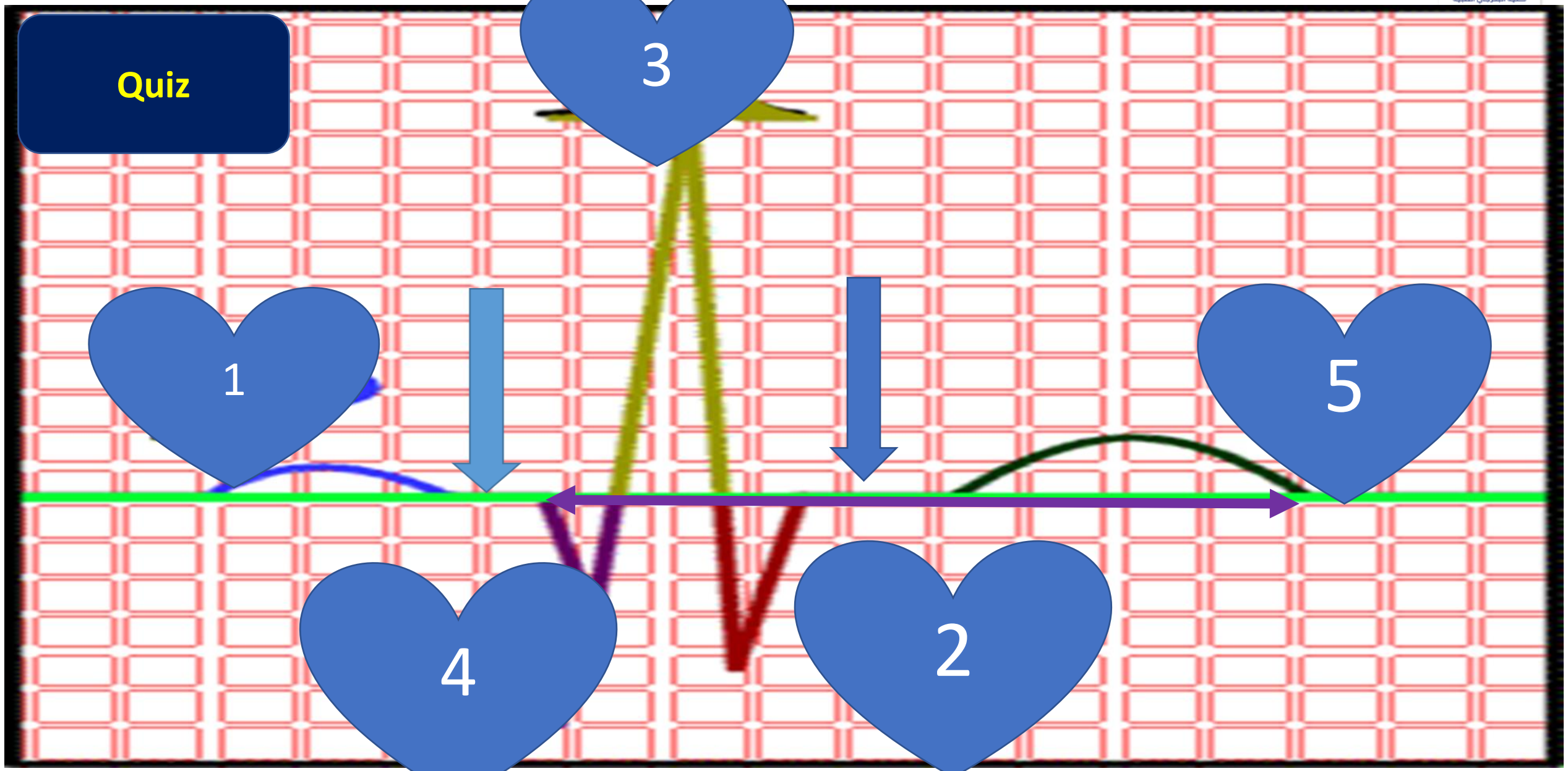
- Apply a systematic approach to the interpretation of a 12-lead ECG.
- Differentiate normal ECG findings from common ECG abnormalities.

Main Subtopics

- Basics of ECG (cardiac conduction system; ECG principles)
- ECG Leads (limb leads; chest leads; bipolar and unipolar leads)
- Systematic ECG Interpretation
  - Identification
  - Standardization
  - voltage
  - Axis
  - Heart Rate
  - P Wave
  - PR Interval
  - QRS Complex
  - ST Segment
  - T Wave

ECG

Quiz



ECG



Einthoven 1893

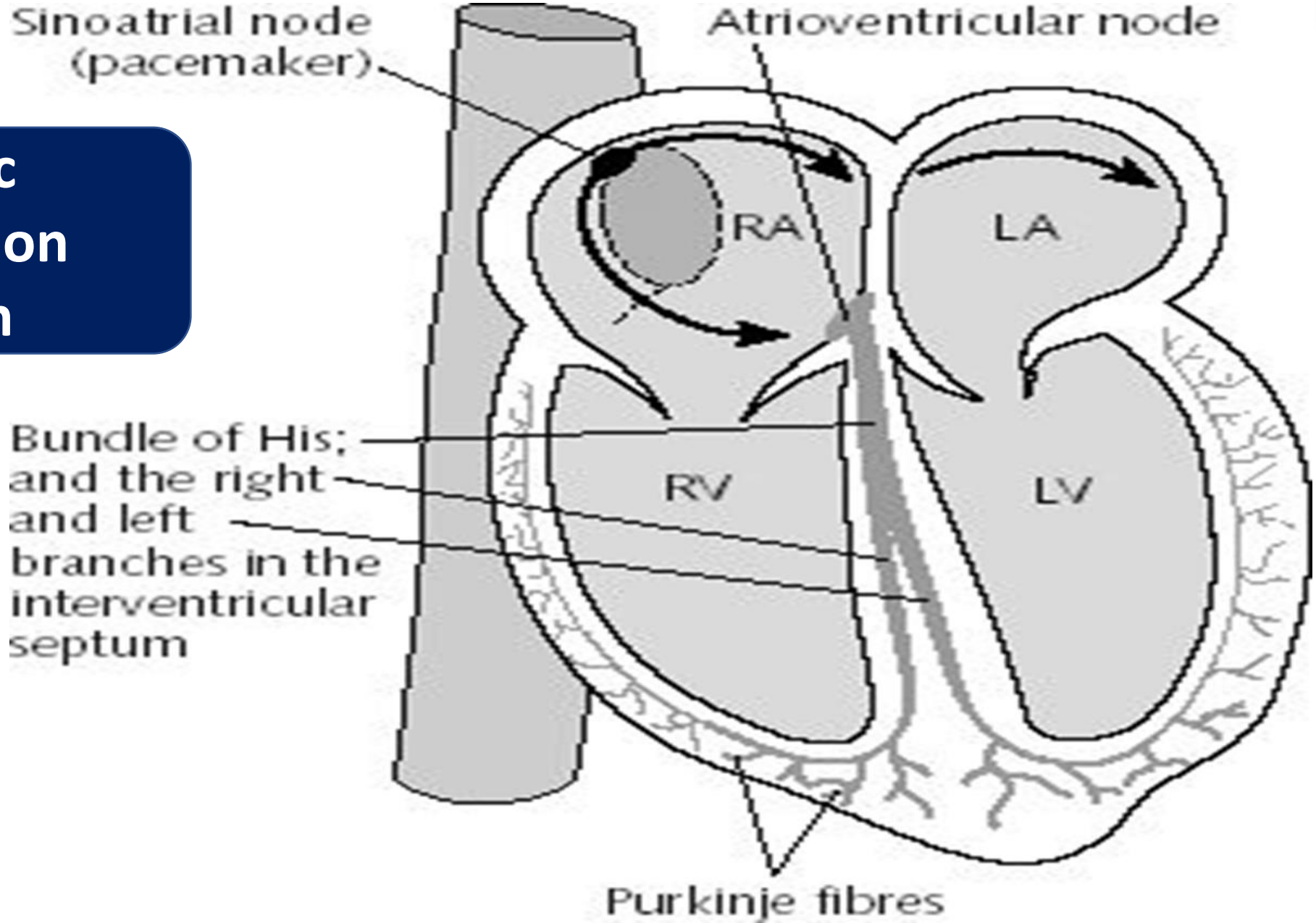


ECG



ECG

Cardiac Conduction System



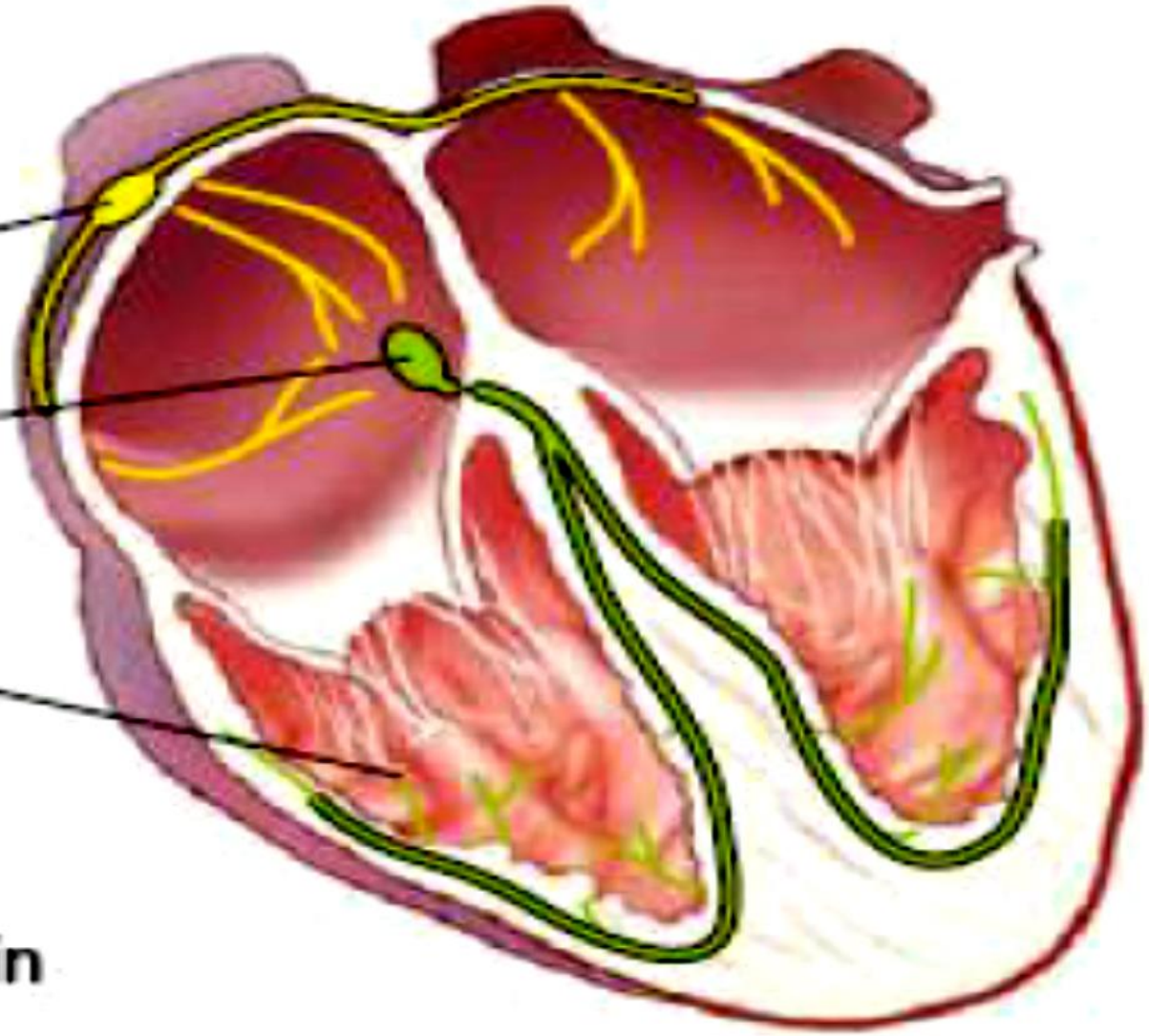
a “fail-safe”

Inherent rates:

SA node (60-100/min)

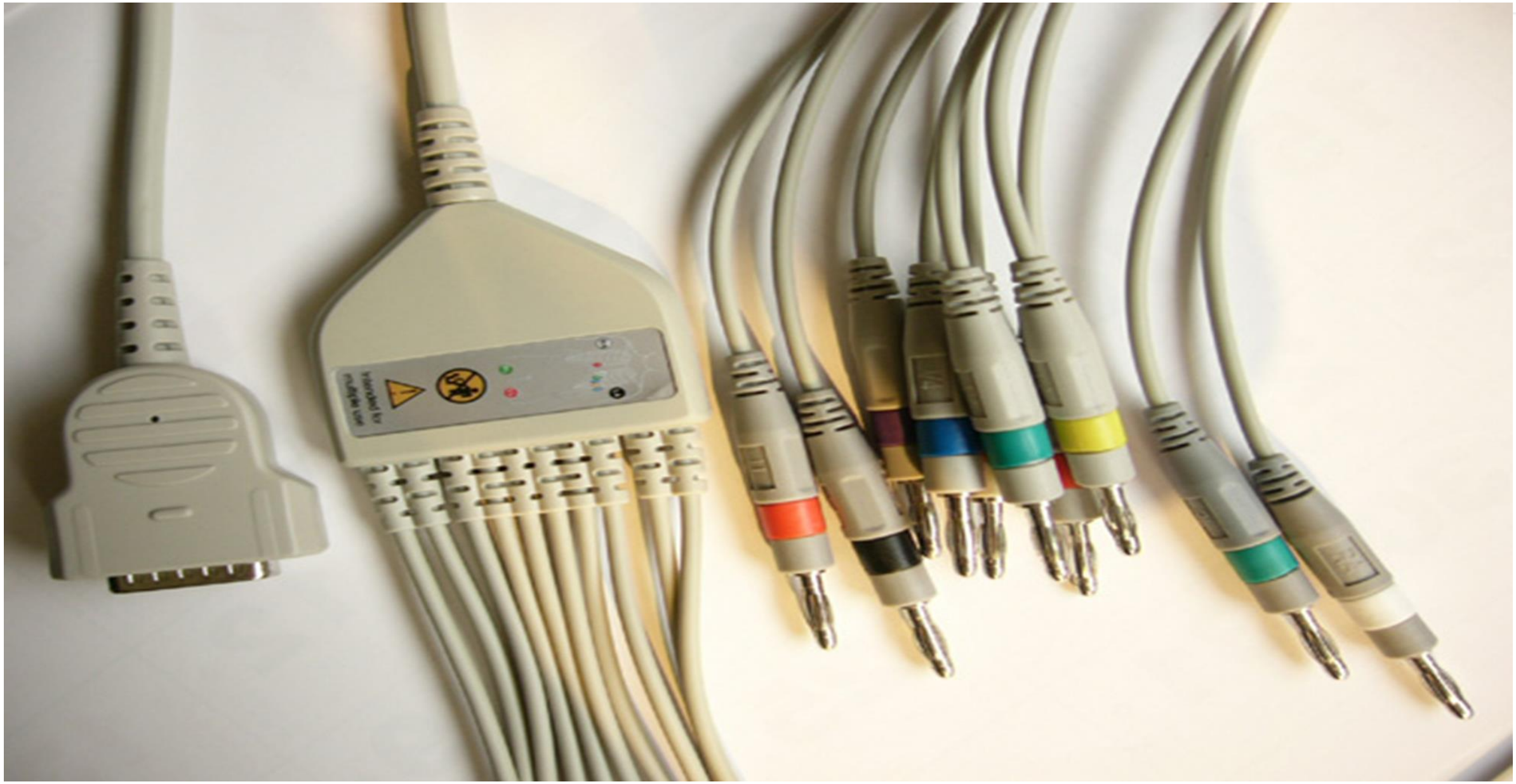
AV node (40-60/min)

Ventricles (20-40/min)

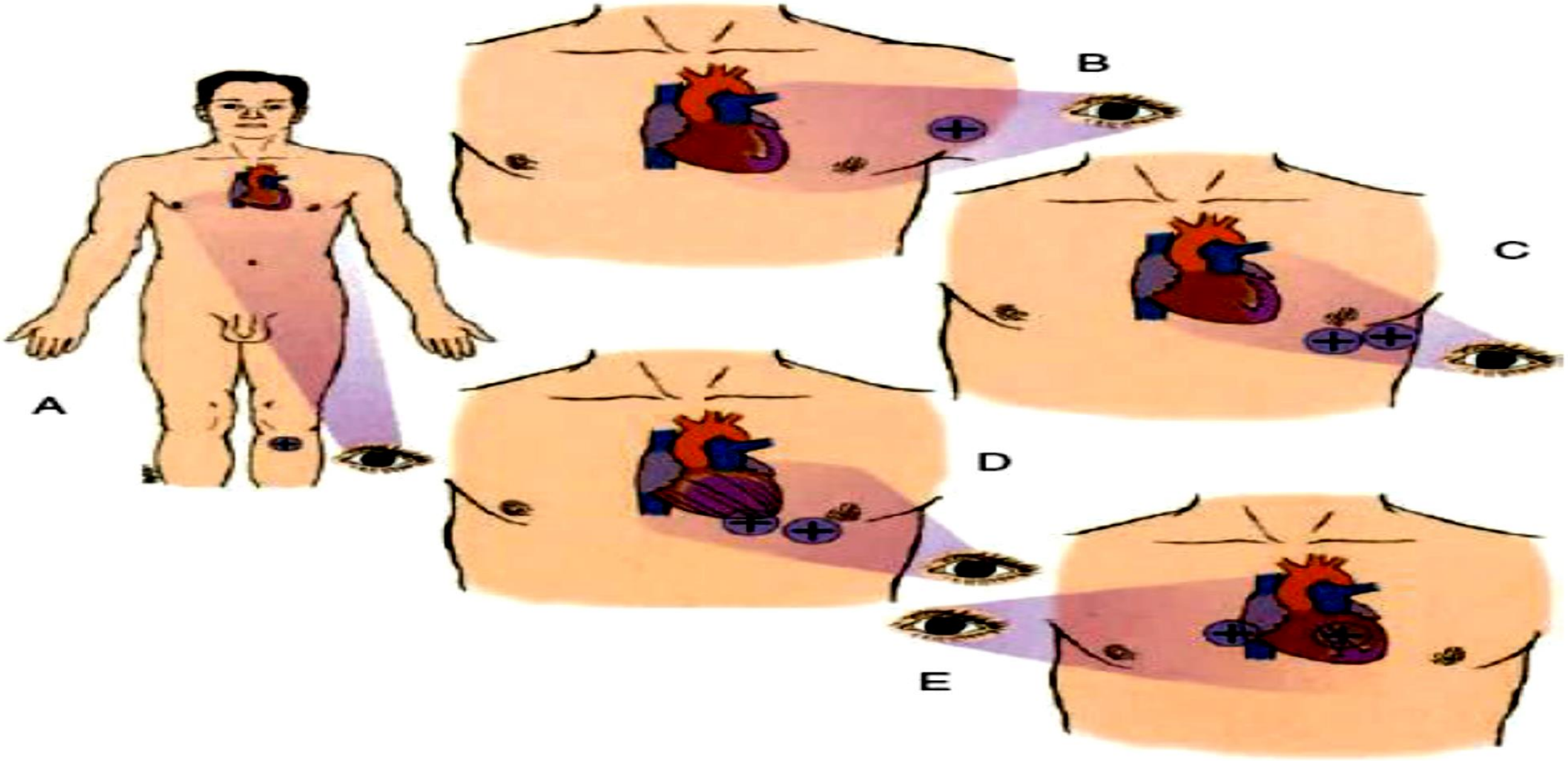


Emergency rates: 150-250/min

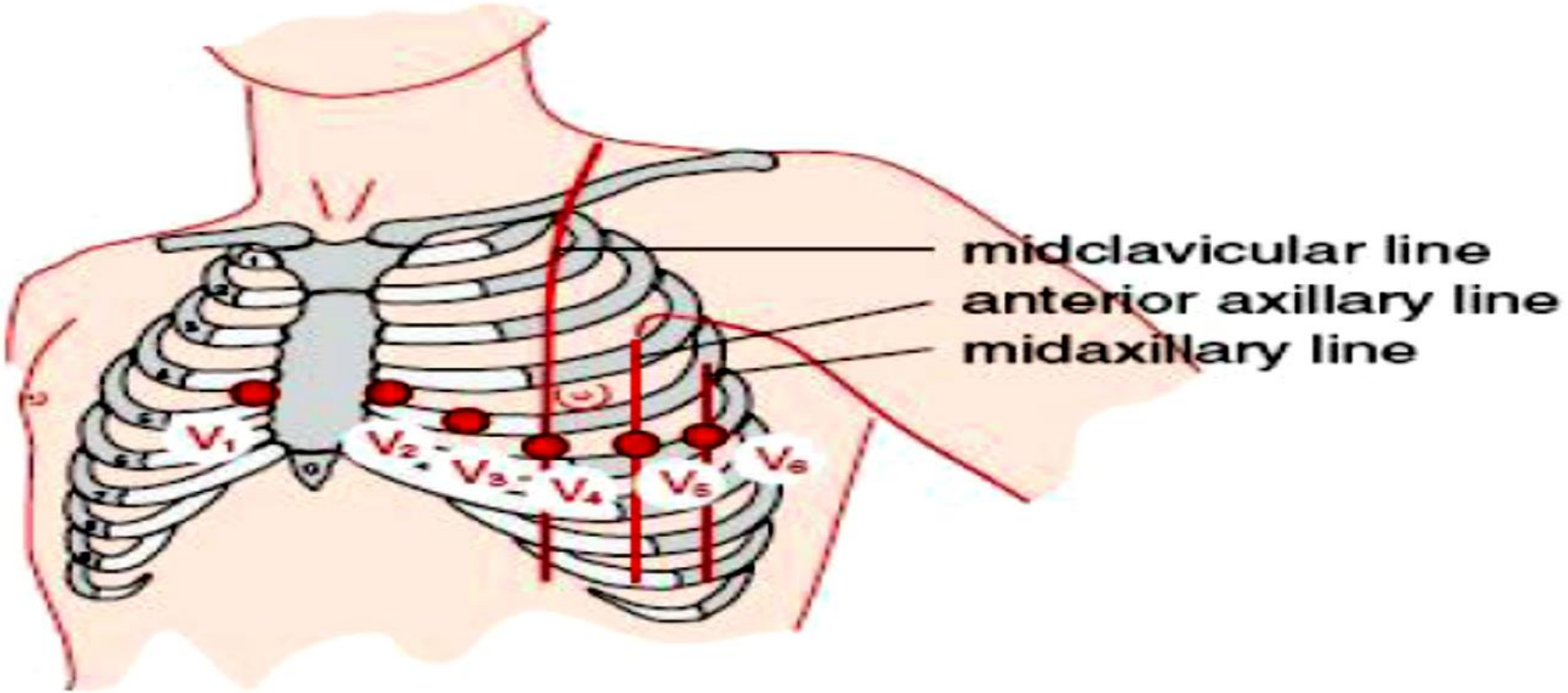
ECG



ECG



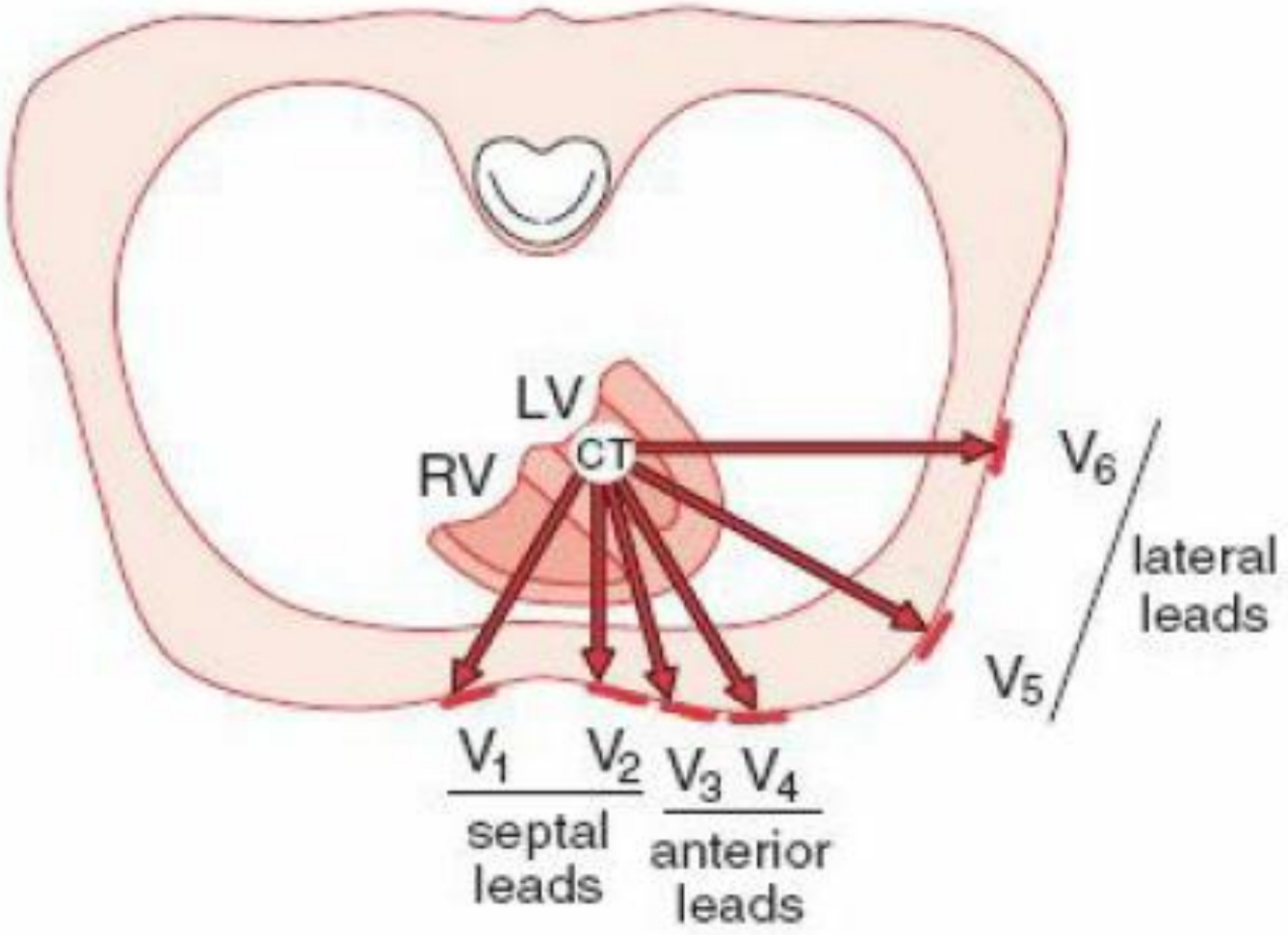
Application of Monitoring Electrodes



Placement of Precordial Electrodes

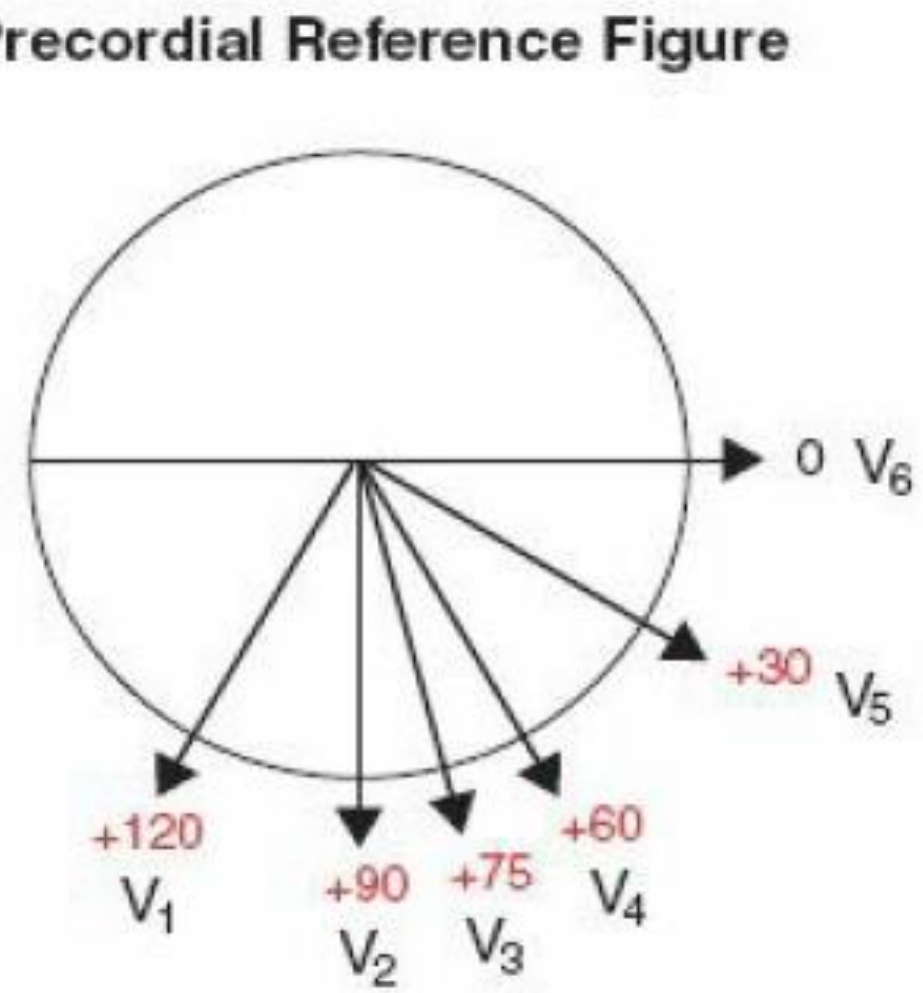
A

ECG

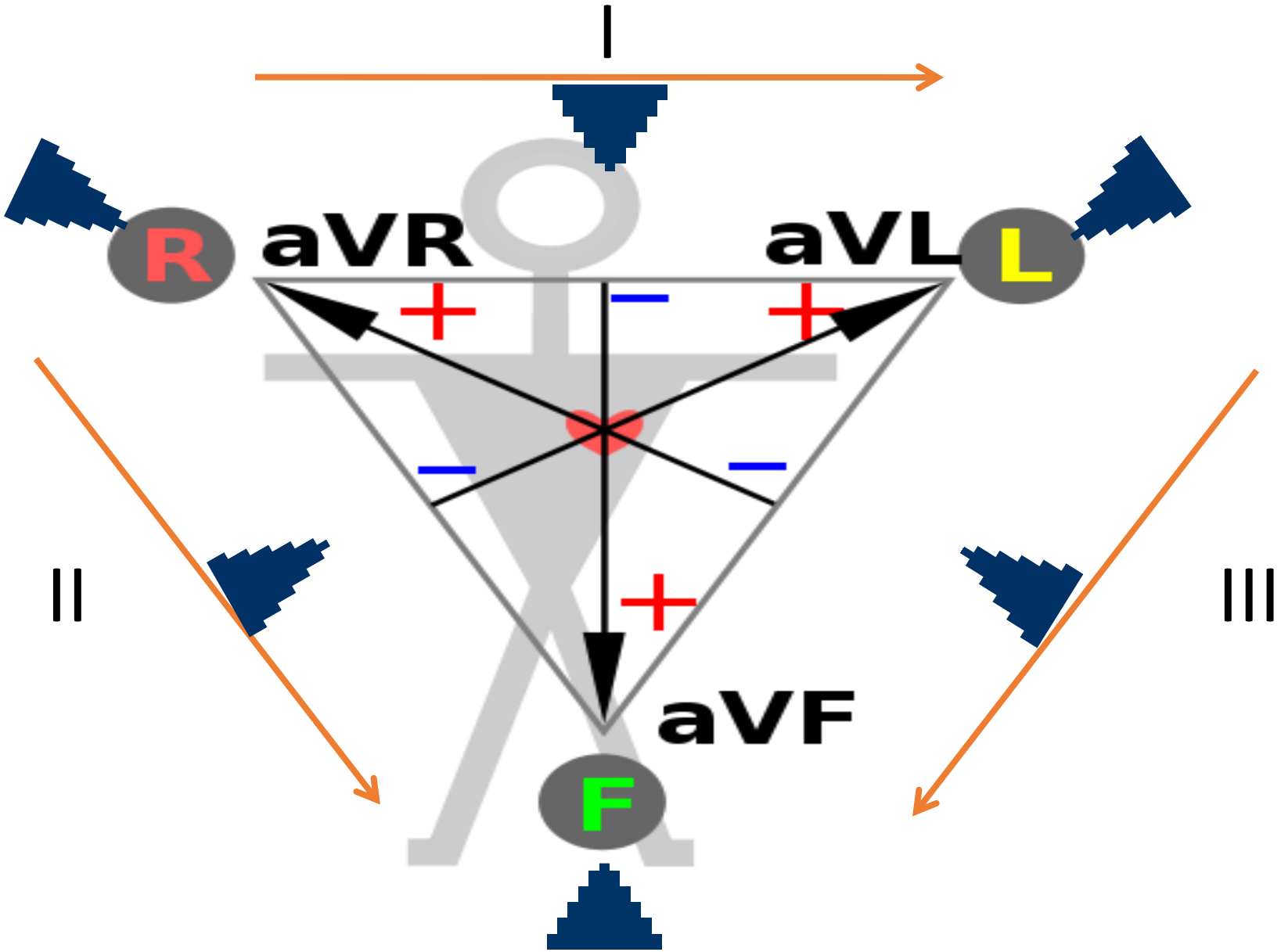


The Precordial Lead Axes

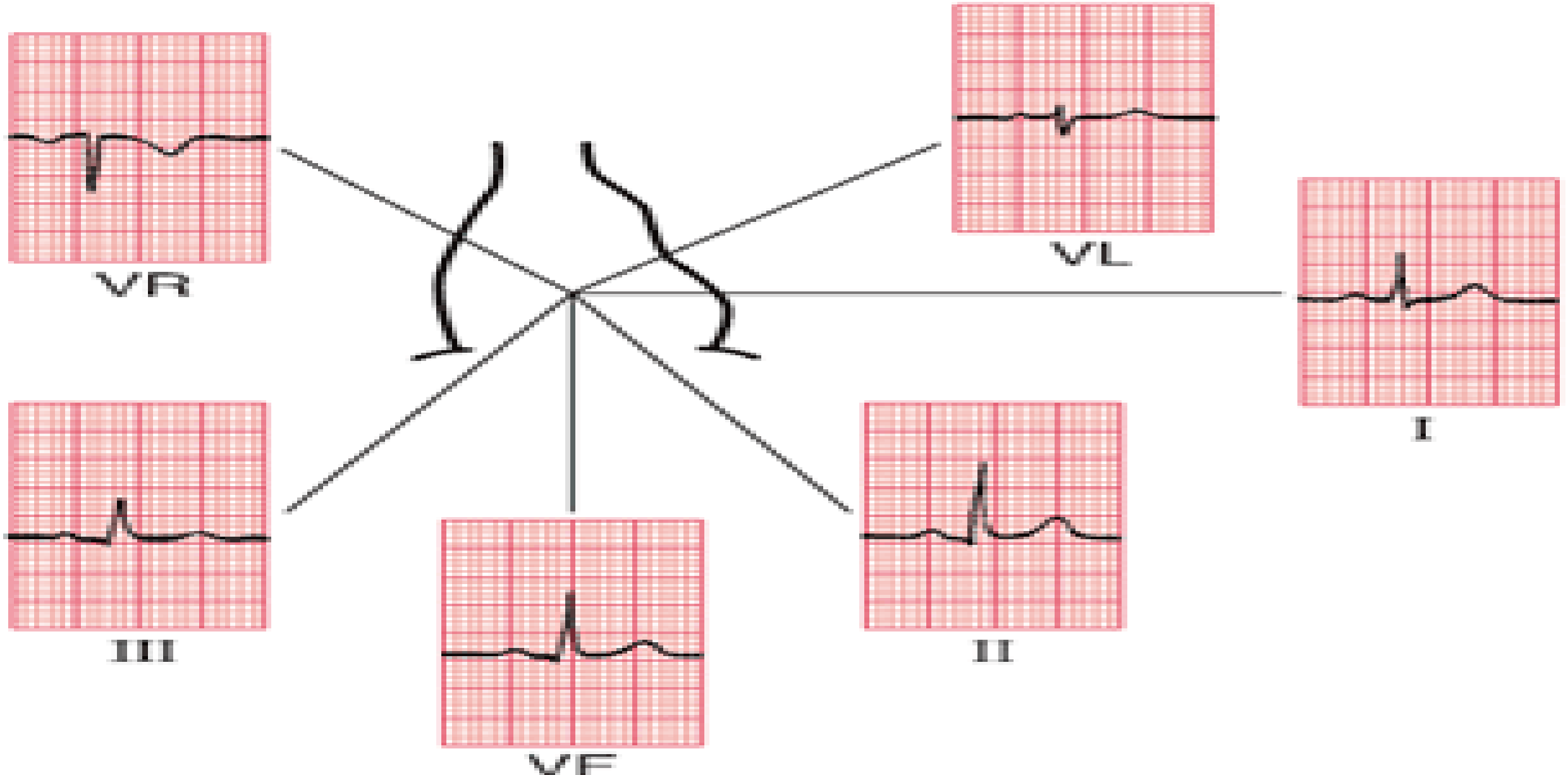
B



ECG



ECG

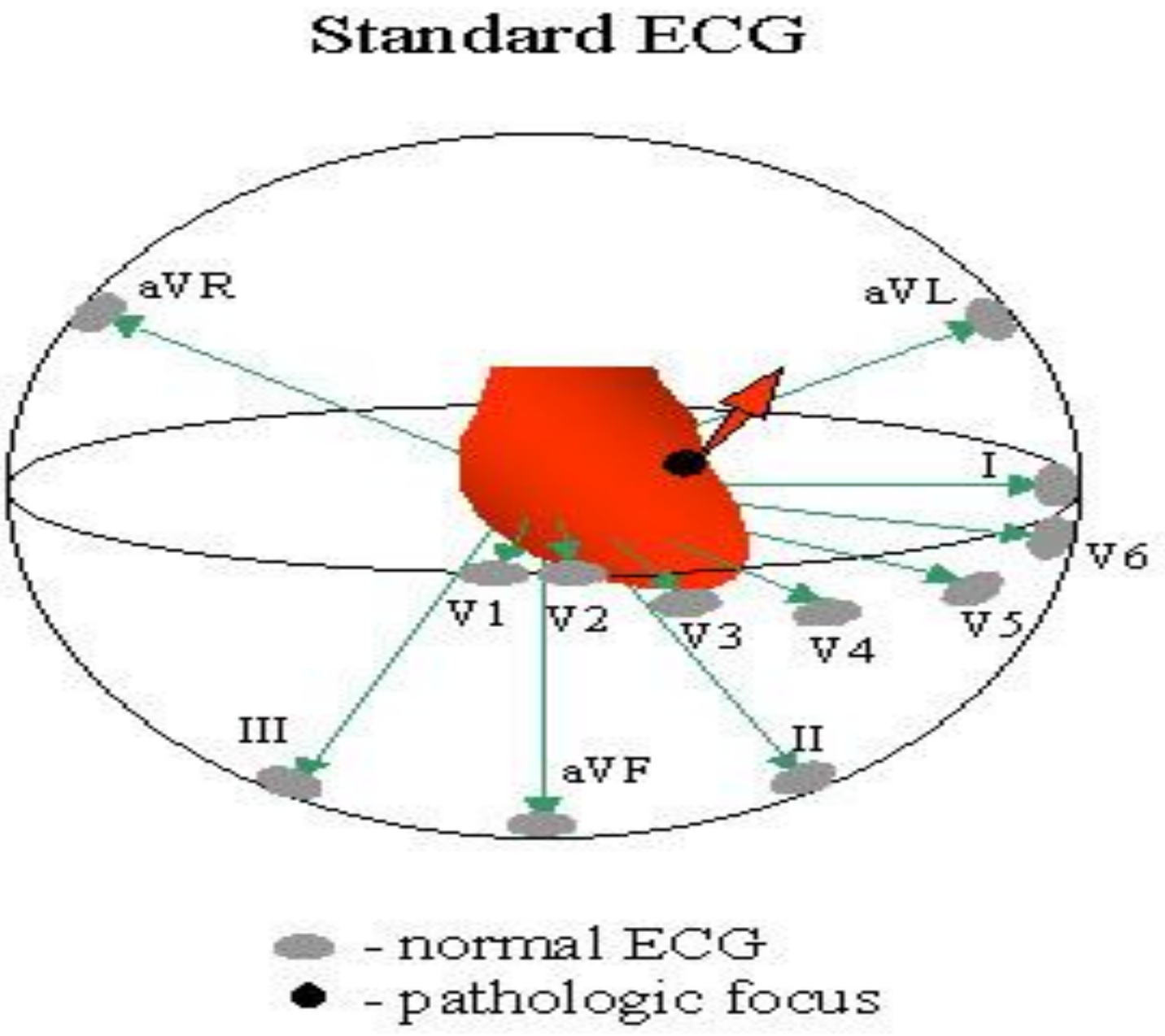


ECG leads

- Limb Leads
  - Bi-polar: I, II, III
  - Uni-polar: aVR, aVL, aVF

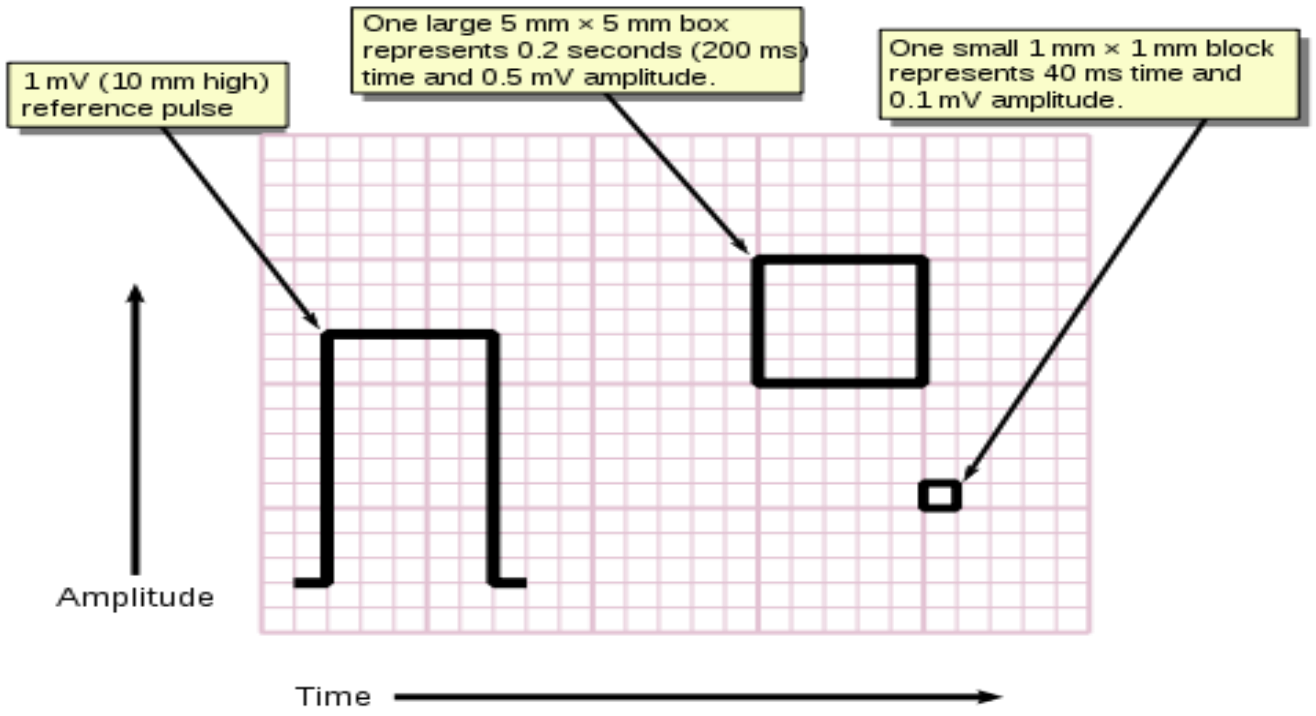
- Chest leads
  - Uni-polar: V1-2-3-4-5-6

ECG

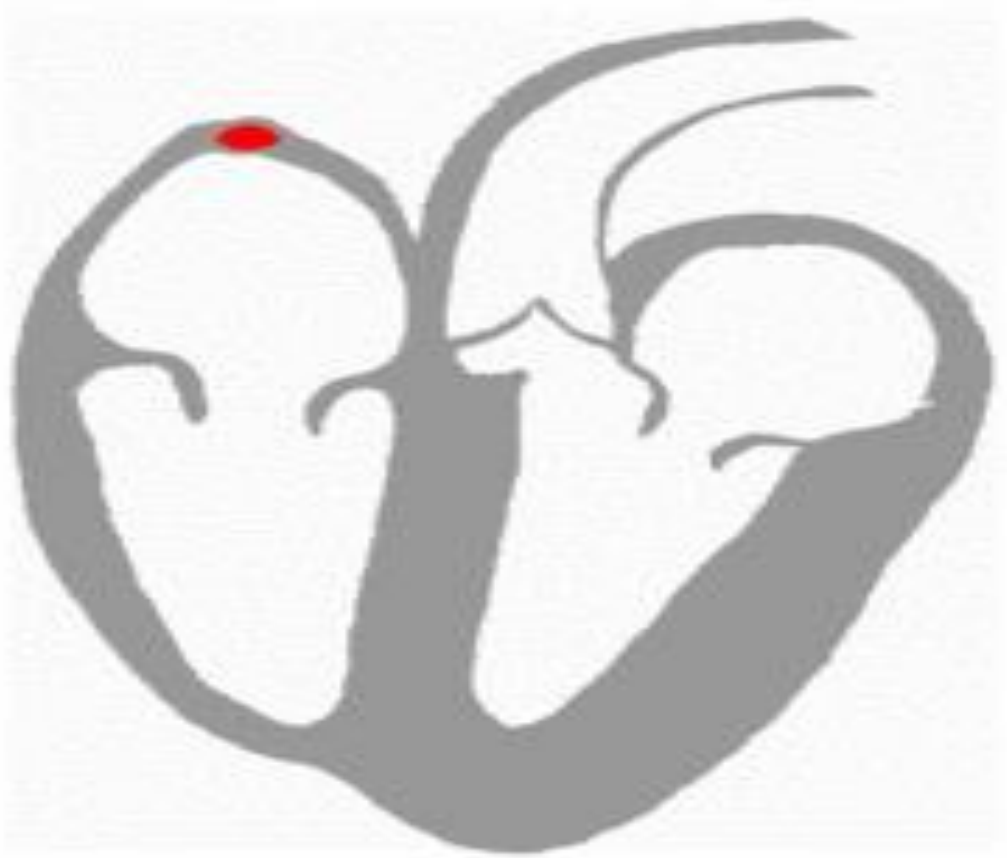

Standard ECG
aVR
aVL
I
V6
V1
V2
V3
V4
V5
III
aVF
II
- normal ECG
- pathologic focus

Electrocardiogram Graph Paper

- ECG paper
- Speed
  - 25mm/sec
  - 25 small Sq.
  - 5 big Sq./sec
- 1 min = 60 sec
- 1 min = 300 b sq
- 1 min = 1500 s sq

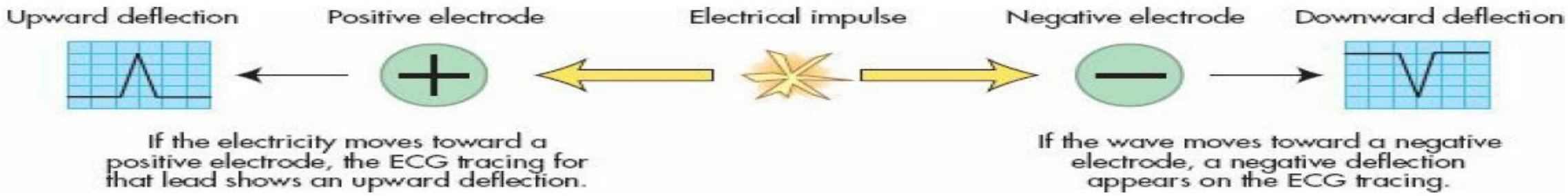


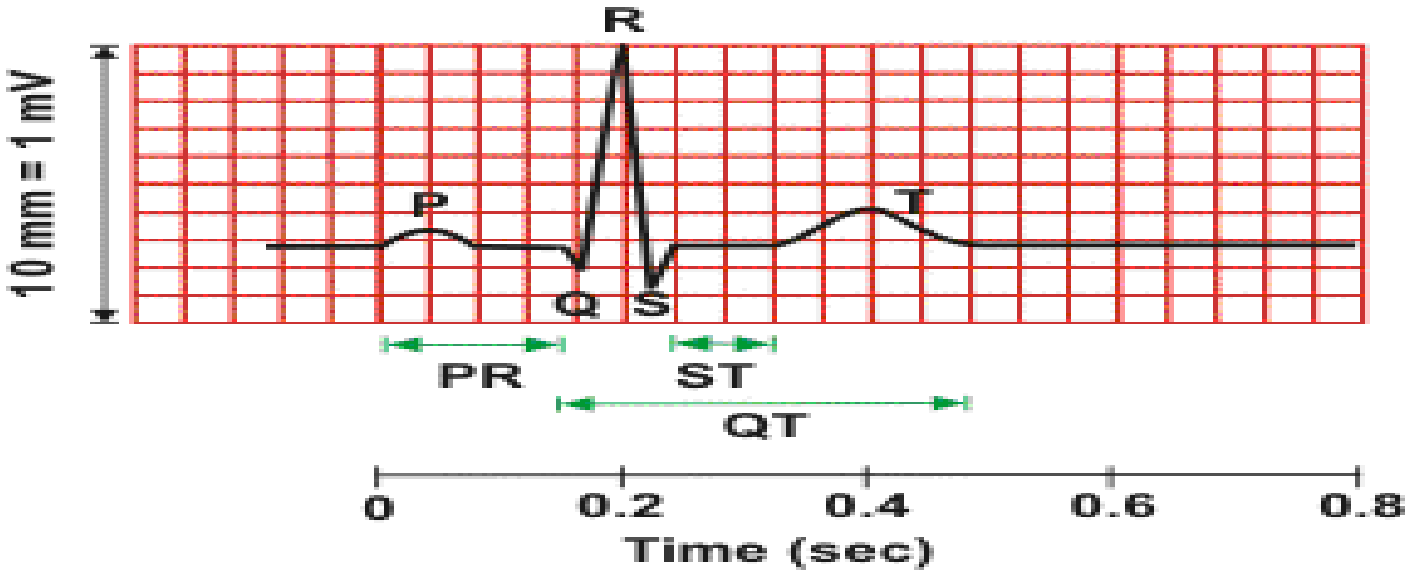
ECG





ECG





P wave (0.08 - 0.10 s)

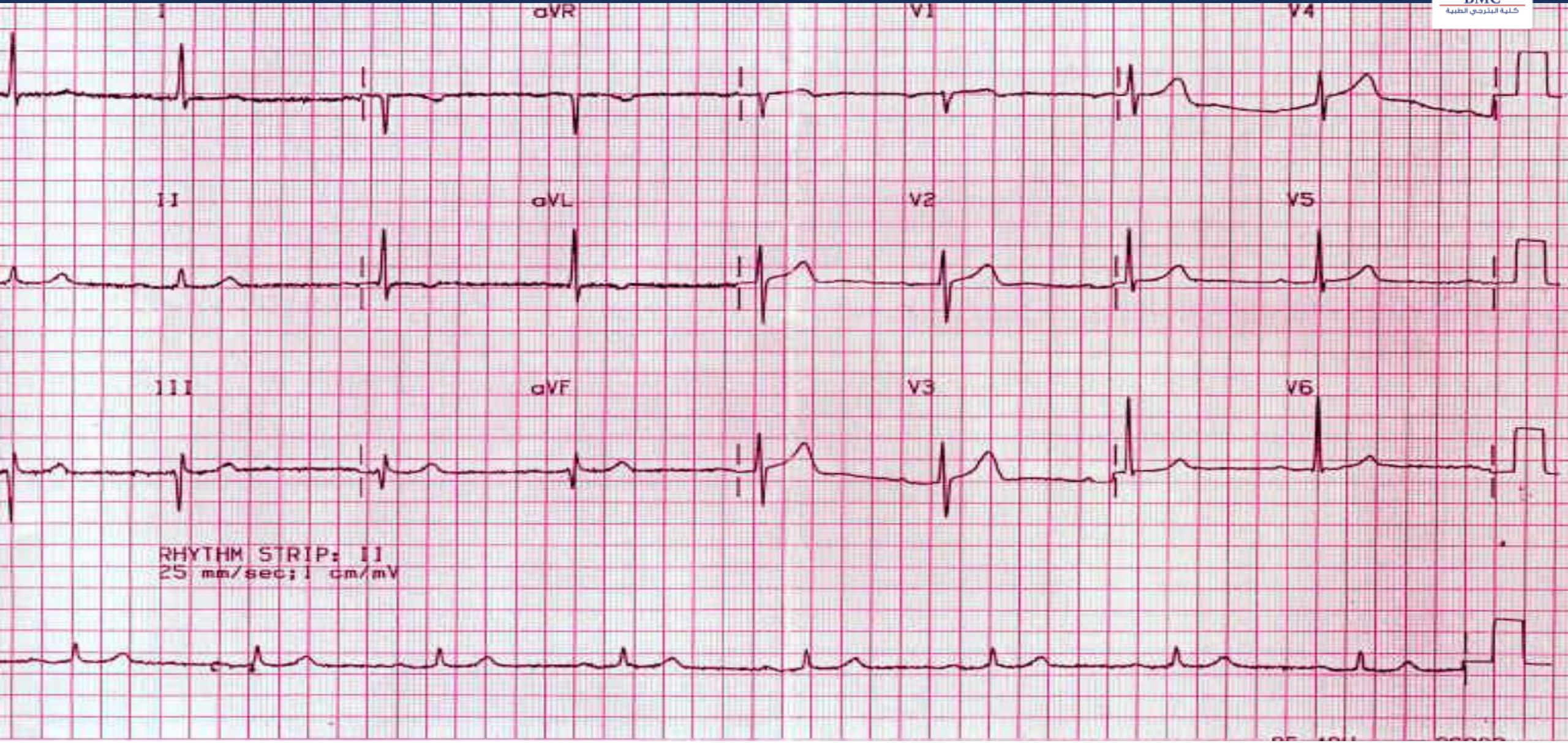
P-R interval (0.12 - 0.20 s)

QRS (0.06 - 0.10 s)

Q-T_c interval ($\leq$ 0.44 s)*

*$QT_c = QT/\sqrt{RR}$

ECG


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
RHYTHM STRIP: II
25 mm/sec; 1 cm/mV

ECG comment

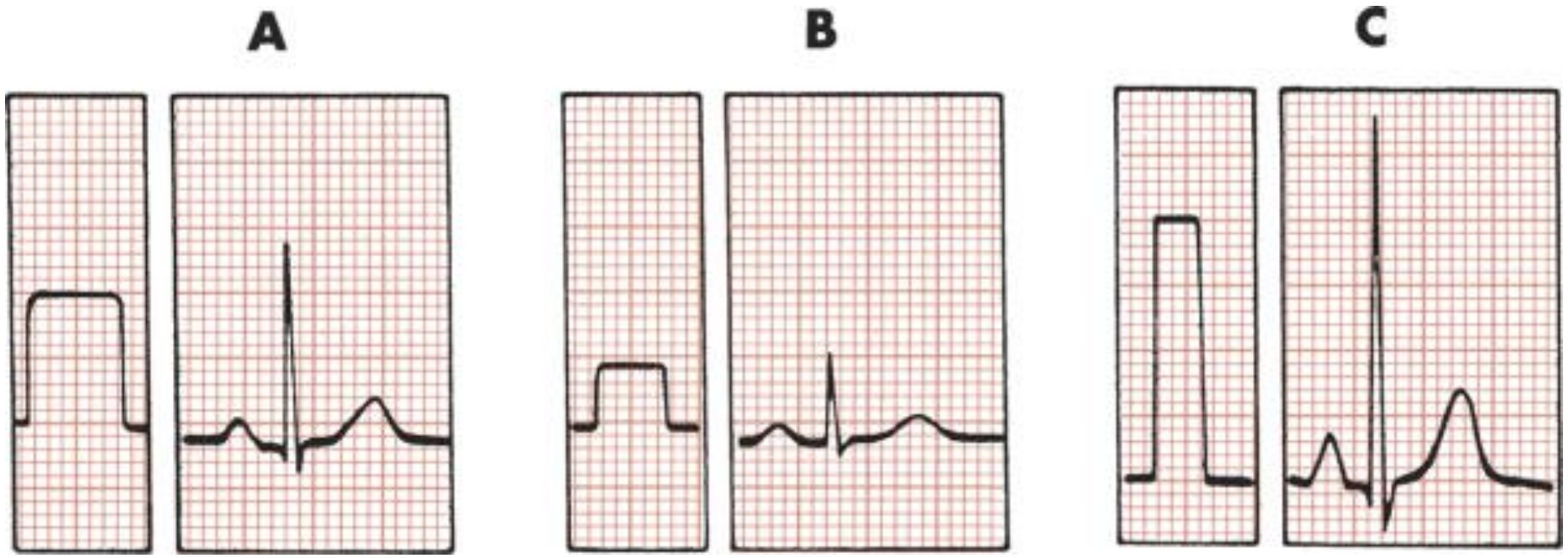
1. Identification
2. Standardization
3. Voltage
4. Axis
5. Rhythm
6. Rate
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

ECG comment

1. Identification
2. **Standardization**
3. Voltage
4. Axis
5. Rhythm
6. Rate
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

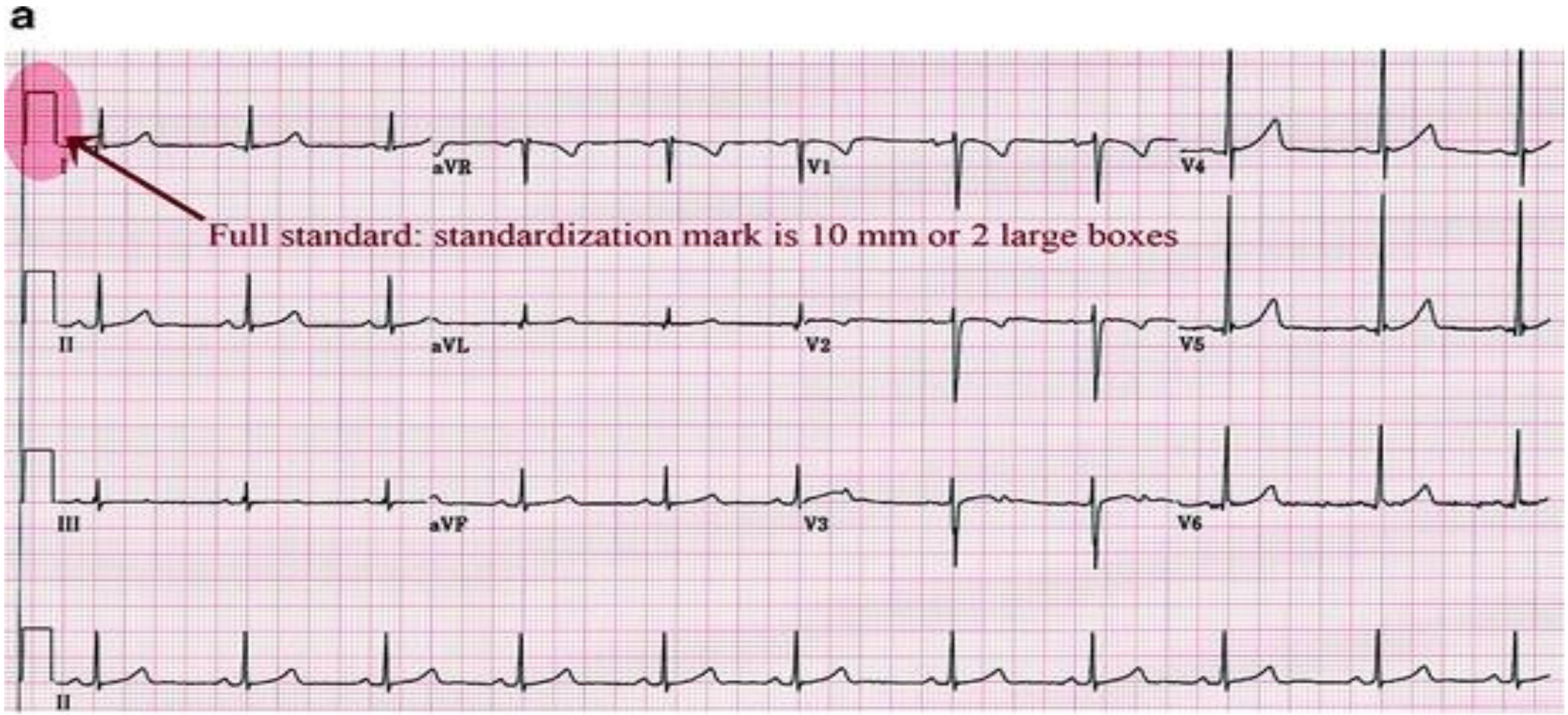
ECG comment

Standardization

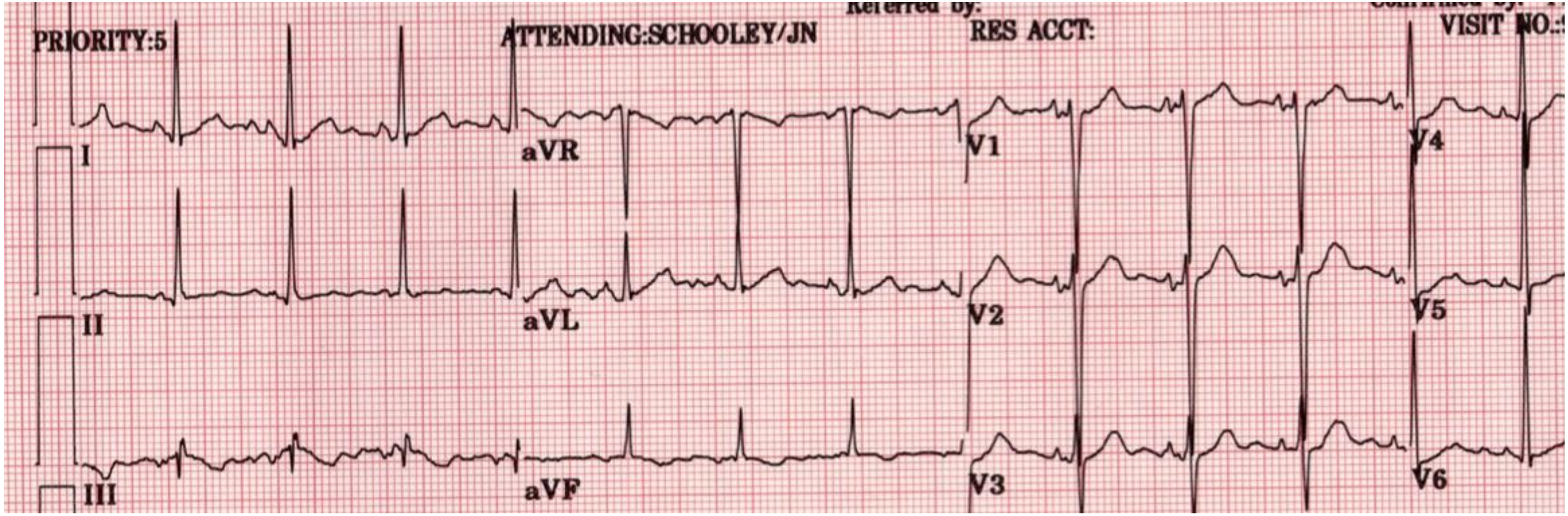


ECG comment

Standardization



Standardization



ECG comment

1. Identification
2. Standardization
3. **Voltage**
4. Axis
5. Rhythm
6. Rate
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

Voltage

- High voltage criteria
- Low voltage criteria
- Normal voltage

Voltage

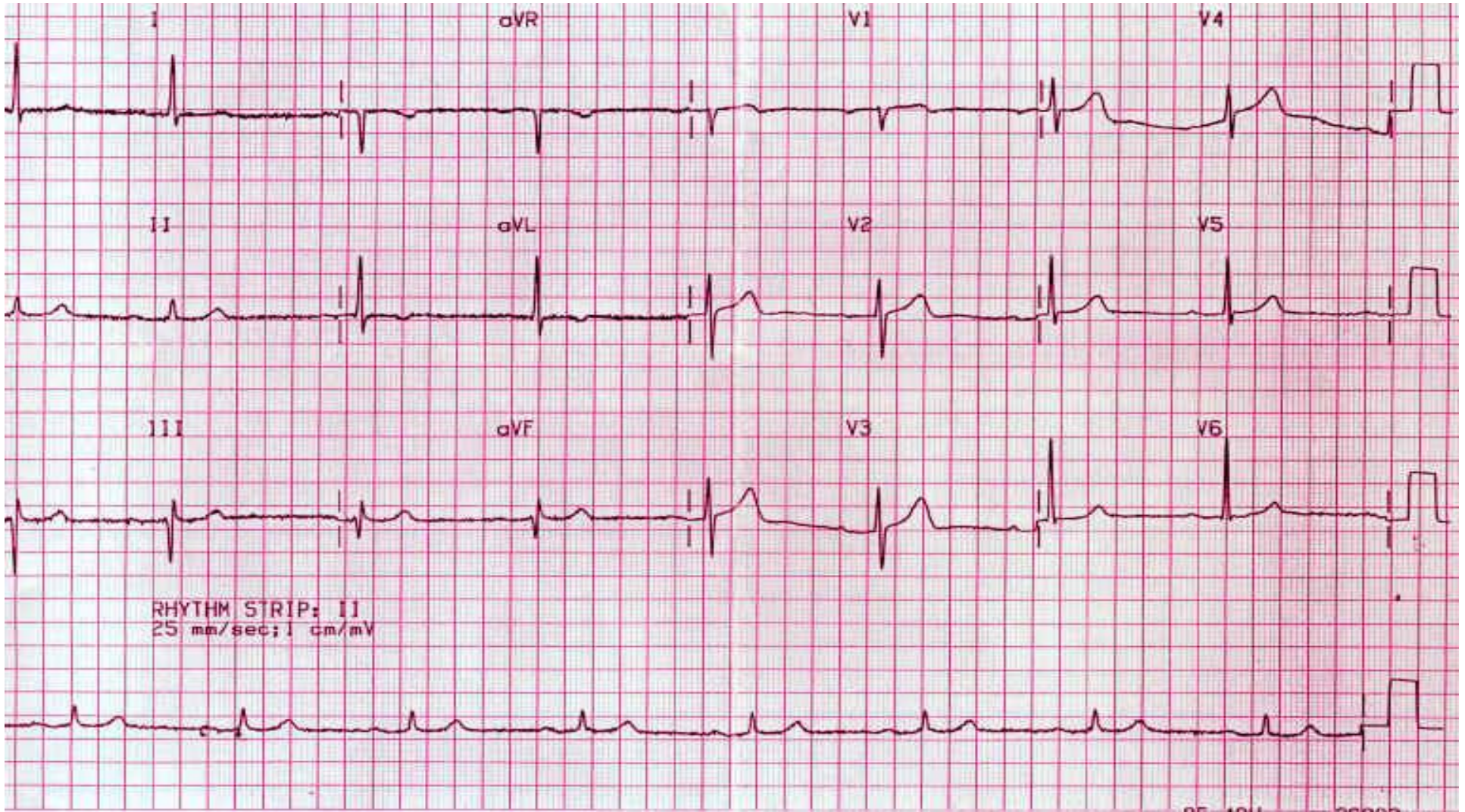
- High voltage criteria

  (SV1 or SV2) + (RV5 or RV6)

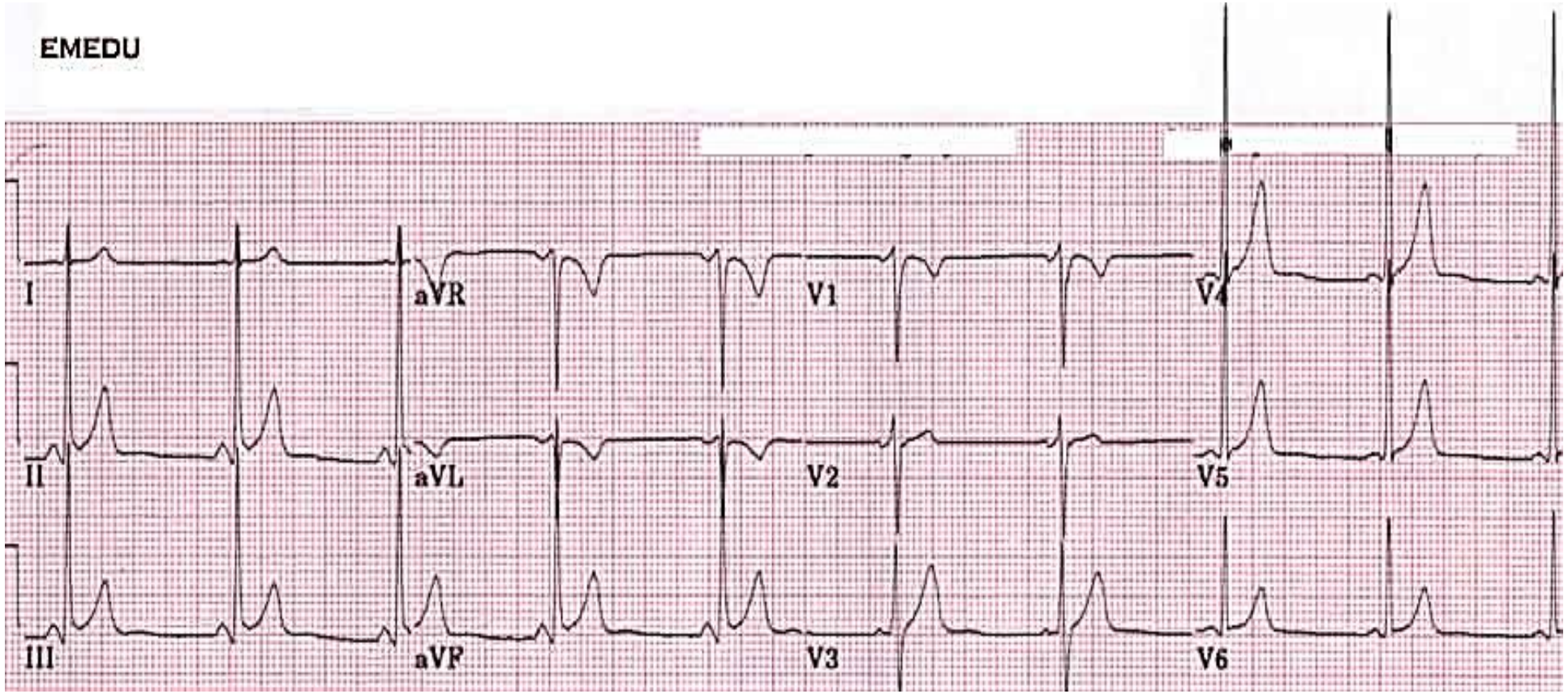
  ≥ 35 mm



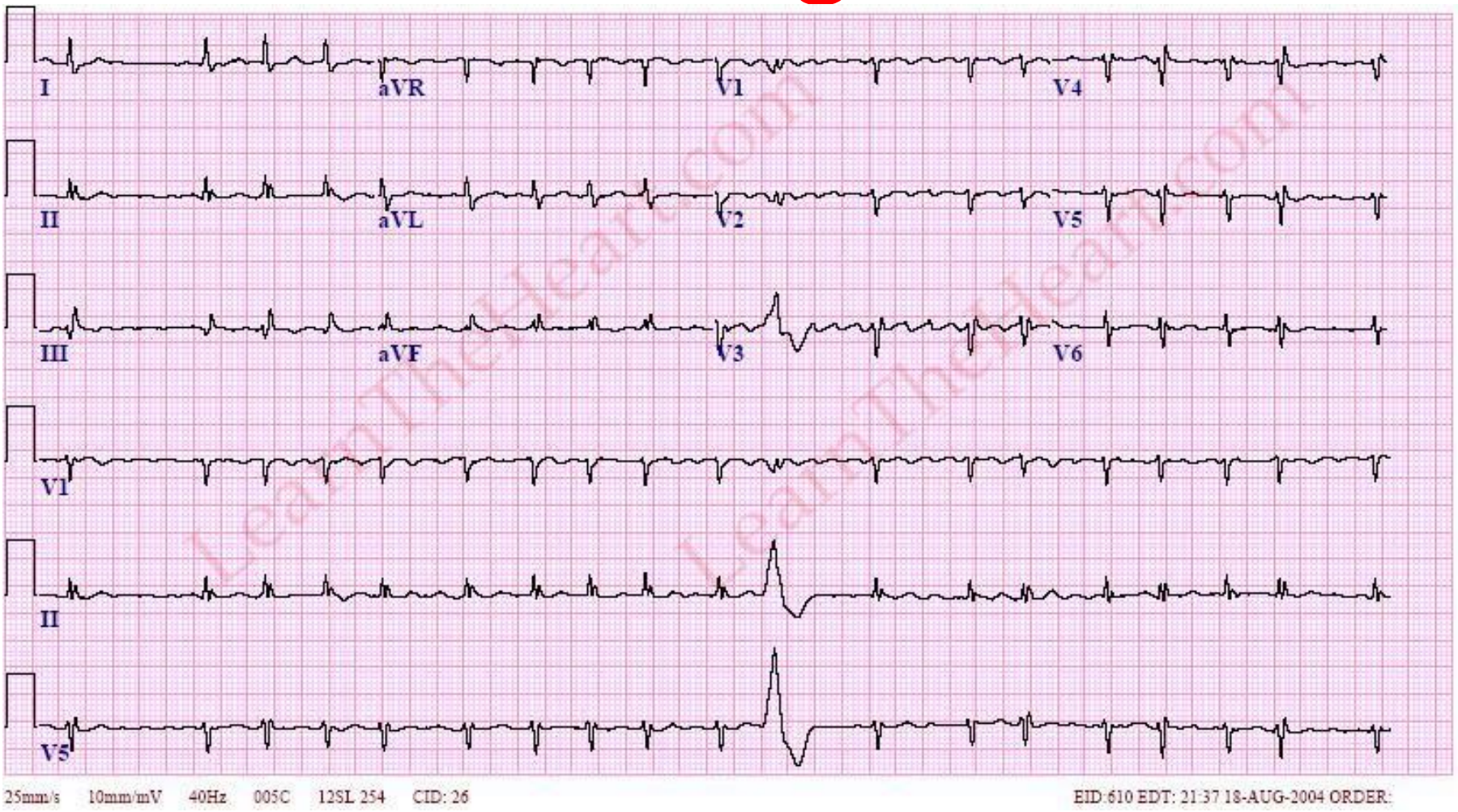
Voltage


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
RHYTHM STRIP: II
25 mm/sec; 1 cm/mV

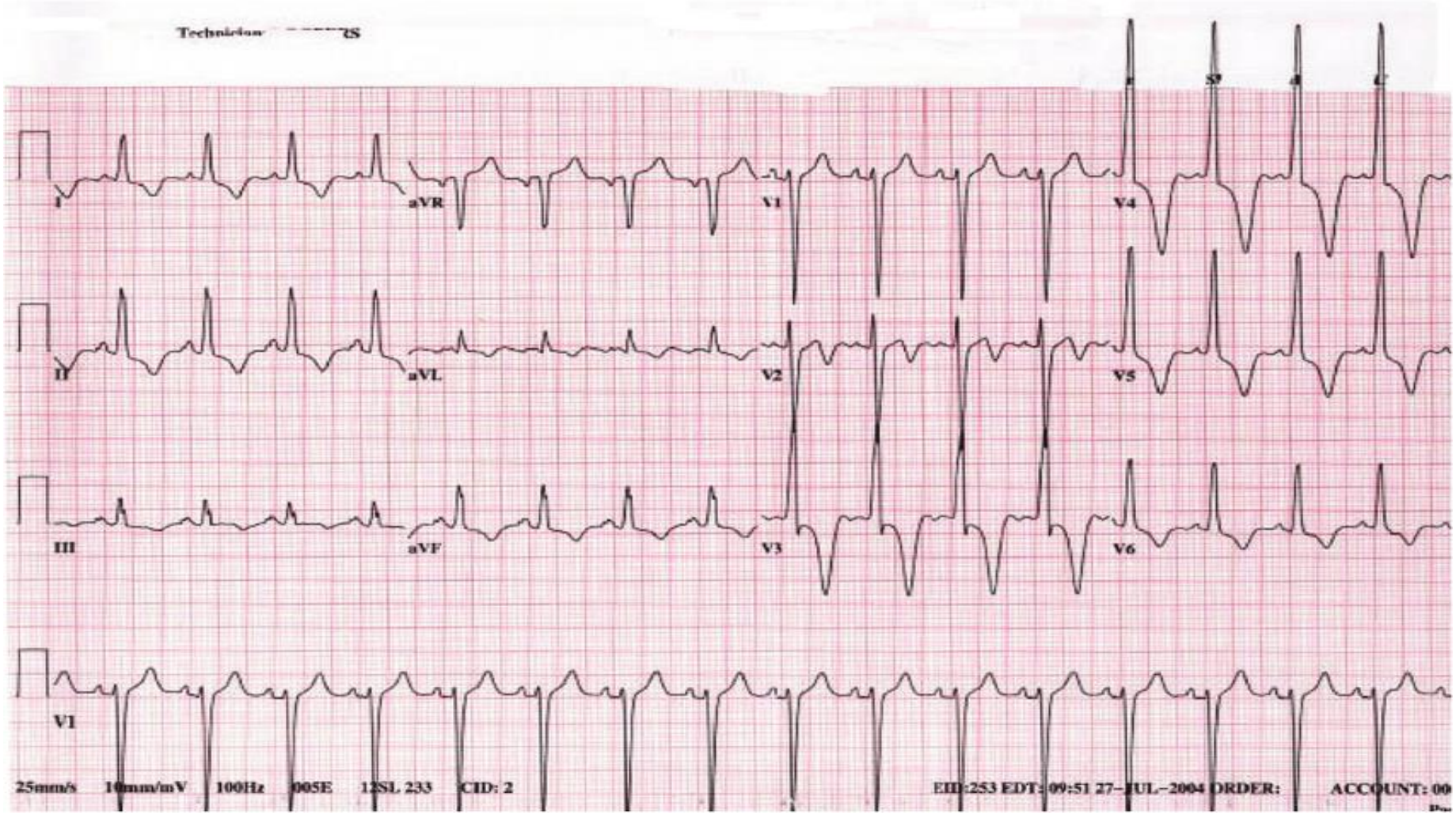
Voltage



Voltage


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
V1
II
V5
25mm/s 10mm/mV 40Hz 005C 12SL 254 CID: 26
EID:610 EDT: 21:37 18-AUG-2004 ORDER:

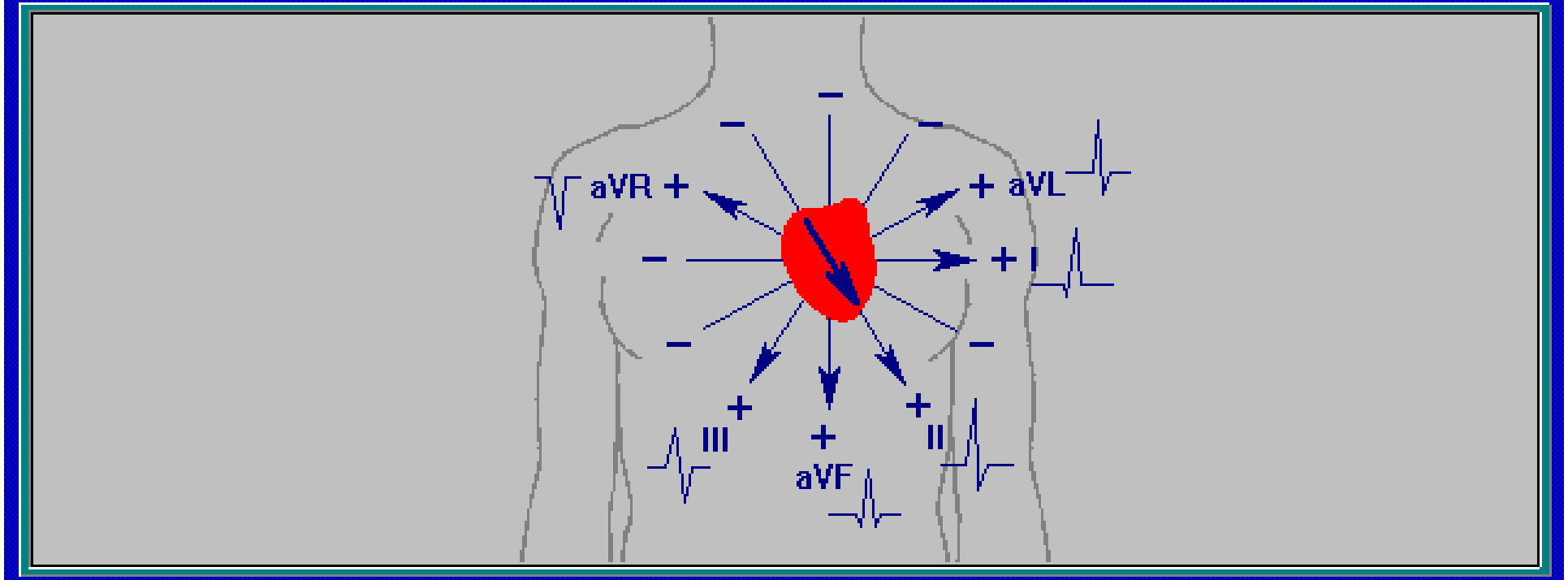
Voltage


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
V1
25mm/s
100Hz
12SL 233
CID: 2
EDT: 09:51 27-JUL-2004 ORDER:
ACCOUNT:

ECG comment

1. Identification
2. Standardisation
3. Voltage
4. **Axis**
5. Rate
6. Rhythm
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

Axis



THE QRS AXIS

Since each lead views ventricular depolarization from a different angle, the shape of the QRS complex is different in each lead. This shape is determined by the direction of ventricular depolarization. The direction of ventricular depolarization as depicted in a lead is called lead axis. Examining the lead axis provides the tool for which the QRS axis is determined. As shown above, each lead axis is oriented in a specific direction in relation to the heart and each lead axis has a positive (+) and negative (-) end.

Axis

1. Normal Axis
2. Right Axis deviation
3. Left Axis deviation
4. Odd Axis (No Man's Land)

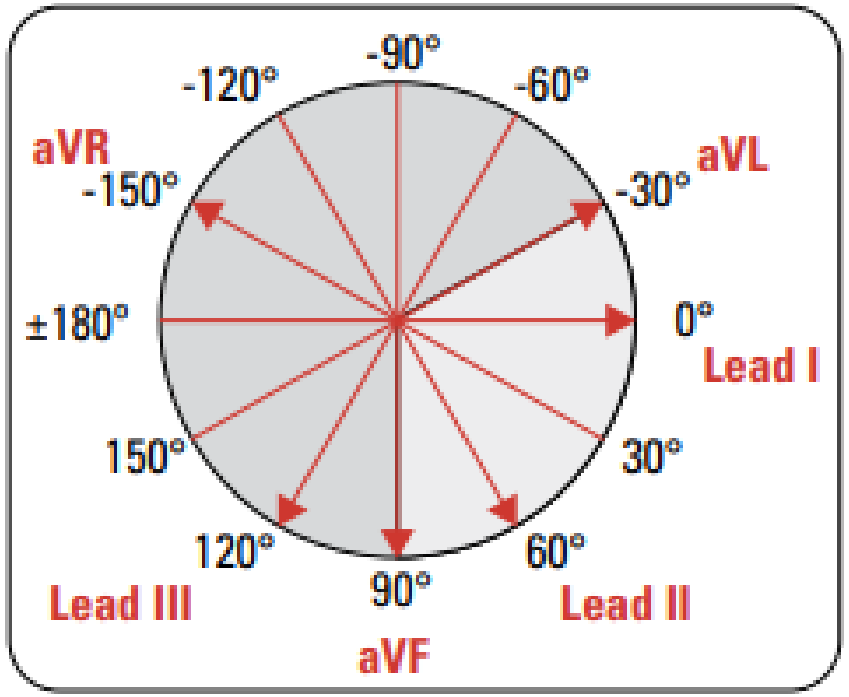
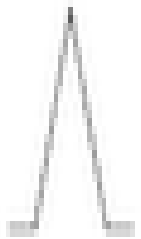
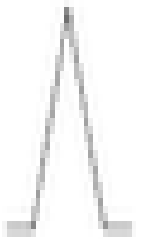
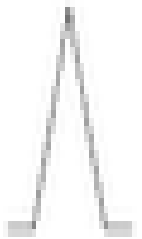
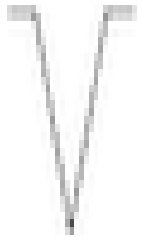
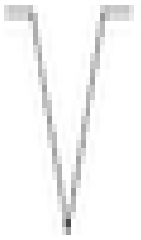
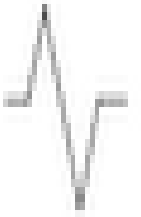
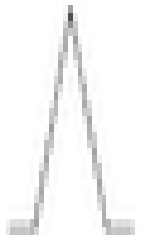
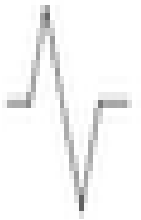
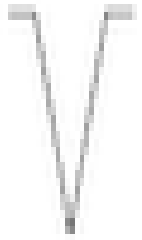
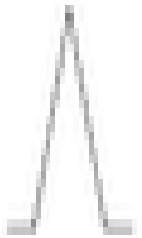
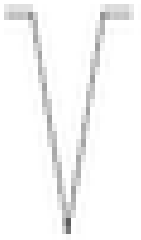
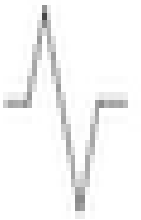
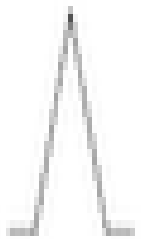
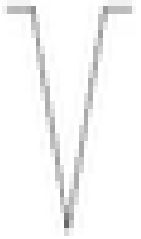
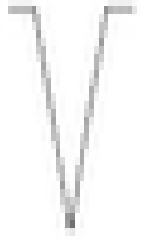
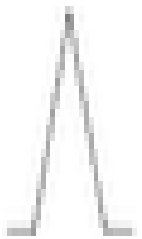
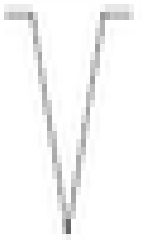
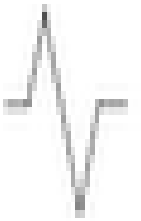


Figure 7. Axial reference system
Each lead contains a (+) area displayed by the bold arrows. Impulses traveling toward the positive region of the lead result in an upward deflection in that lead. Normal QRS axis is between -30° and +90°

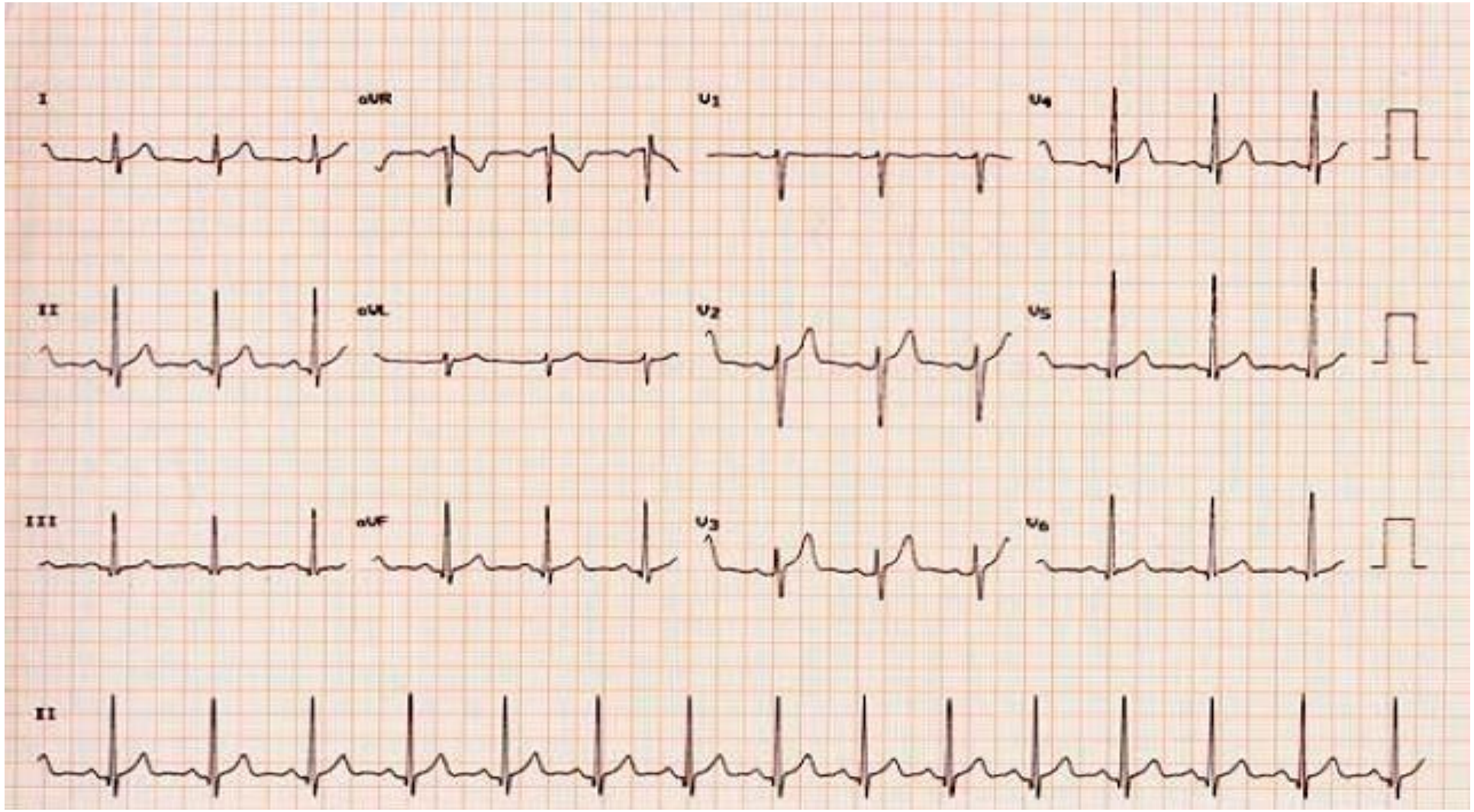
ECG

Axis

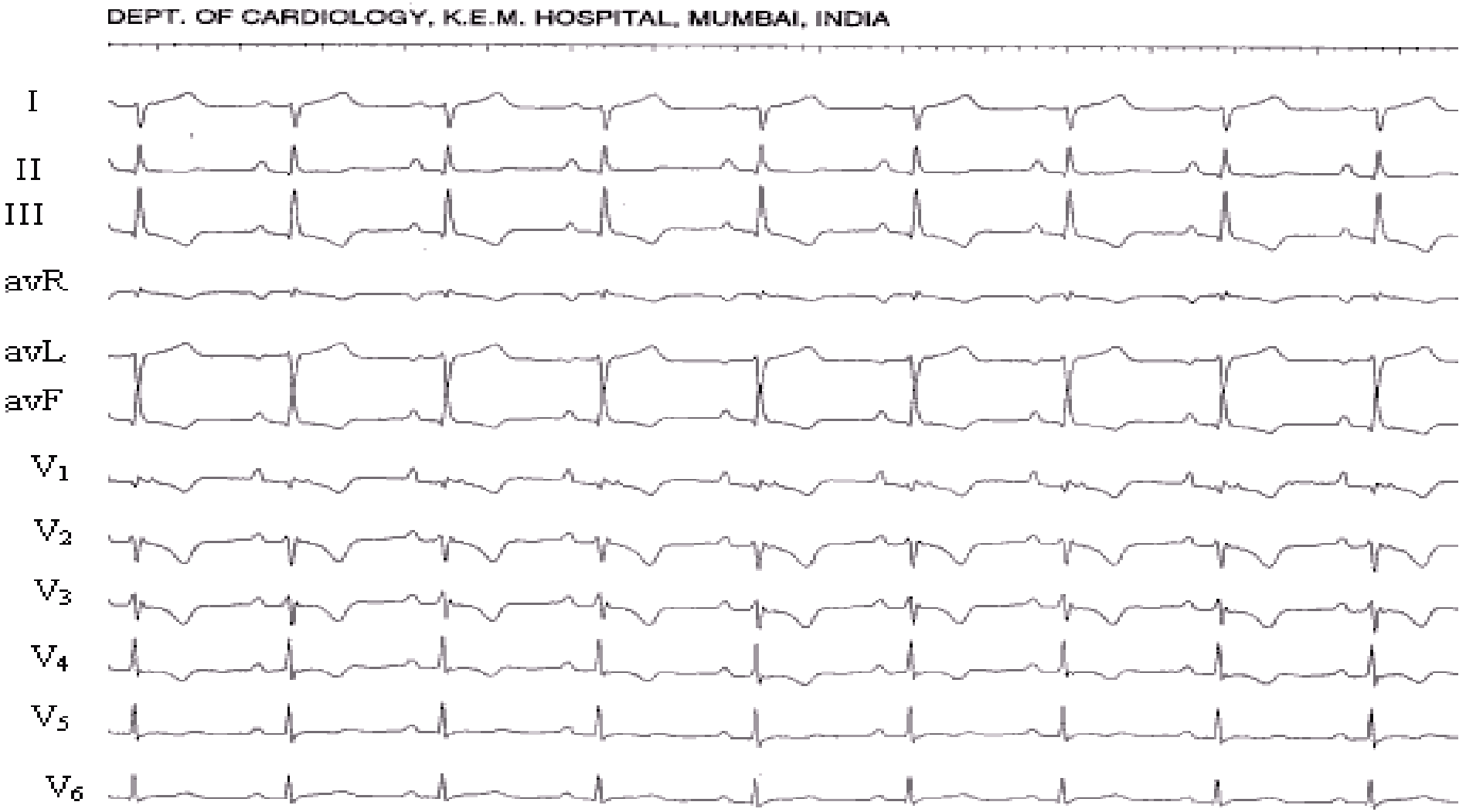
	Normal Axis 0 to 90	Left Axis Physiological 0 to -30	Left Axis Pathological -30 to -90	Right Axis 90 to 180	Extreme Axis -90 to -180	Indeterminate Axis ?
Lead I						
Lead II						
Lead III						

ECG

Axis

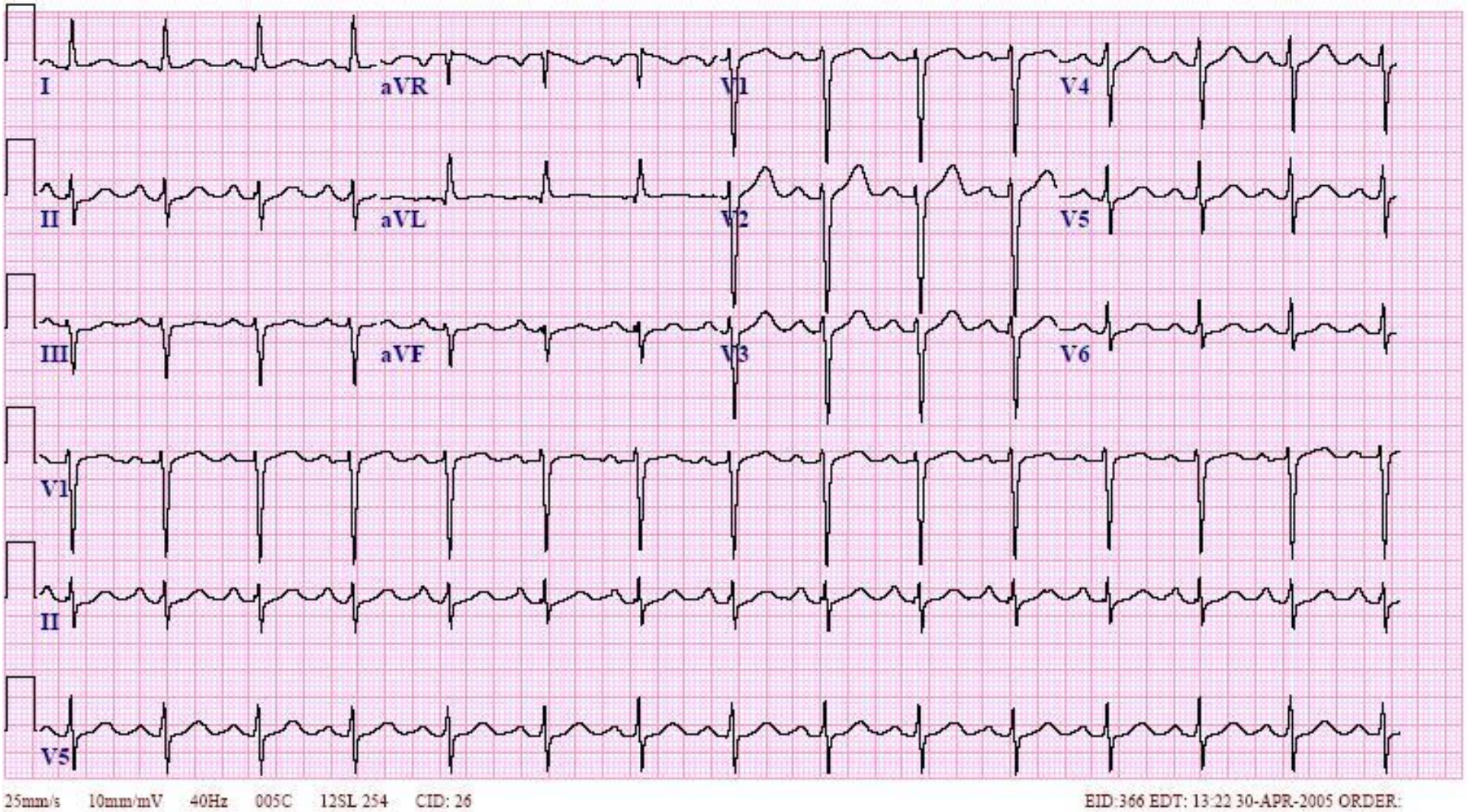


Axis

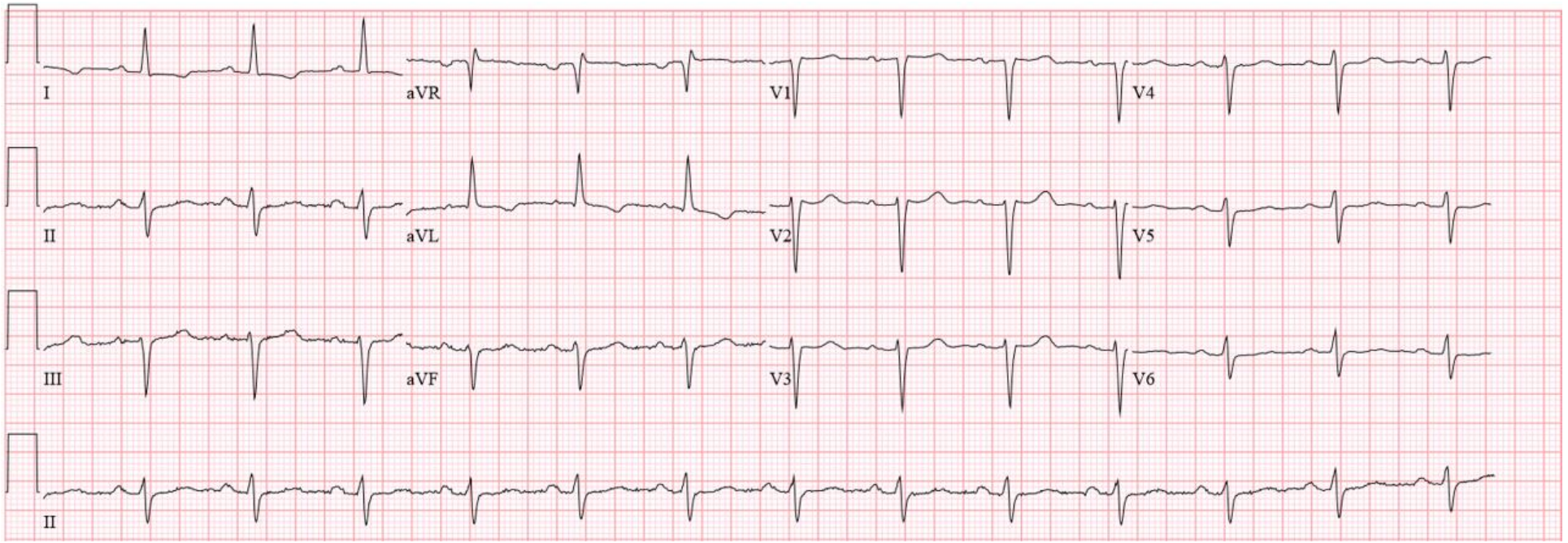

DEPT. OF CARDIOLOGY, K.E.M. HOSPITAL, MUMBAI, INDIA
I
II
III
avR
avL
avF
V1
V2
V3
V4
V5
V6

ECG

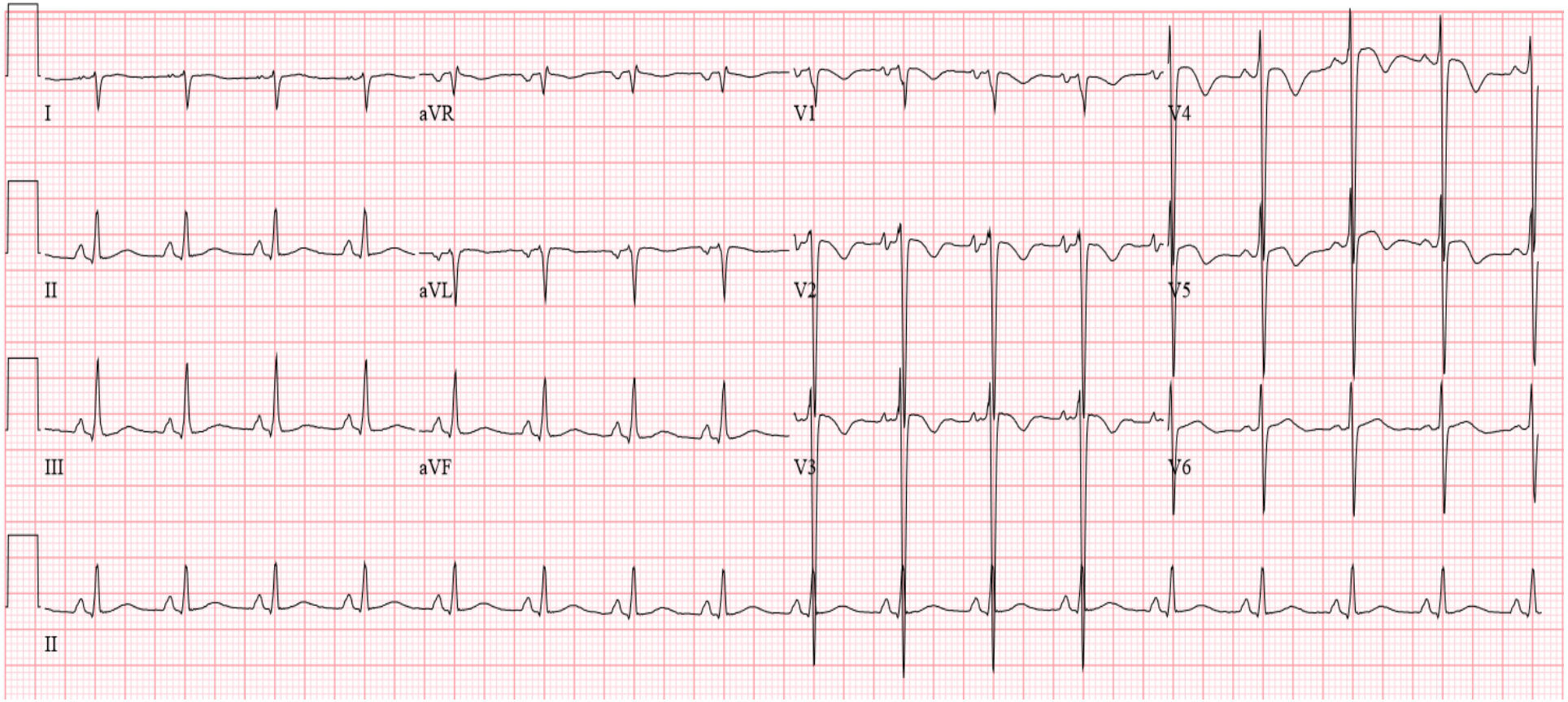
Axis


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
V1
II
V5
25mm/s 10mm/mV 40Hz 005C 12SL 254 CID: 26
EID:366 EDT: 13:22 30-APR-2005 ORDER:

Axis



Axis



ECG comment

1. Identification
2. Standardization
3. Voltage
4. Axis
5. **Rhythm**
6. Rate
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

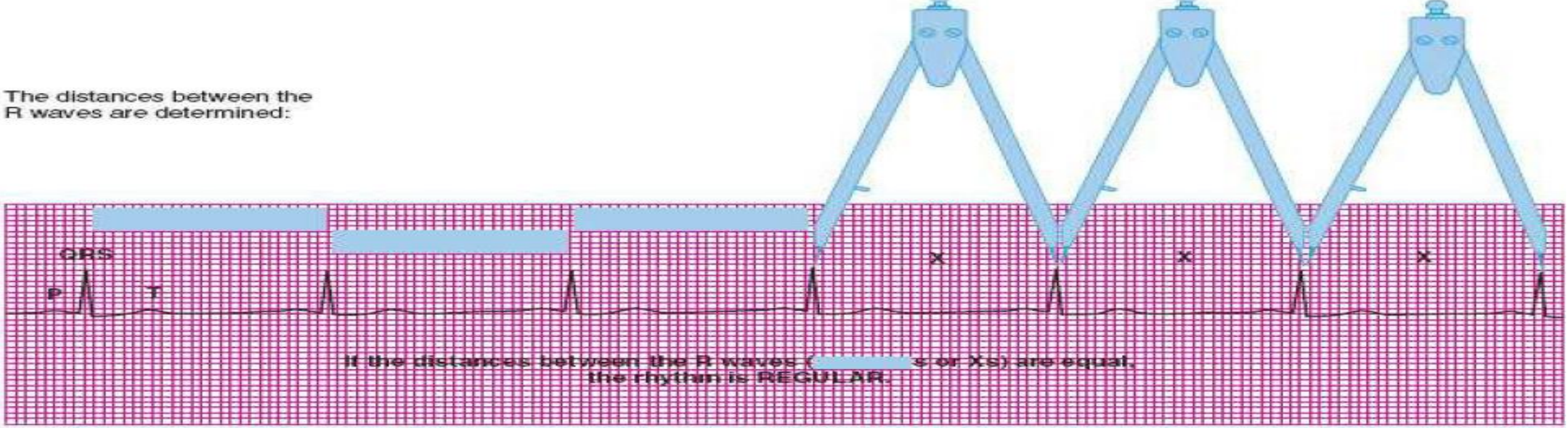
ECG comment

Rhythm

1. Regular

2. Irregular

The distances between the R waves are determined:



1. by estimating the R-R intervals,

2. by measuring the R-R intervals with ECG calipers,* or

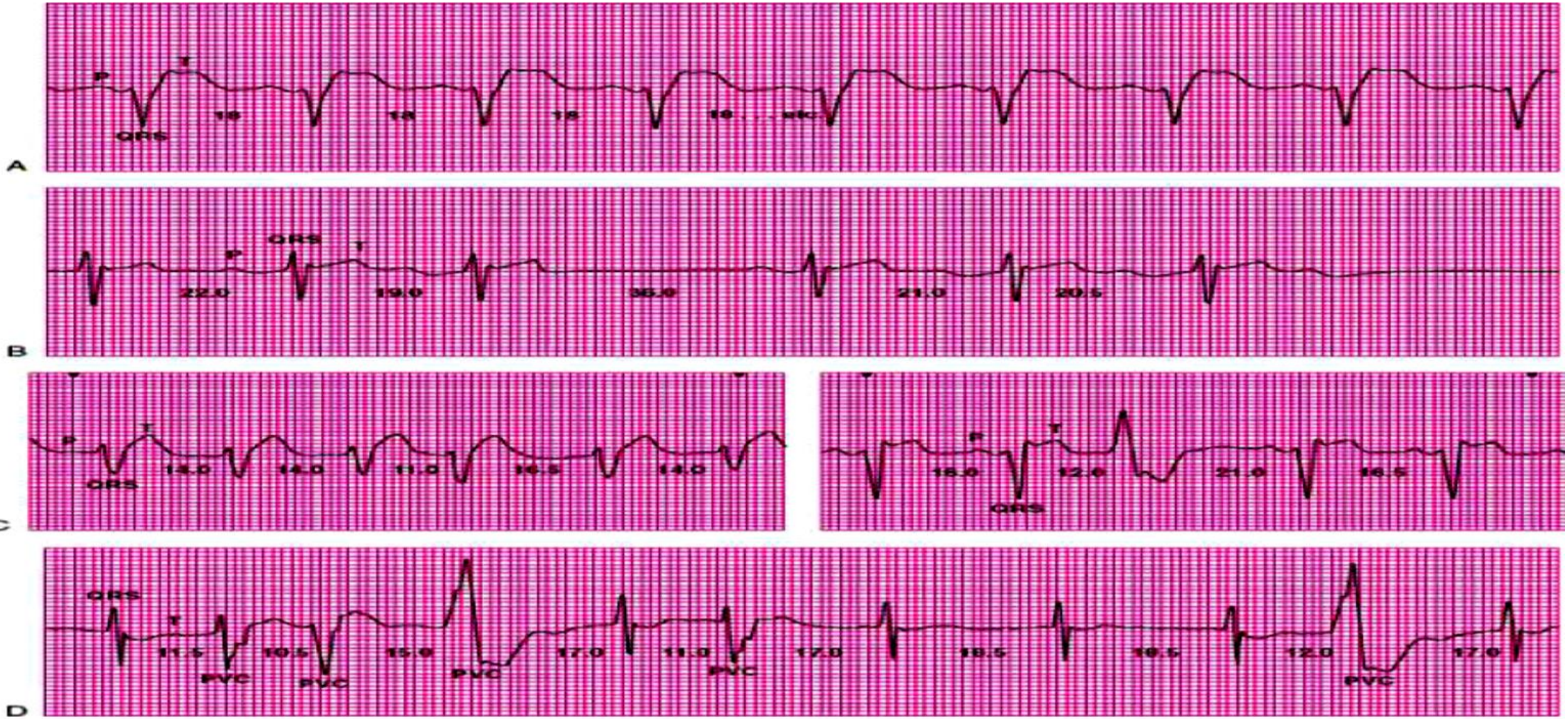


3. by counting the small squares between the R waves.

* If calipers are not available, mark off the distance between two R waves on a piece of paper and compare this distance with the other R-R intervals.

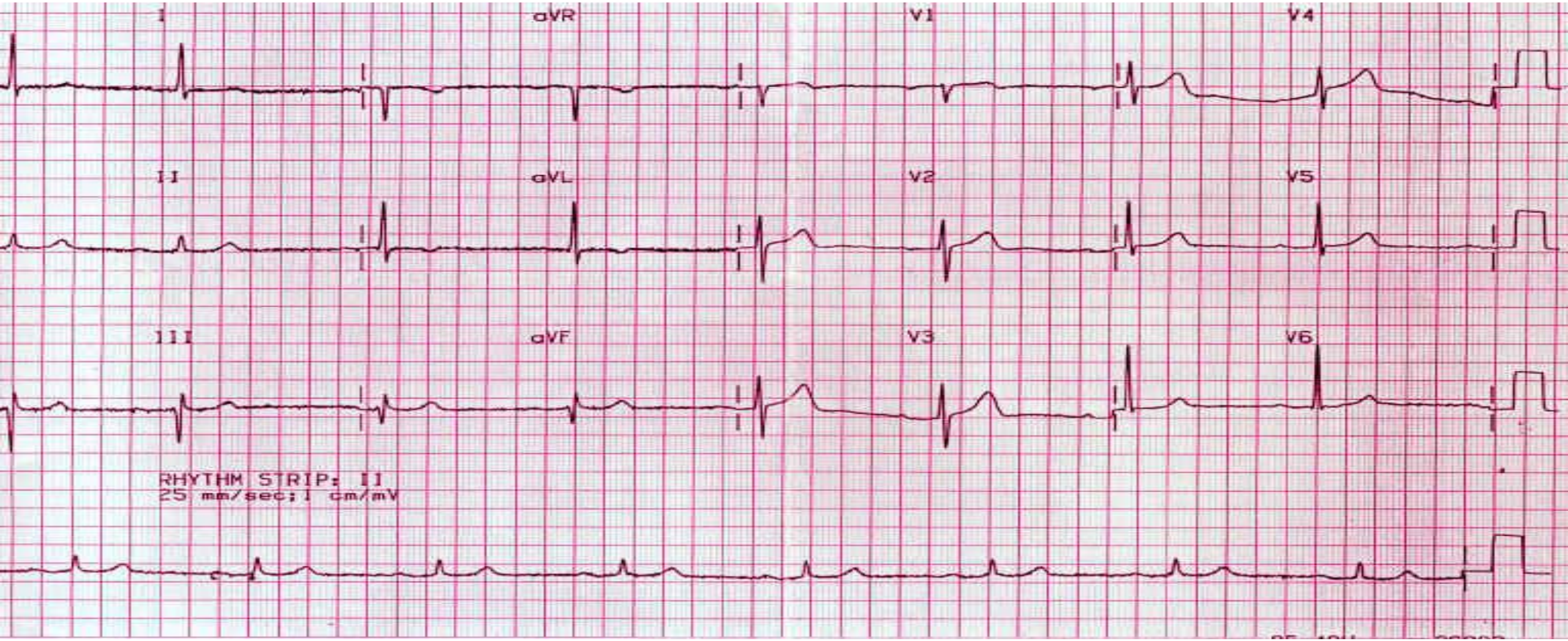
ECG comment

Rhythm



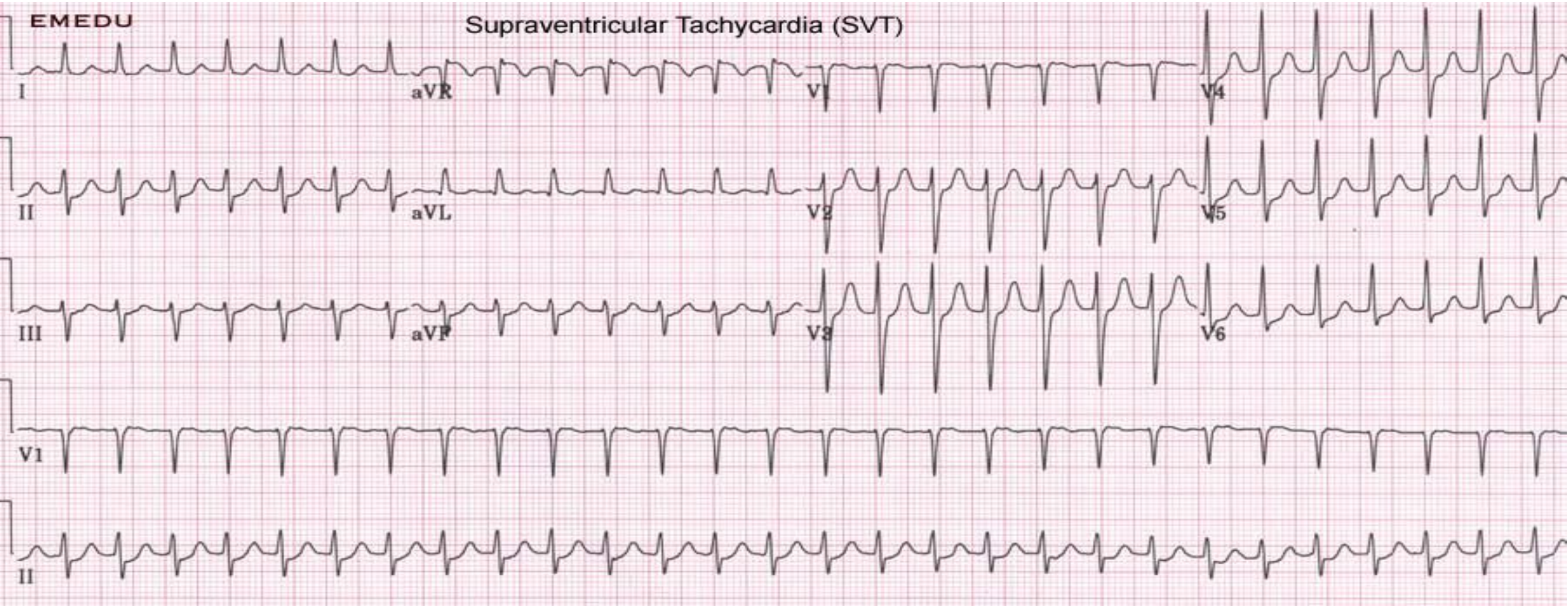
ECG comment

Rhythm



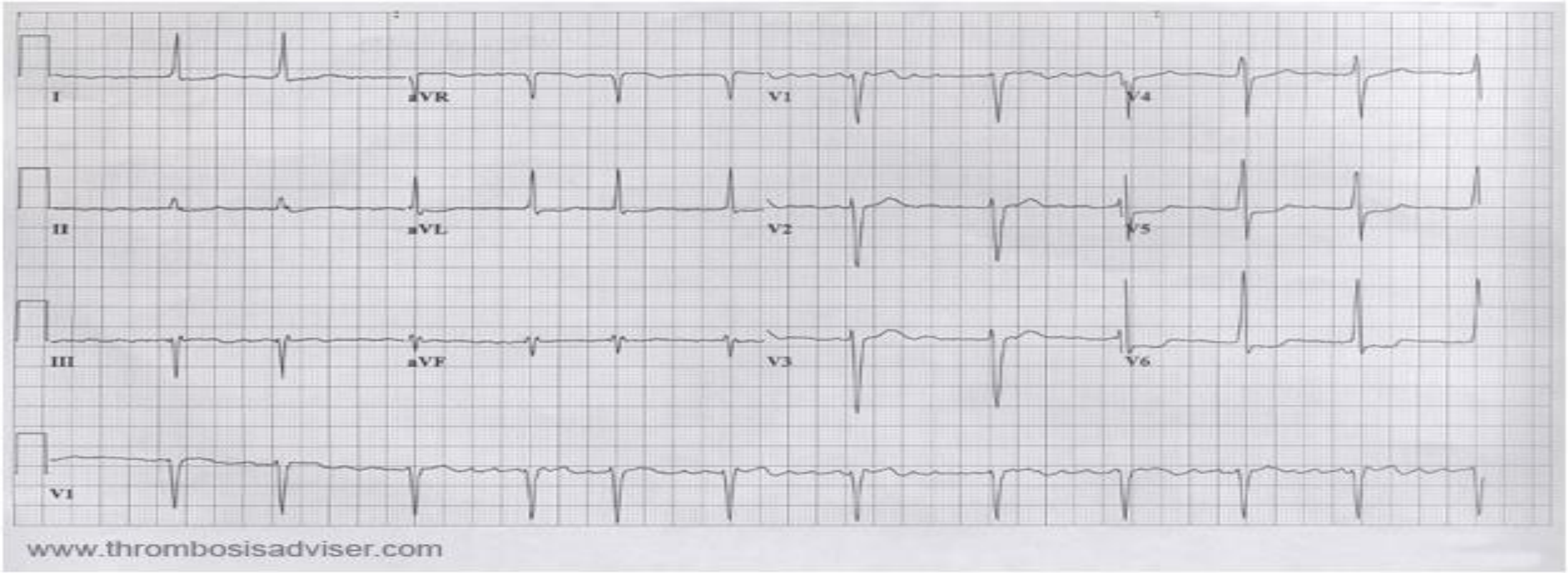
ECG comment

Rhythm



ECG comment

Rhythm



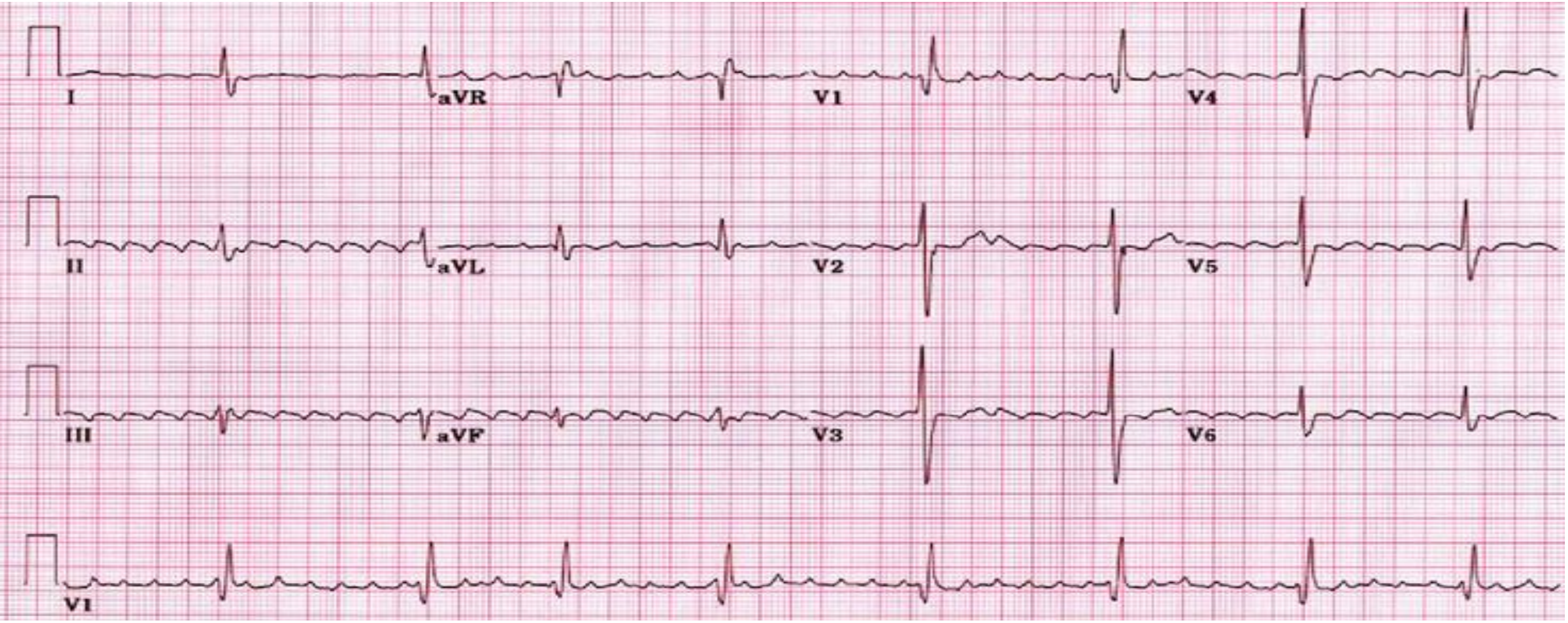
ECG comment

Rhythm



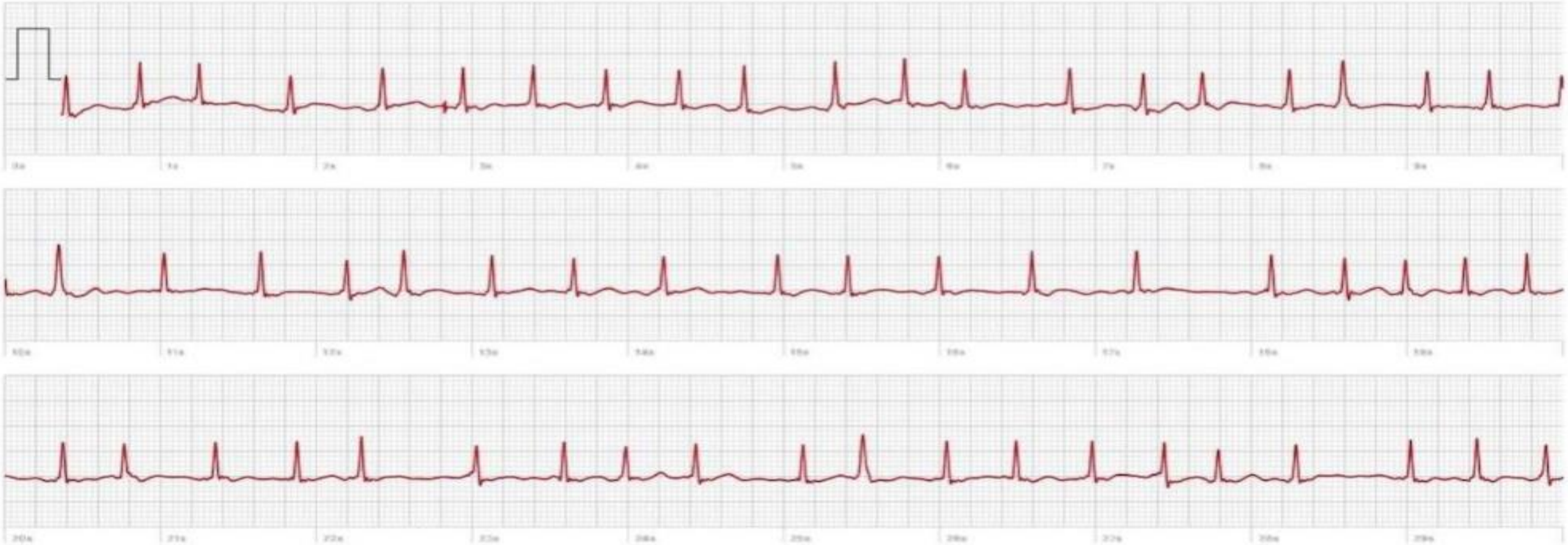
ECG comment

Rhythm



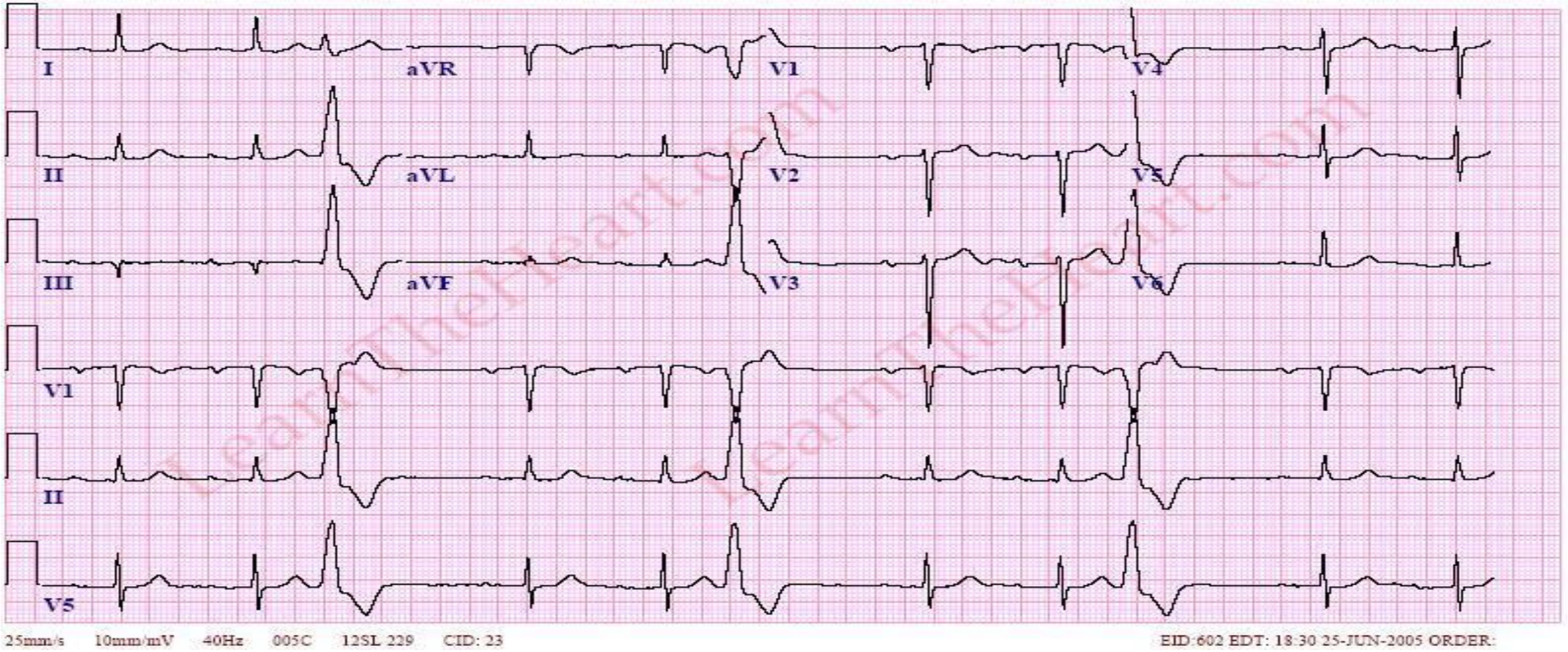
ECG comment

Rhythm



ECG comment

Rhythm

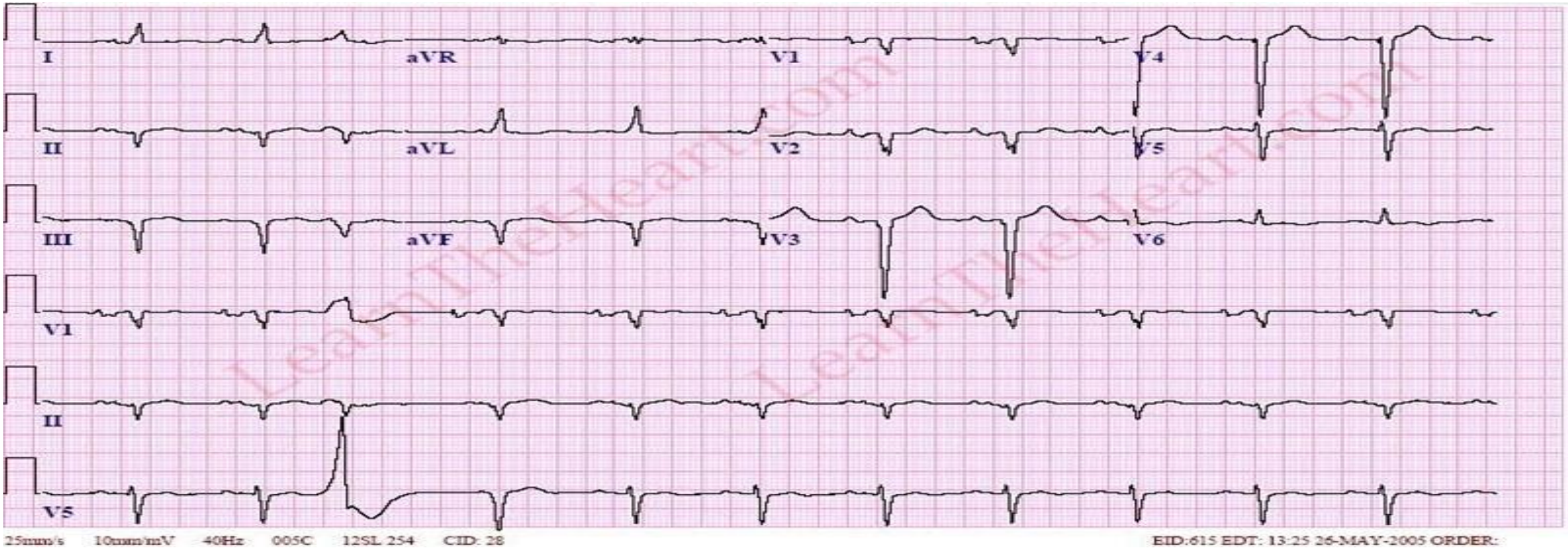


ECG comment

Rhythm



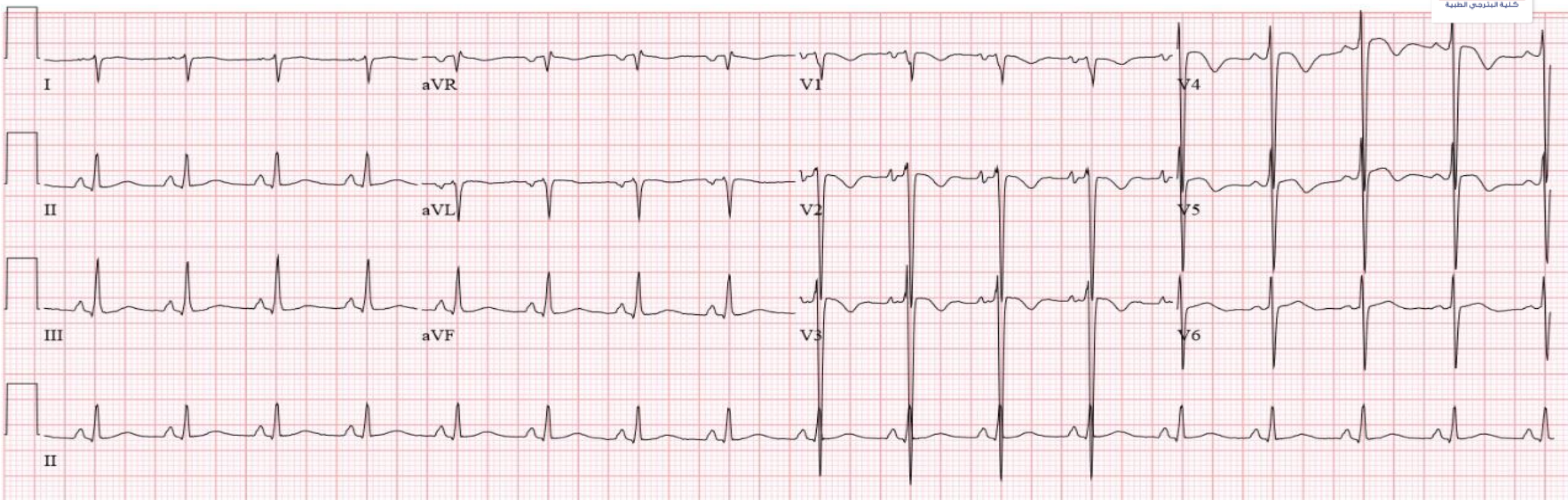
ECG comment



1. Standardization : Normal
2. Voltage: Normal
3. Axis: Left
4. Rhythm: occasional irregularity

Quiz

ECG comment



1. Standardization : Normal
2. Voltage: Normal-high
3. Axis: Right
4. Rhythm: Regular

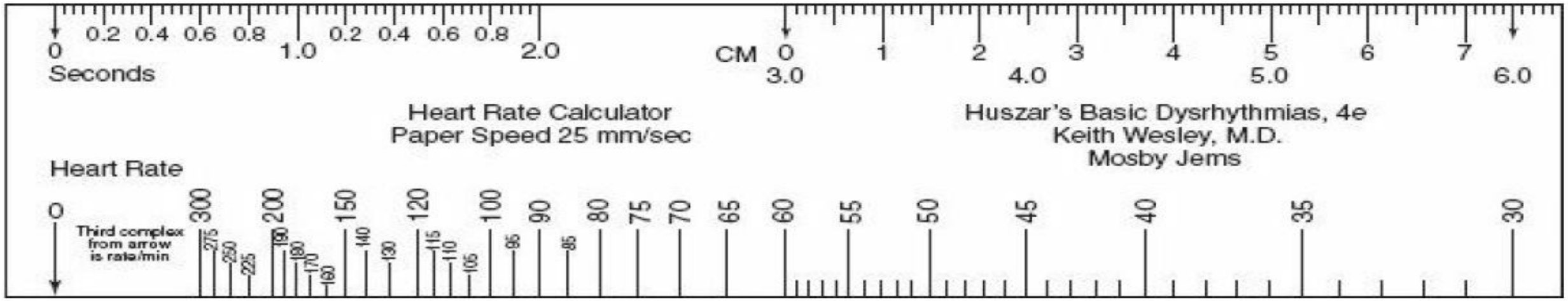
Quiz

ECG comment

1. Identification
2. Standardization
3. Voltage
4. Axis
5. Rhythm
6. **Rate**
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. T wave

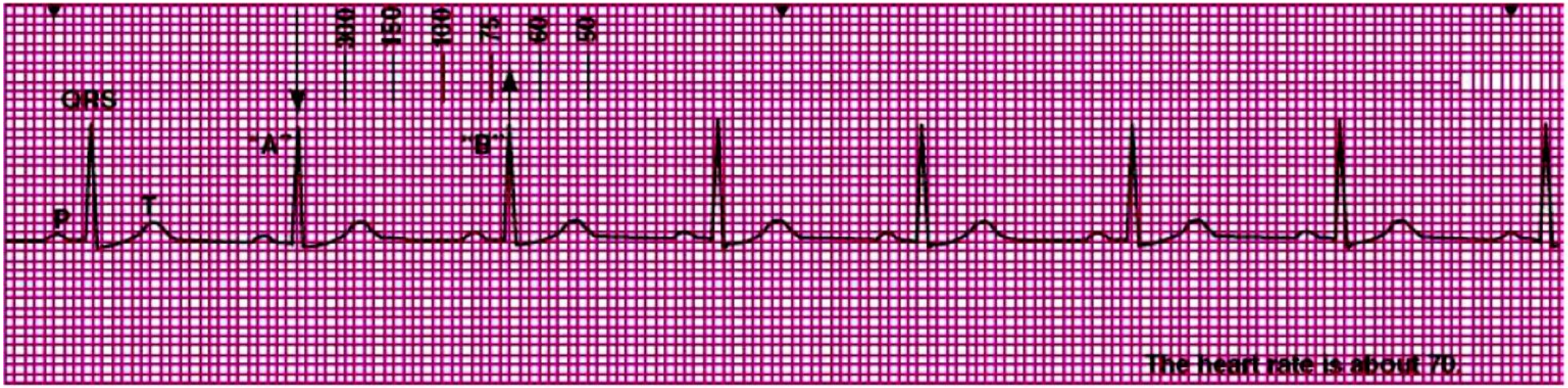
ECG comment

Heart rate calculator ruler



ECG comment

TRIPLICATE METHOD



ECG comment

R-R method

Heart Rate

Definition

Calculation

No. of QRS waves in 1 min. * 1

No. of QRS waves in 1/2 min. * 2

No. of QRS waves in 1/3 min. * 3

No. of QRS waves in 1/4 min. * 4

No. of QRS waves in 1/5 min. * 5

No. of QRS waves in 1/10 min. *10

ECG comment

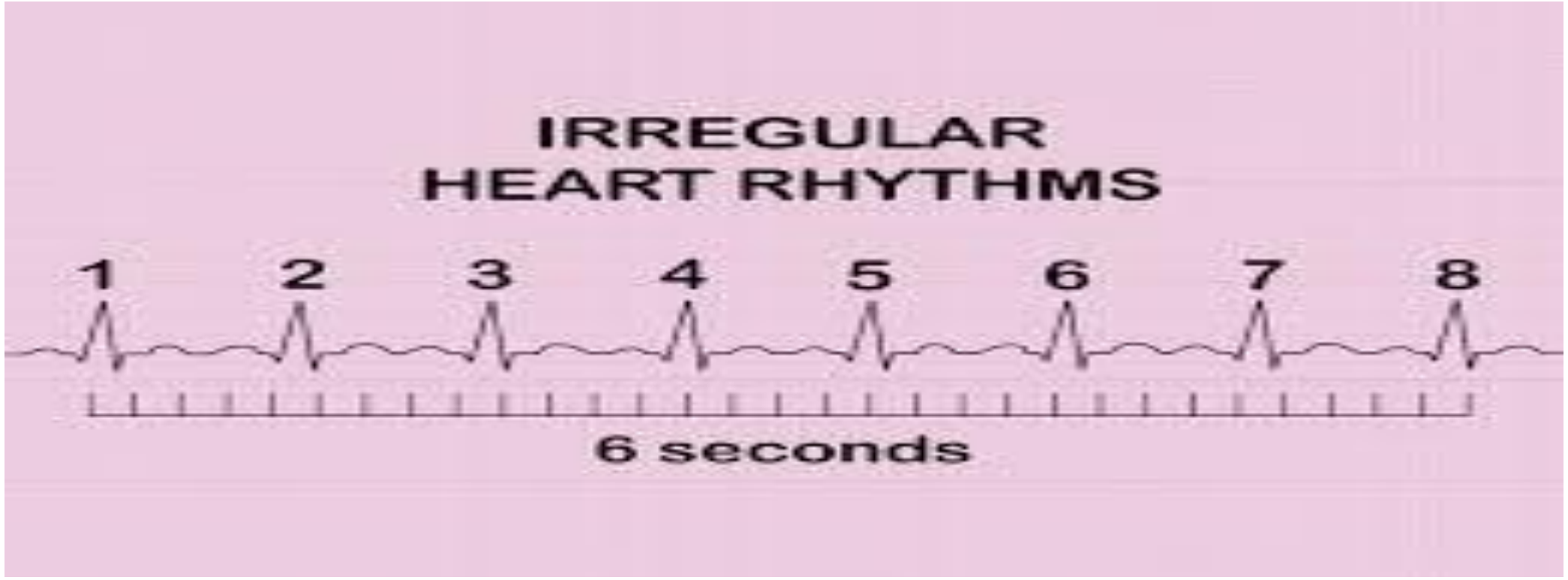
Rate

HR (REG.) = 300/RR interval in big Sq.

HR (REG.) = 1500/RR interval in small Sq.

SIX-SECOND COUNT METHOD

- The least accurate method of determining the heart rate



ECG comment

Rate

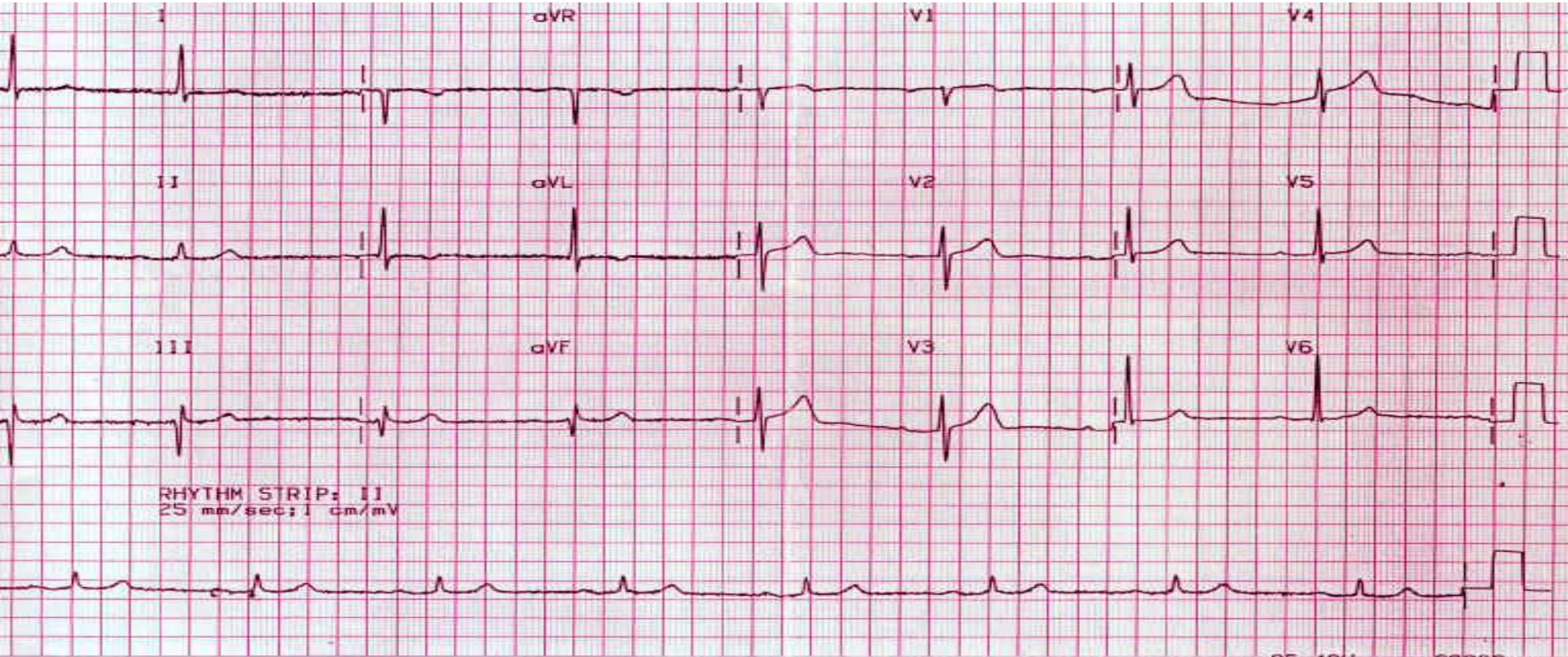
HR (IRREG.) = **No. of QRS waves in 1/10 min.** ***10**

HR (IRREG.) = **No. of QRS waves in 6 sec.** ***10**

HR (IRREG.) = **No. of QRS waves in 30 big Sq.** ***10**

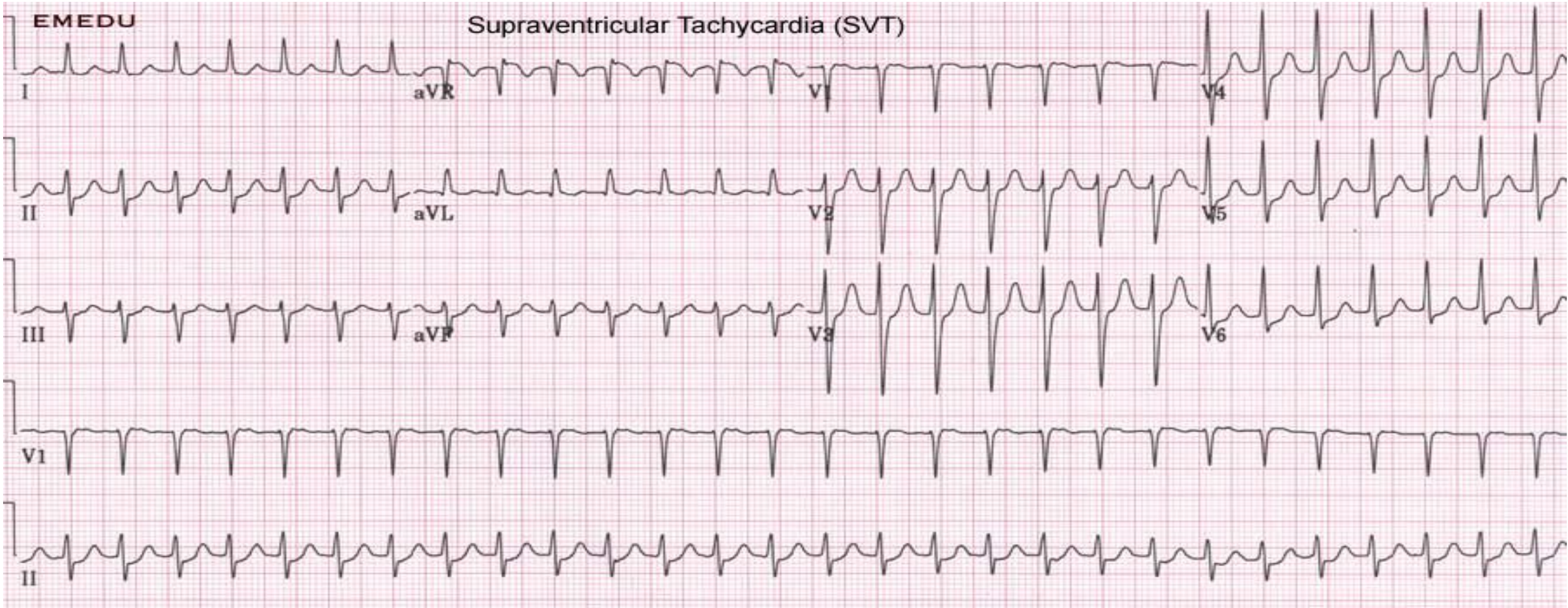
ECG comment

Rate



ECG comment

Rate



ECG comment

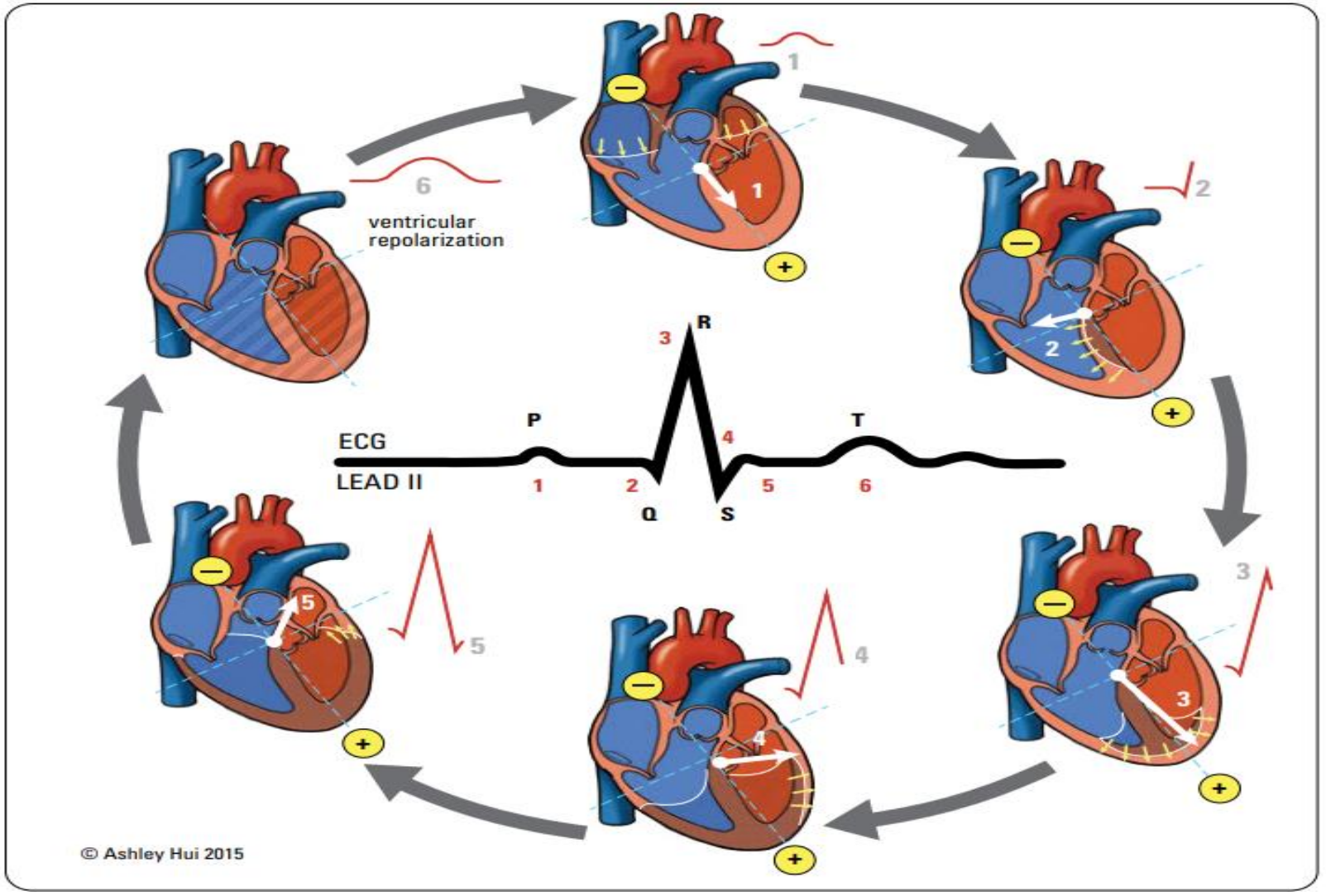
Rate



ECG Waves

Alternative PQRSTU Approach to ECGs

Note: the PQRSTU Approach is organized a different way based on the anatomy of the ECG



PQRSTU Approach to ECGs
P wave
P-R interval
QRS complex
ST segment
T wave
Q-T interval
U wave

Wave:
direction, amplitude, duration, shape

Segment:
duration, deviation

Interval:
duration

ECG comment

1. Identification
2. Standardisation
3. Voltage
4. Axis
5. Rate
6. Rhythm
7. **P-wave**
8. PR interval
9. QRS wave
10. ST segment
11. T wave

ECG comment

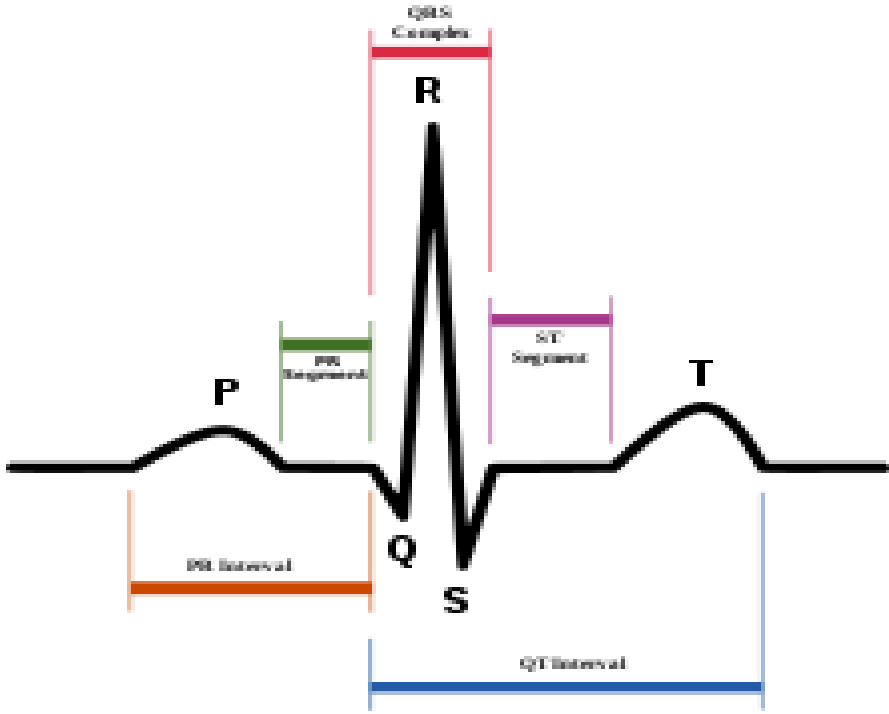
P-wave

1. Direction:
   In I, aVR

2. Amplitude: RA
   2.5 mm

3. Duration: LA
   2.5 mm = 0.1 sec

4. Shape:
   In II, V1

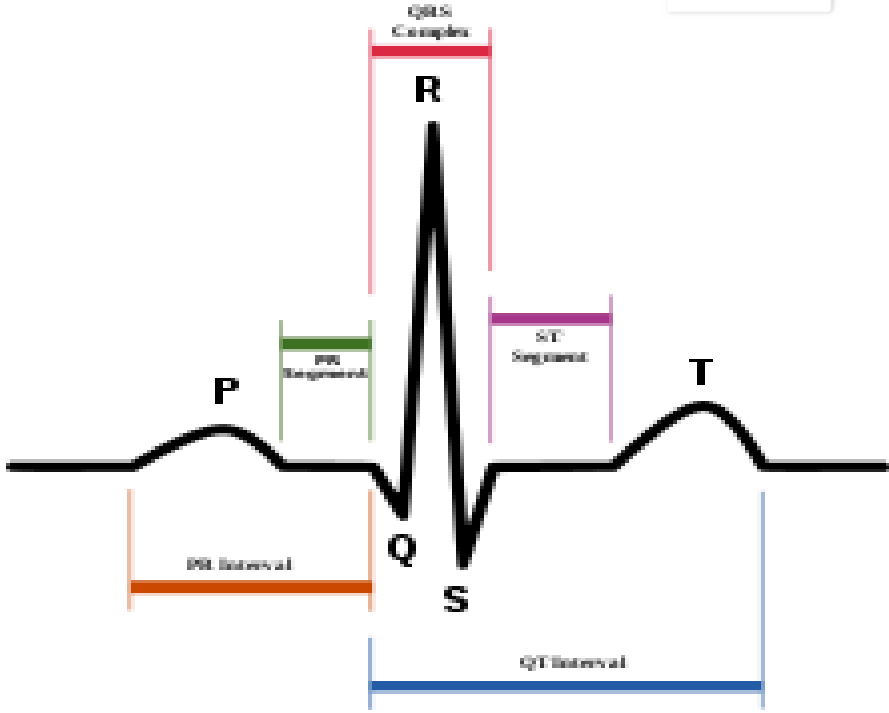


Analyze the P Waves

- ECG reader should observe the following five components when evaluating P waves:

1. Are P waves present?
2. Are the P waves occurring at regular intervals?
3. Is there one P wave for each QRS complex, and is there a QRS complex after each P wave?
4. Are the P waves upright or inverted?
5. Do they all look alike? (P waves that look alike and are regular are likely from the same

P-wave abnormalities



1. Abnormal direction:
   In I, aVR

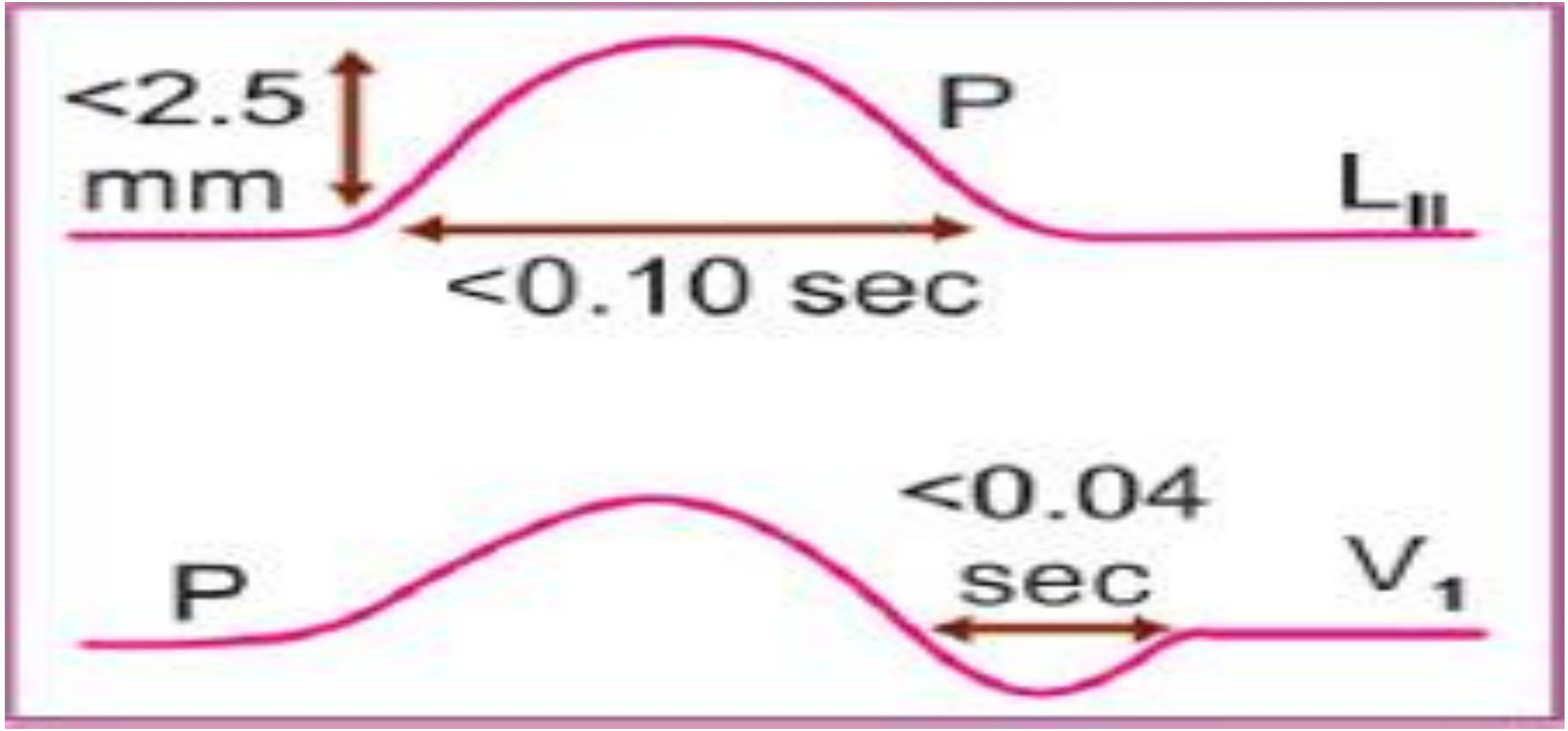
2. Abnormal Amplitude: RA
   2.5 mm, if increased = RA^{++} (p pulmonale)

3. Abnormal Duration: LA
   2.5 mm = 0.1 sec, if increased = LA^{++} (p mitrale)

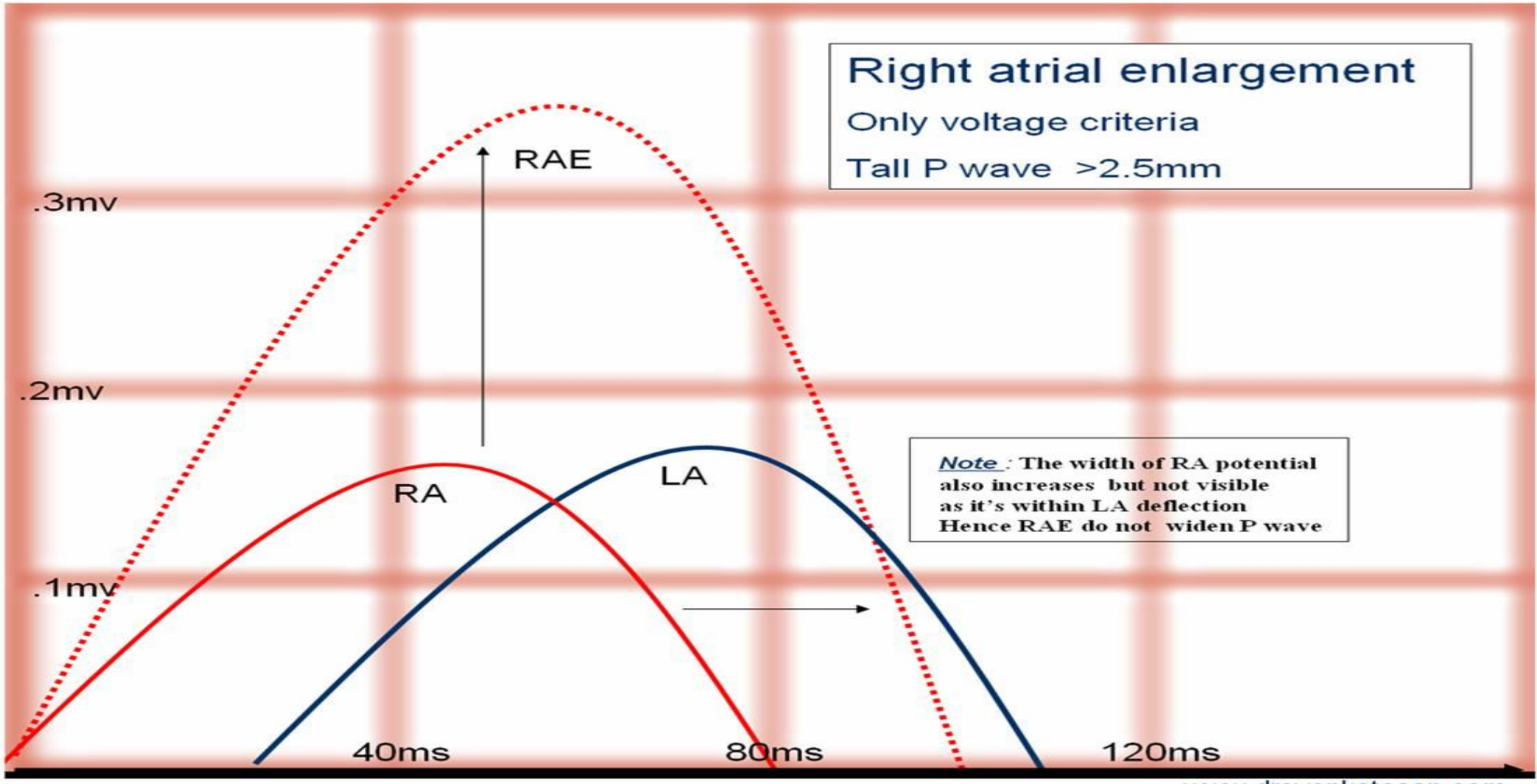
4. Absent or replaced by abnormal wave (fibrillatory, flutter, naudal rhythm, SVT)

ECG comment

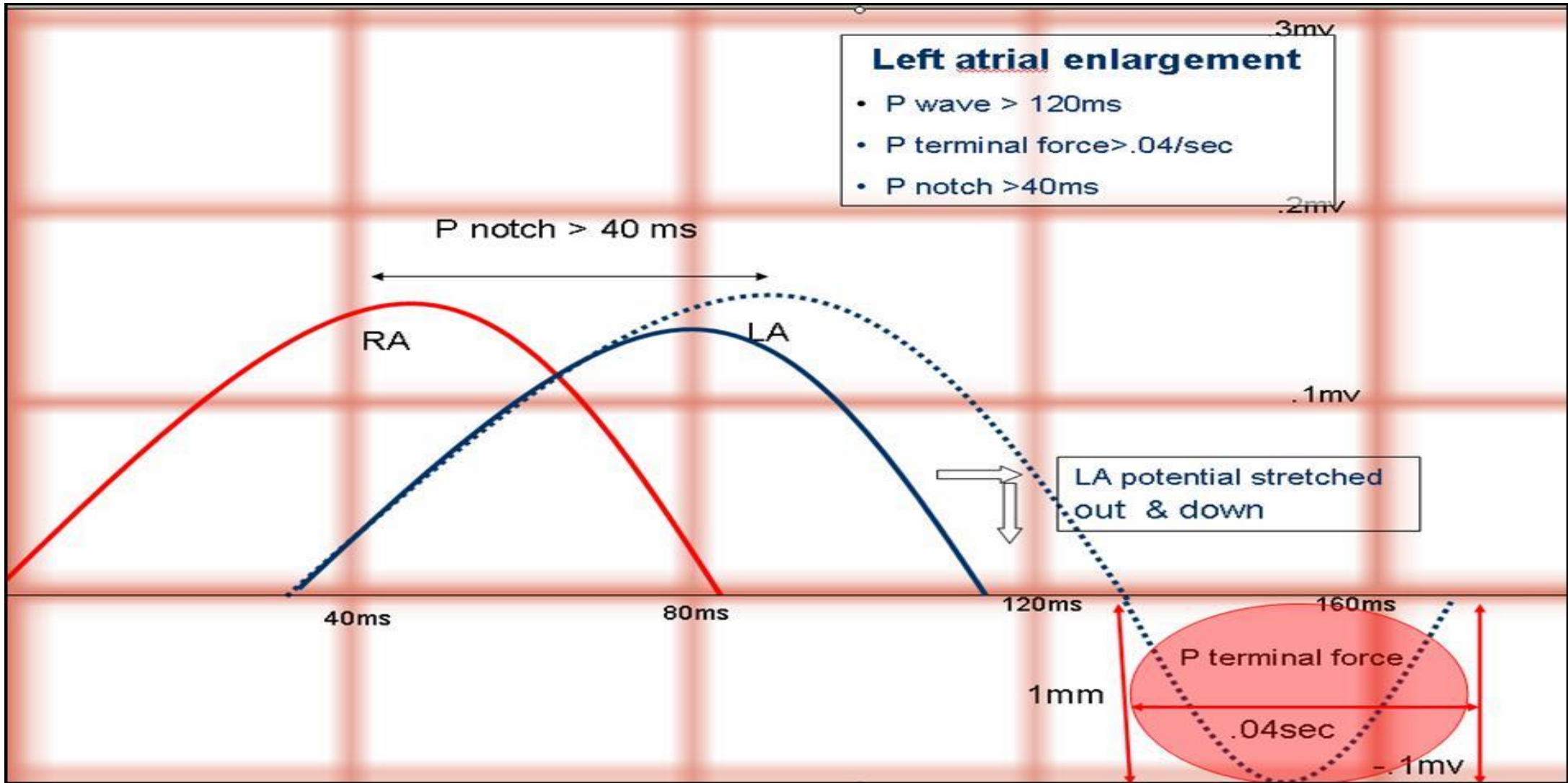
Normal P wave



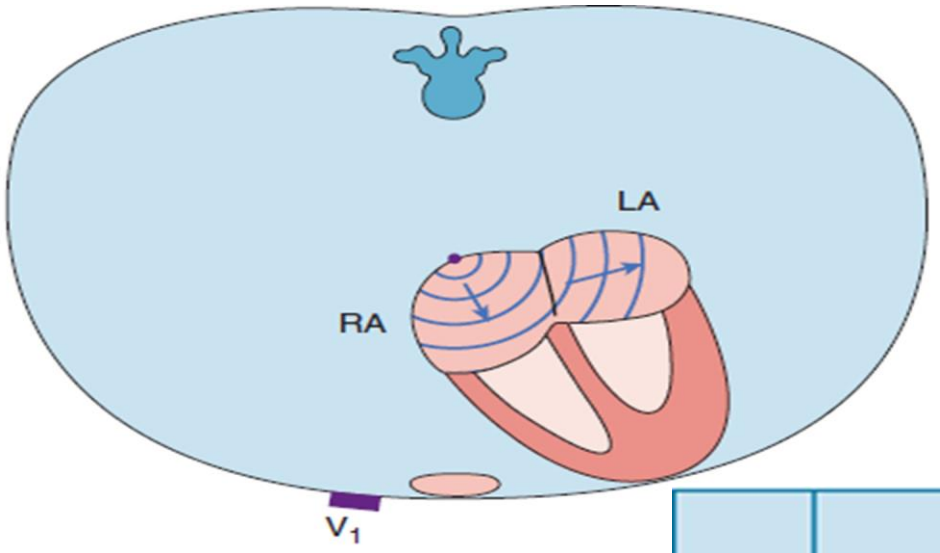
P-wave abnormalities

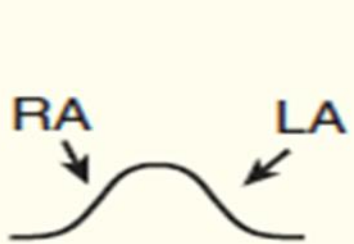
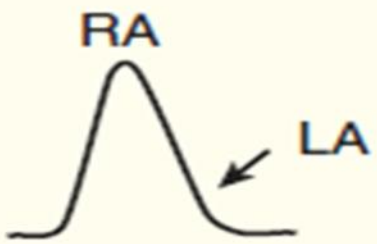
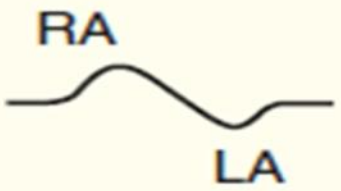
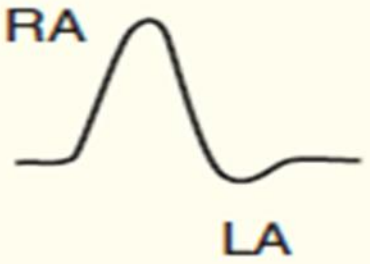
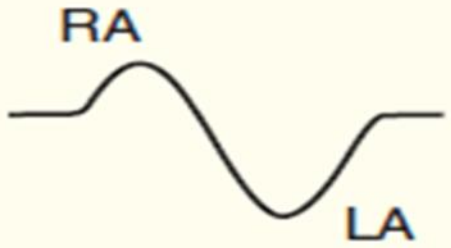


P-wave abnormalities

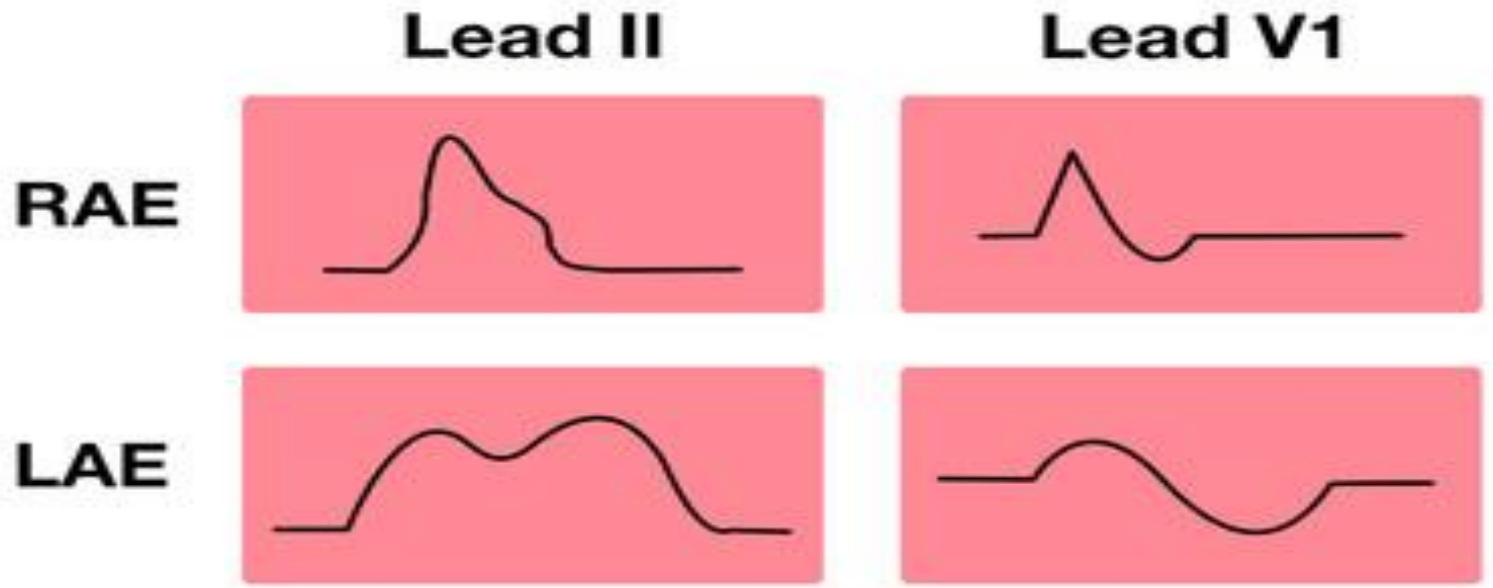


ECG



	Normal	Right	Left
II			
V_1			

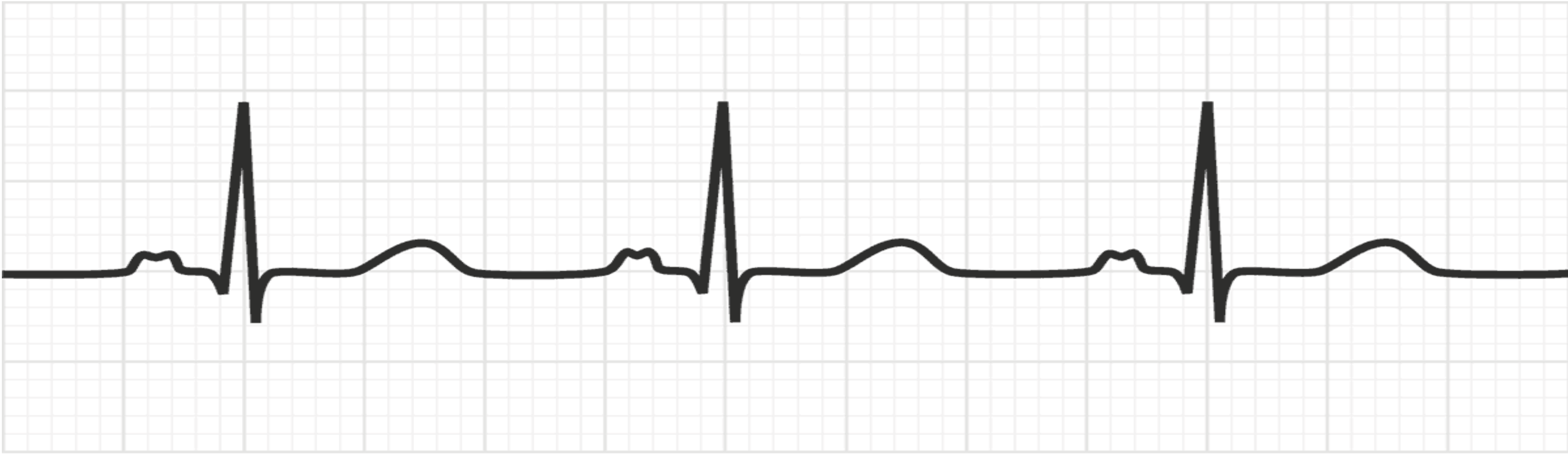
P-wave abnormalities



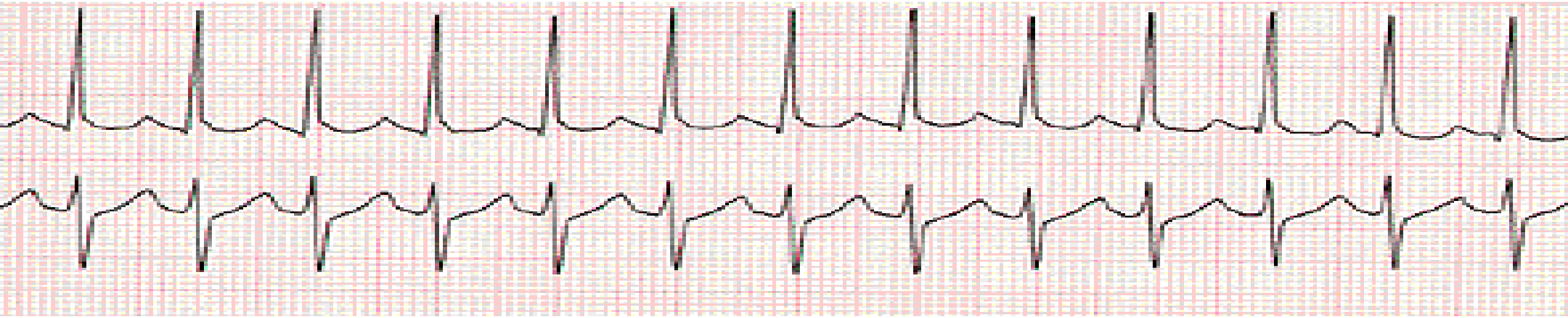
P-wave abnormalities



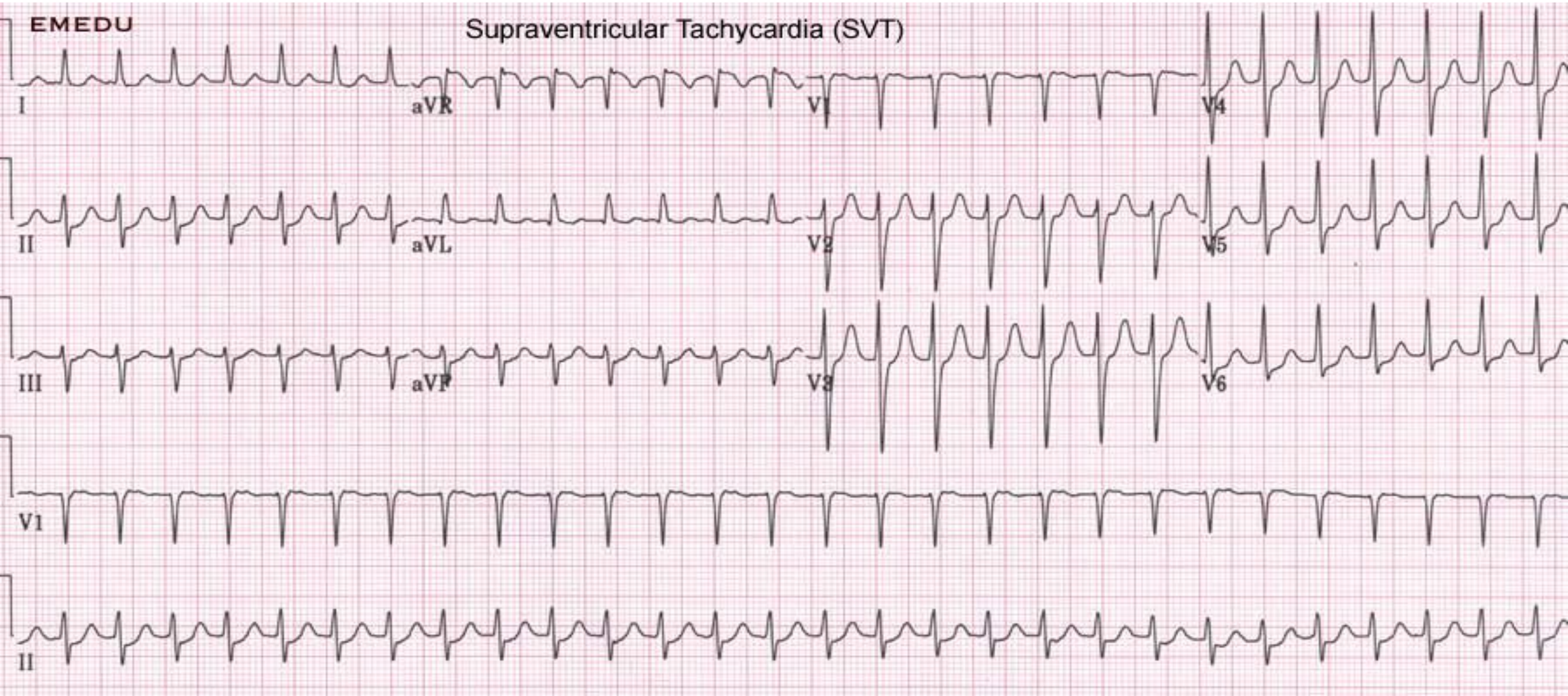
P-wave abnormalities



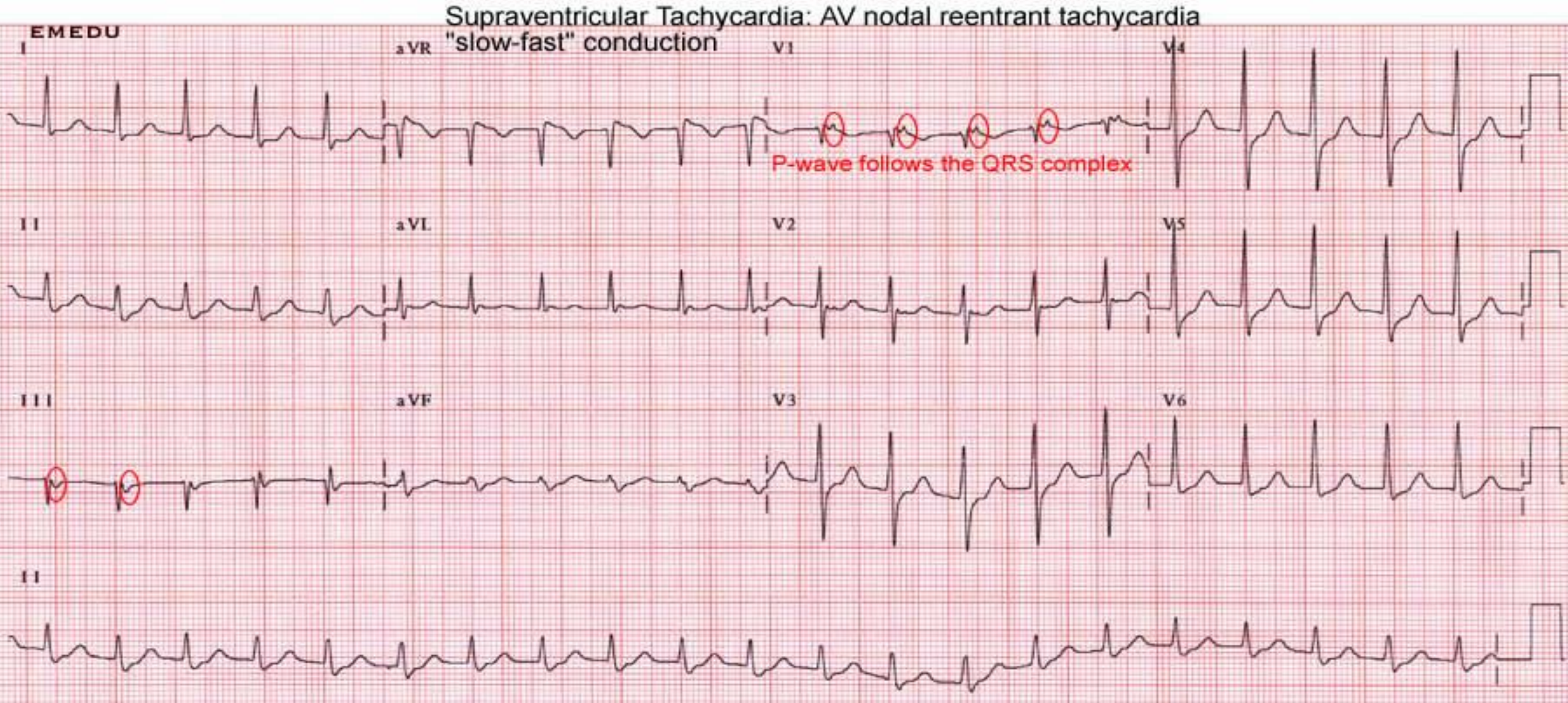
P-wave abnormalities



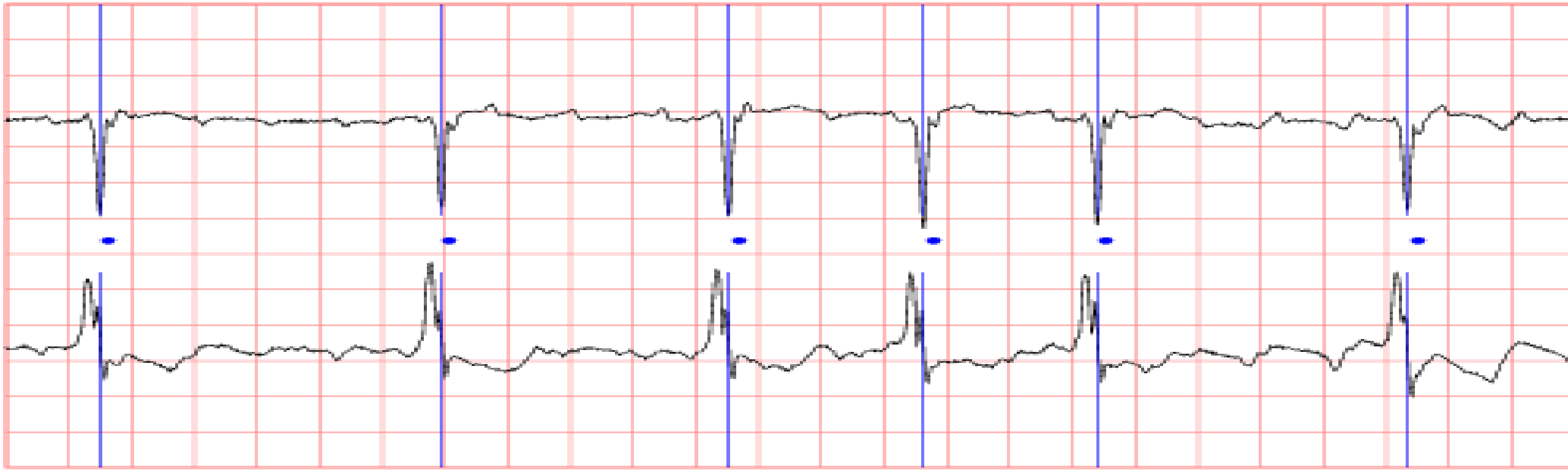
P-wave abnormalities



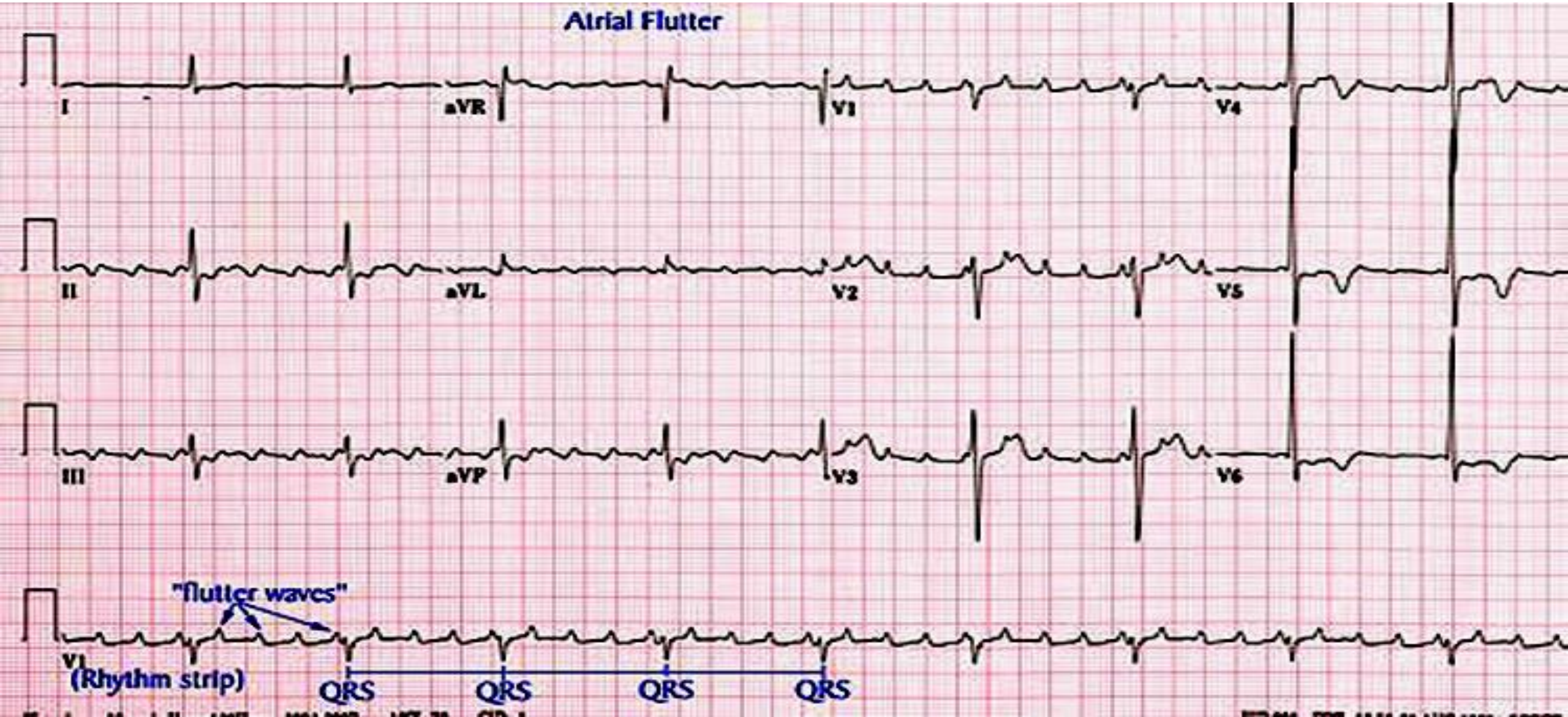
P-wave abnormalities



P-wave abnormalities



P-wave abnormalities



ECG comment

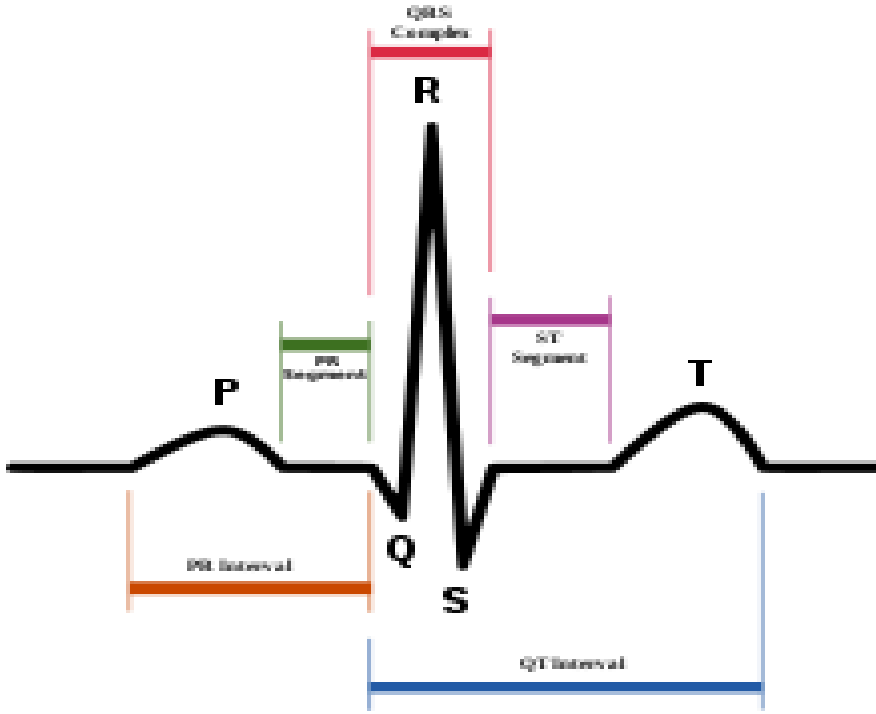
1. Identification
2. Standardisation
3. Voltage
4. Axis
5. Rate
6. Rhythm
7. P-wave
8. **PR interval**
9. QRS wave
10. ST segment
11. T wave

PR interval

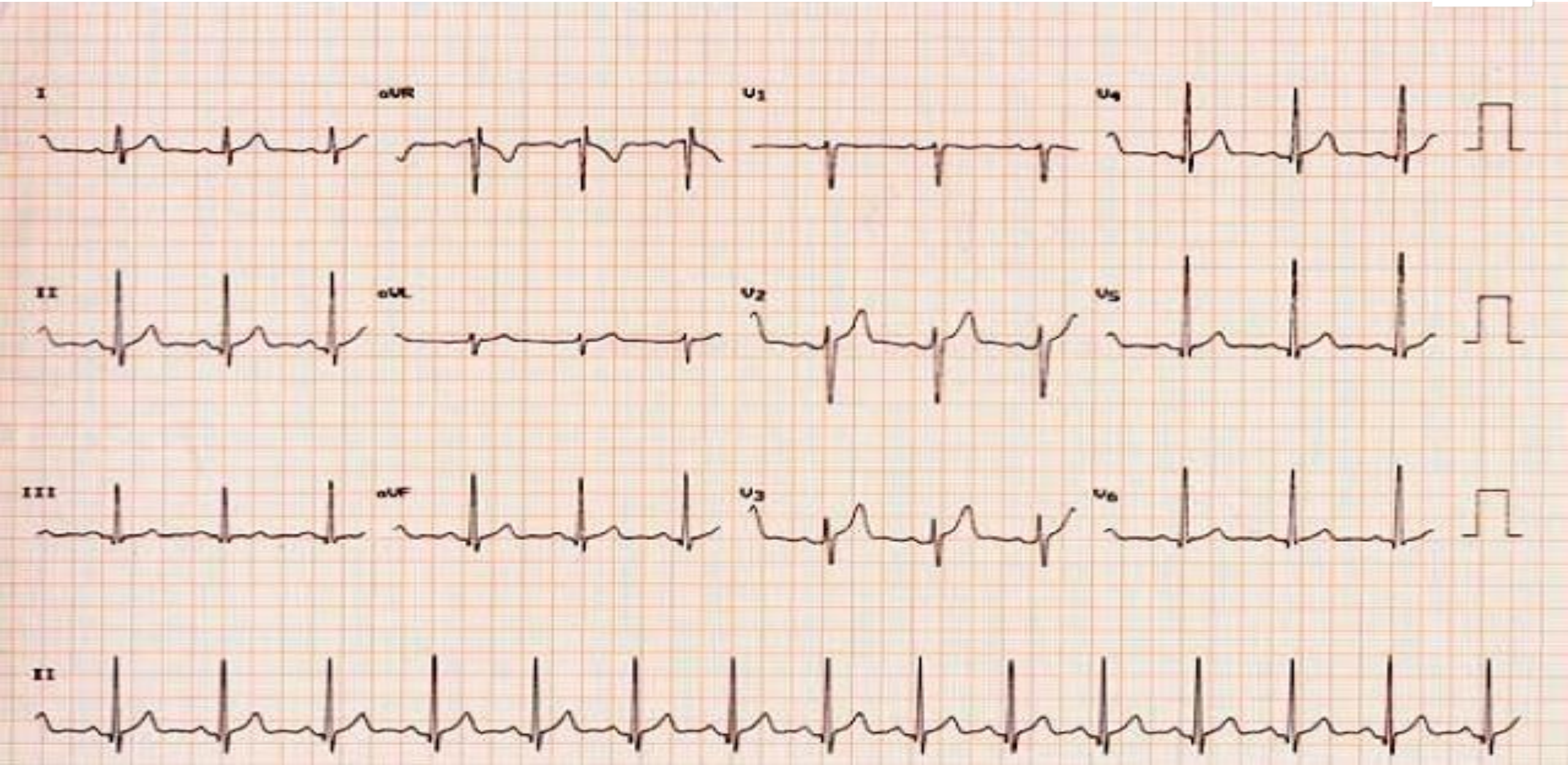
1. Def.:
2. It represents …
3. Duration:

   3-5 small Sq. = 0.12-0.20 sec

   If prolonged
   If shortened



ECG



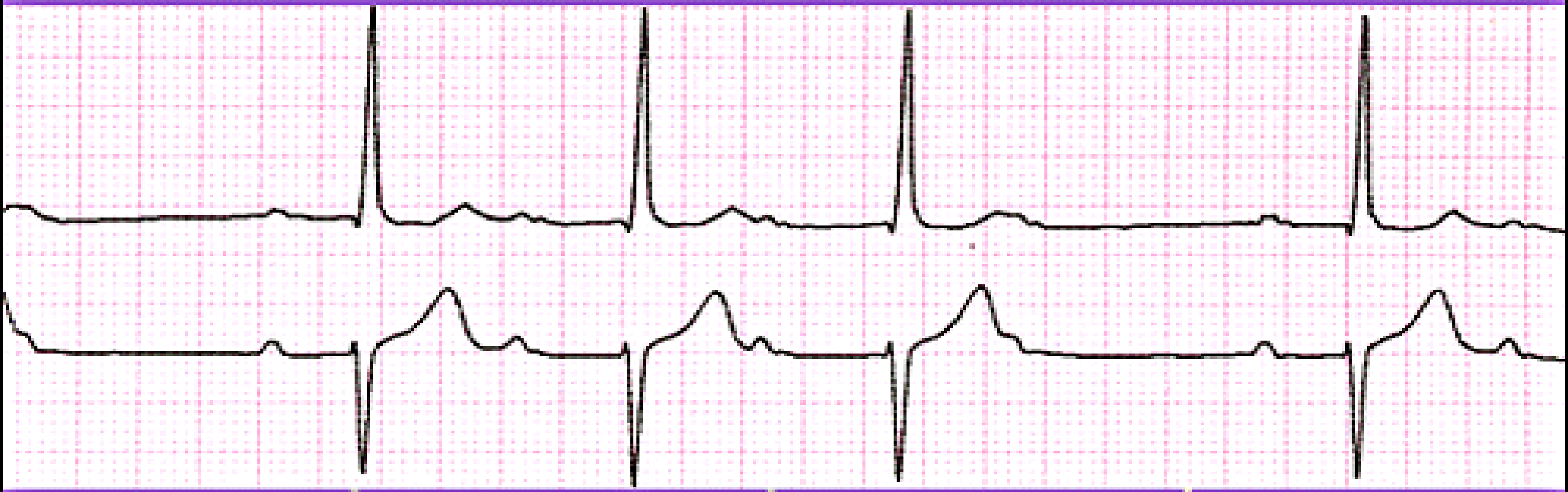
ECG

First-Degree AV Block



ECG

Second Degree AV Block • Mobitz 1 (Wenckebach)

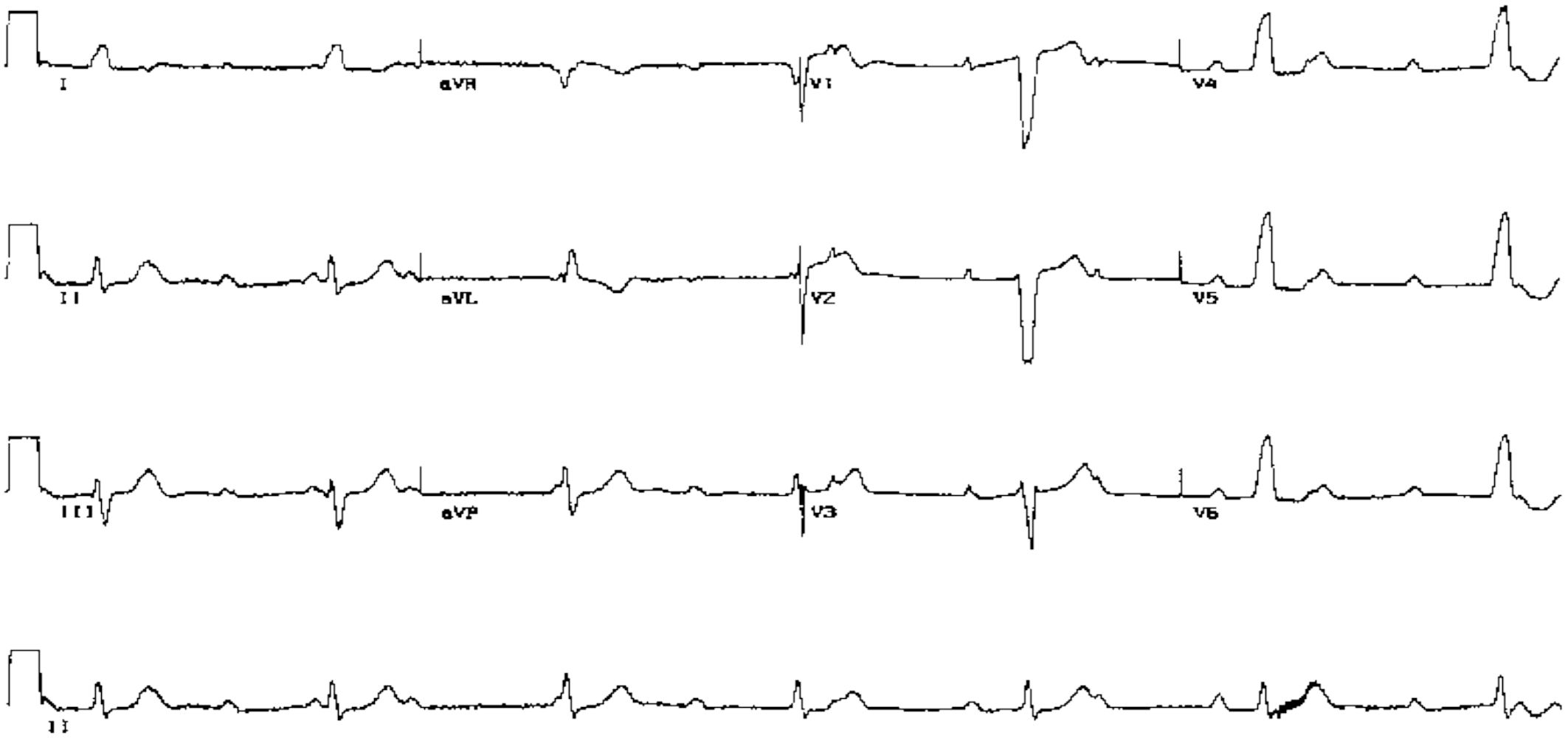


P Wave	PR Interval (in seconds)	QRS (in seconds)	Characteristics
Conduction intermittant	Increasingly Prolonged	<.12	QRS dropped in a repeating pattern

ECG



ECG



ECG comment

1. Identification
2. Standardisation
3. Voltage
4. Axis
5. Rate
6. Rhythm
7. P-wave
8. PR interval
9. **QRS wave**
10. ST segment
11. T wave

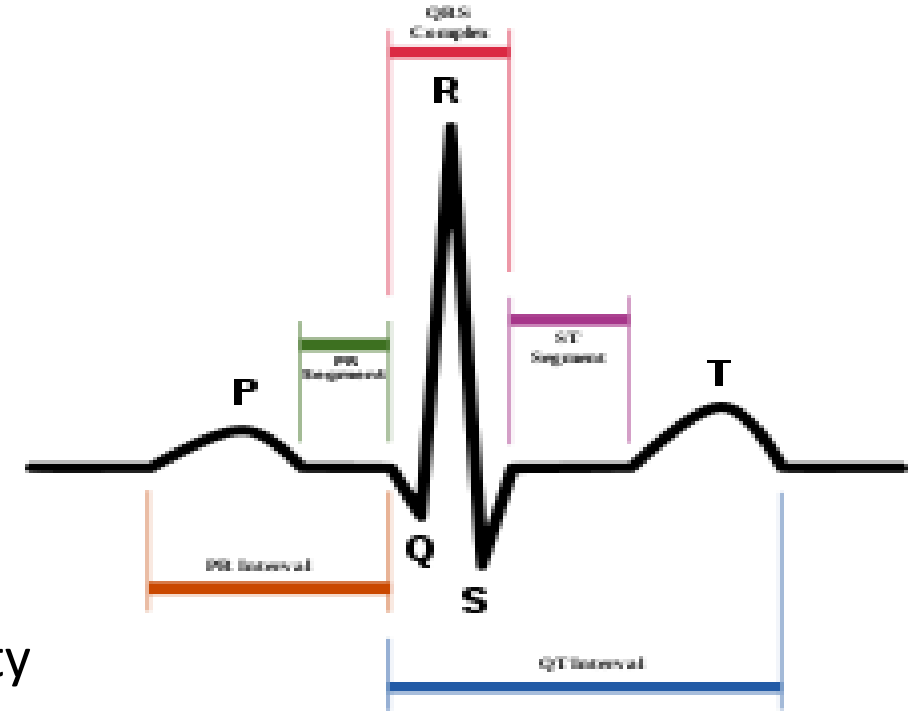
QRS wave

1. Direction:
   In I, aVR, aVF, V1-6

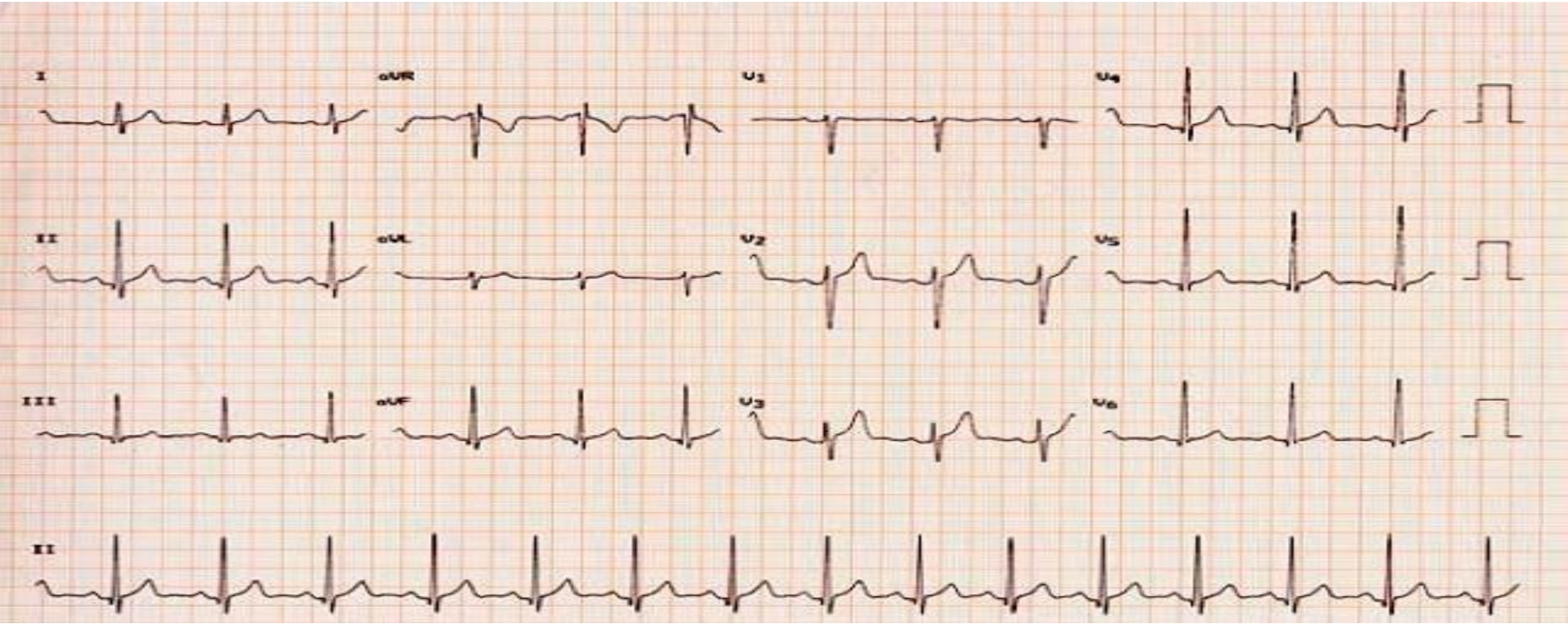
2. Amplitude: Ventricles
   See Voltage

3. Duration: Ventricles
   2 mm = 2 small Sq. = 0.08 sec
   If increased = abnormal electrical activity

4. Shape and naming:
   QS, qR, Qr, qRs, R, Rs, rS

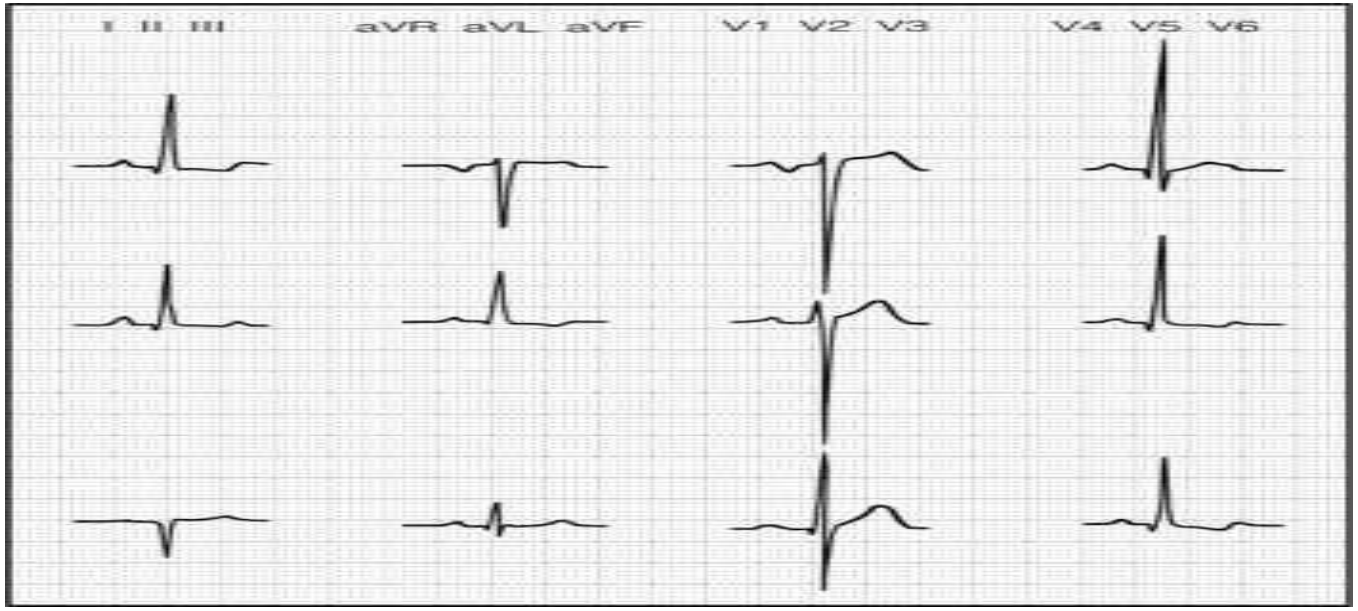


Normal QRS wave

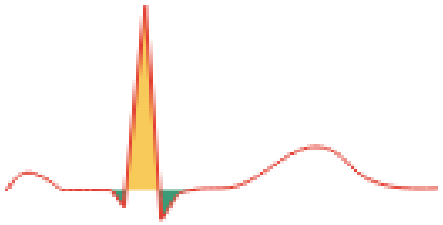


How to measure the QT interval

QRS complex synonyms

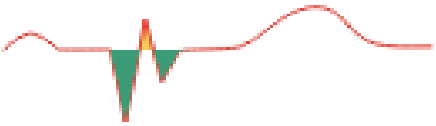


NET POSITIVE QRS-COMPLEX



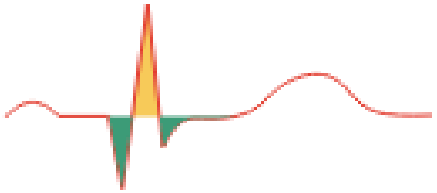
More positive than negative

NET NEGATIVE QRS-COMPLEX



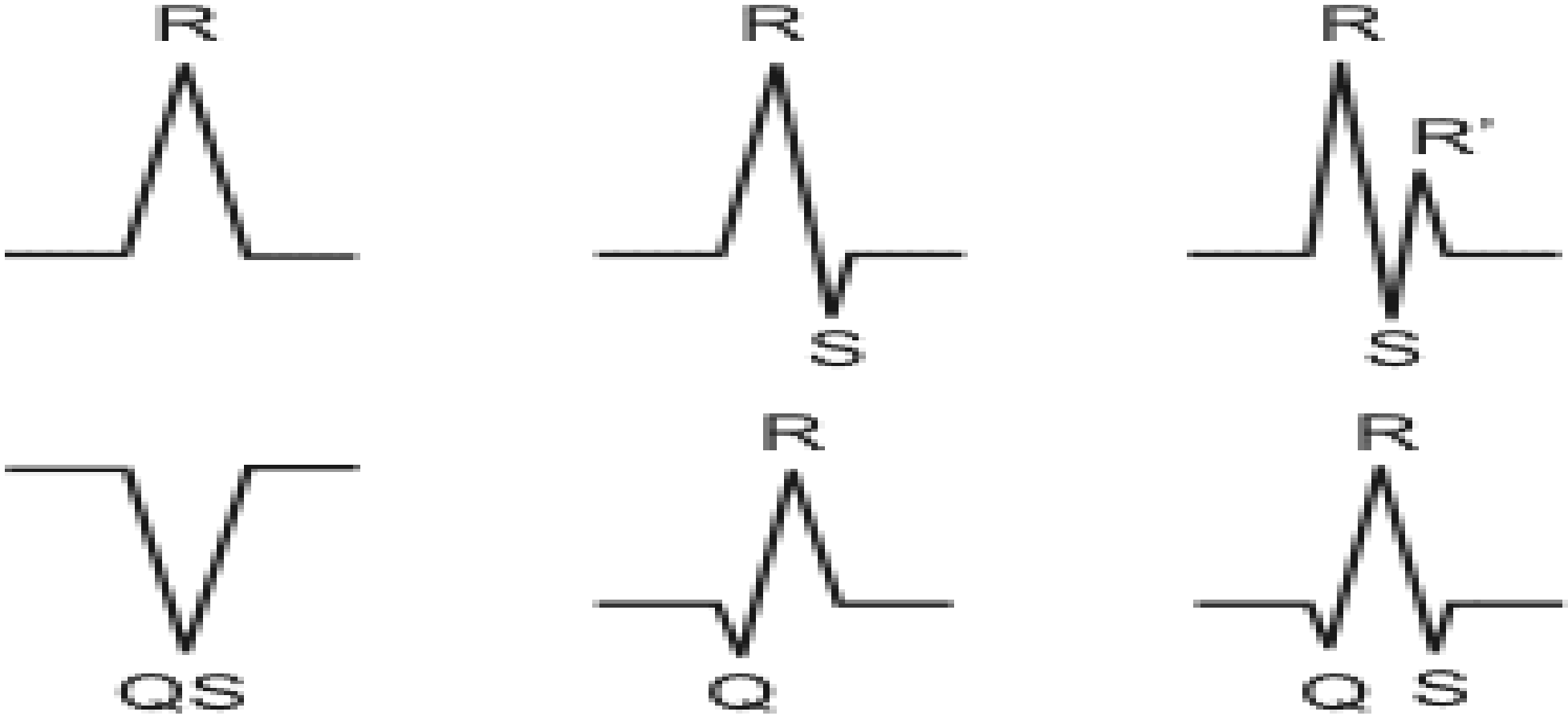
More negative than positive

NO NET DIRECTION



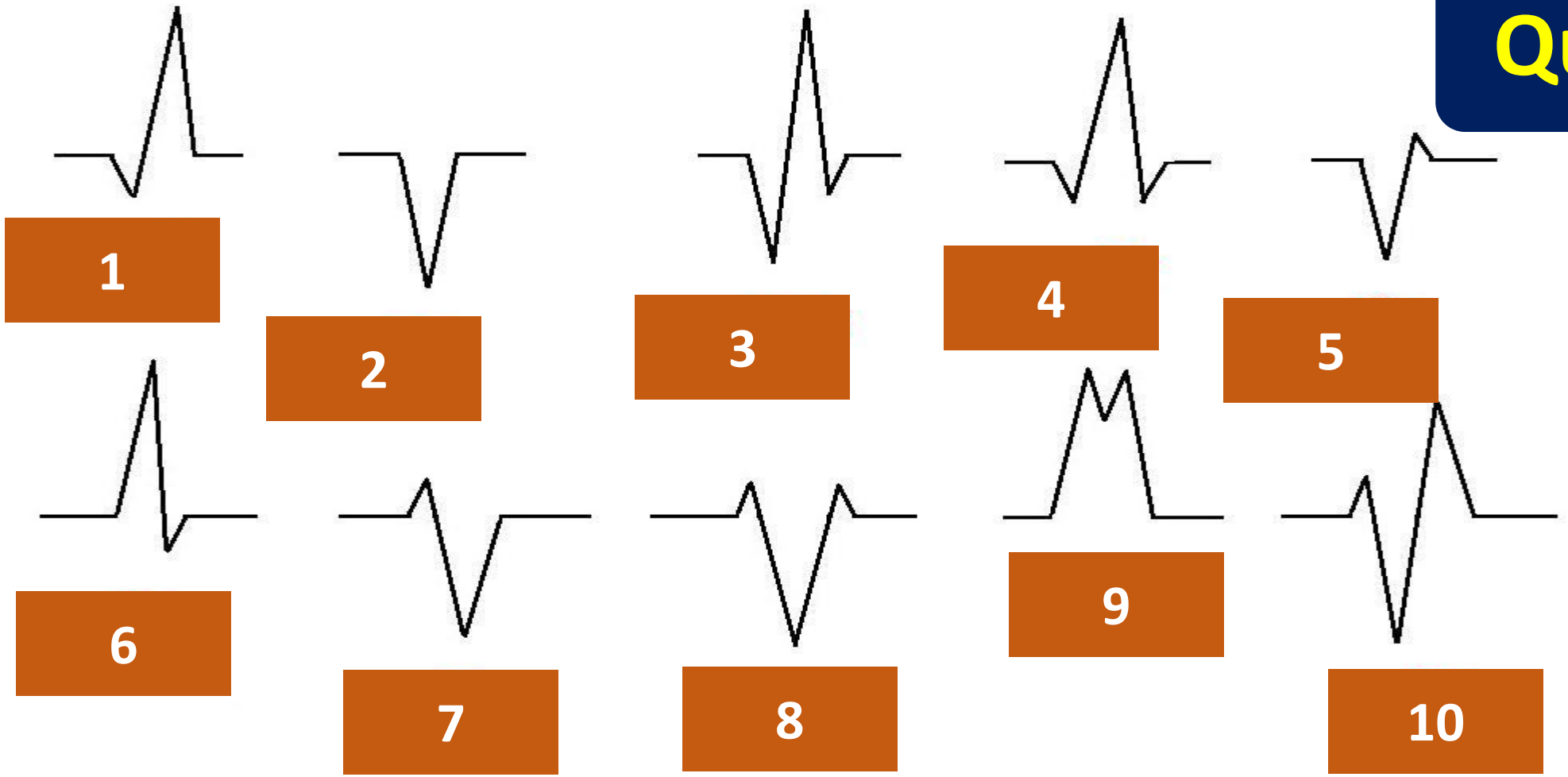
Roughly similar amounts of negative and positive area

ECG

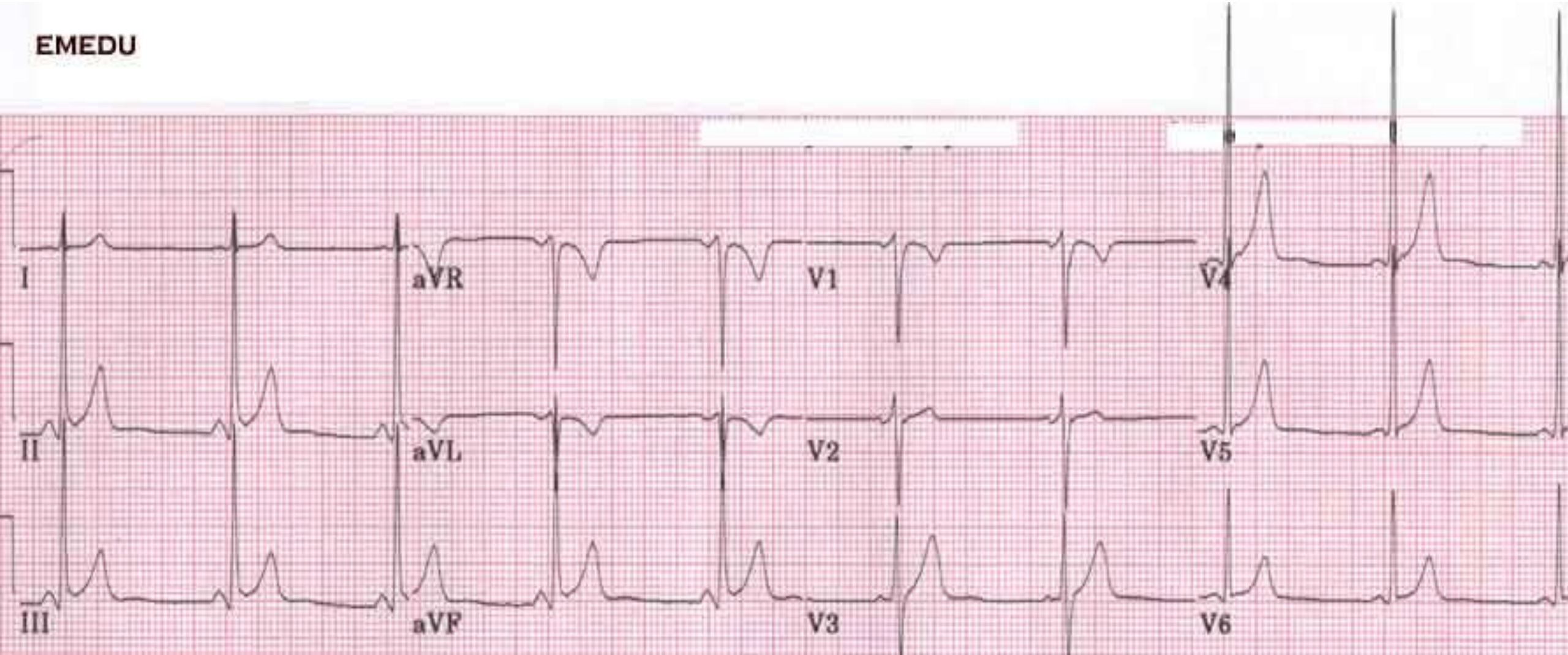


ECG

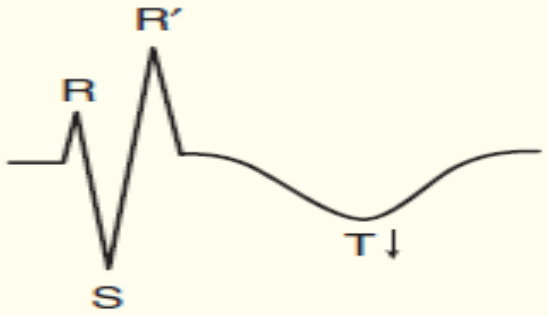
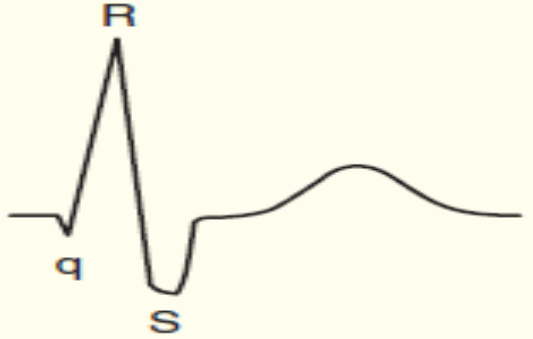
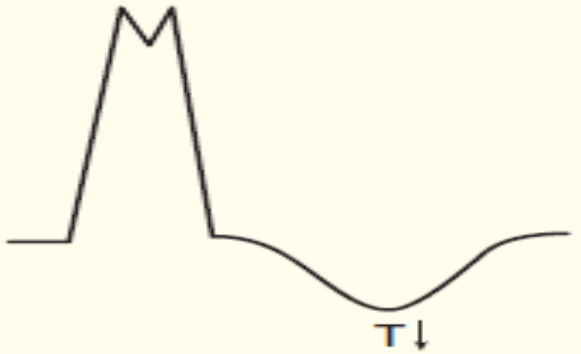
Quiz 5



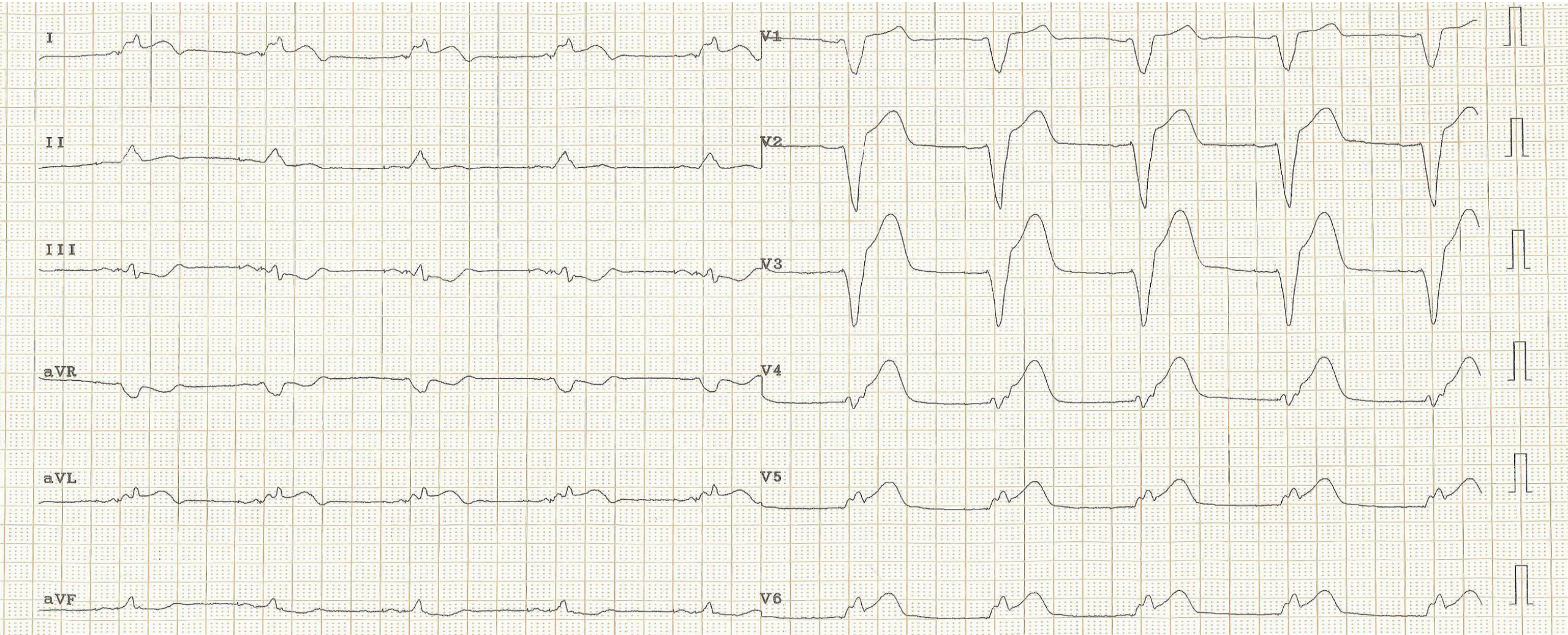
QRS wave abnormalities



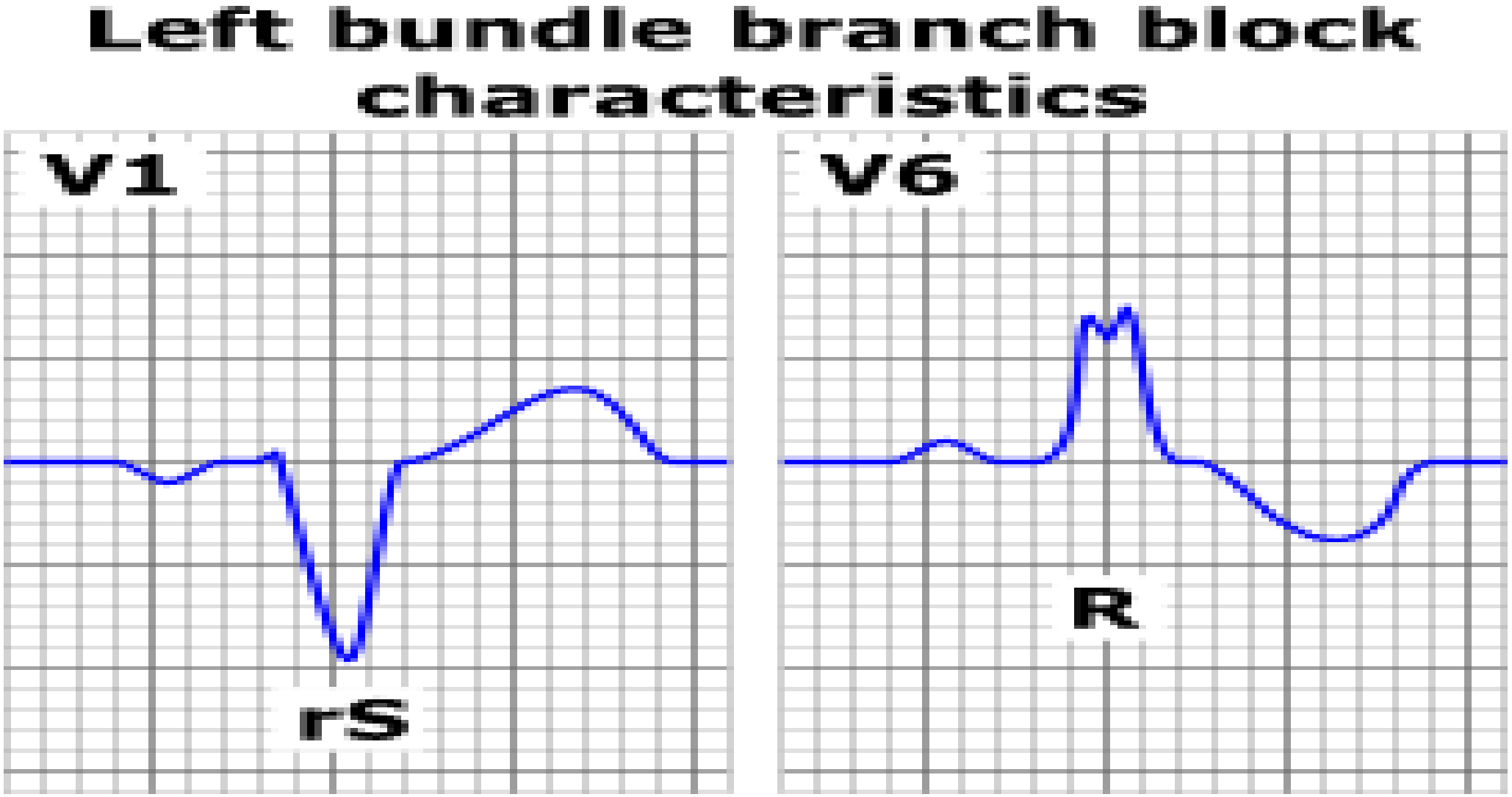
QRS wave abnormalities

	V_1	V_6
Normal		
RBBB		
LBBB		

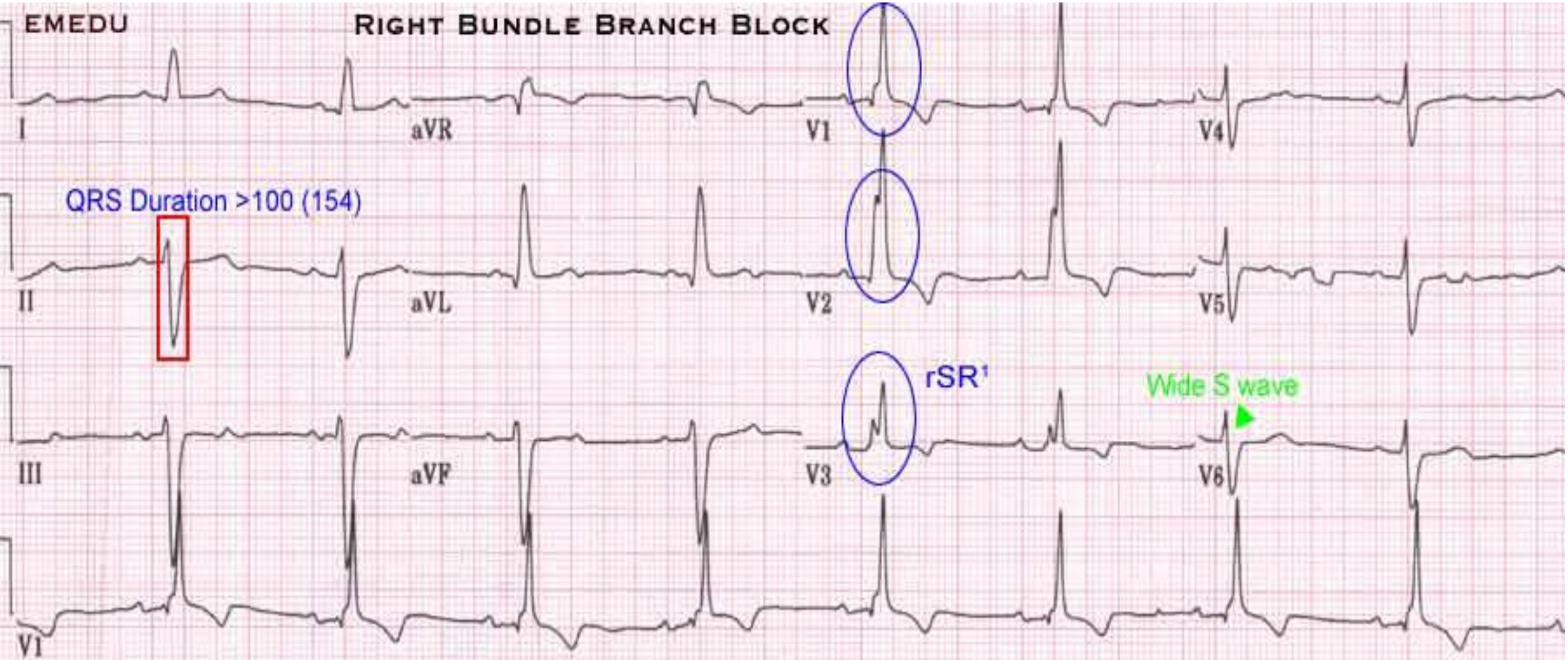
QRS wave abnormalities



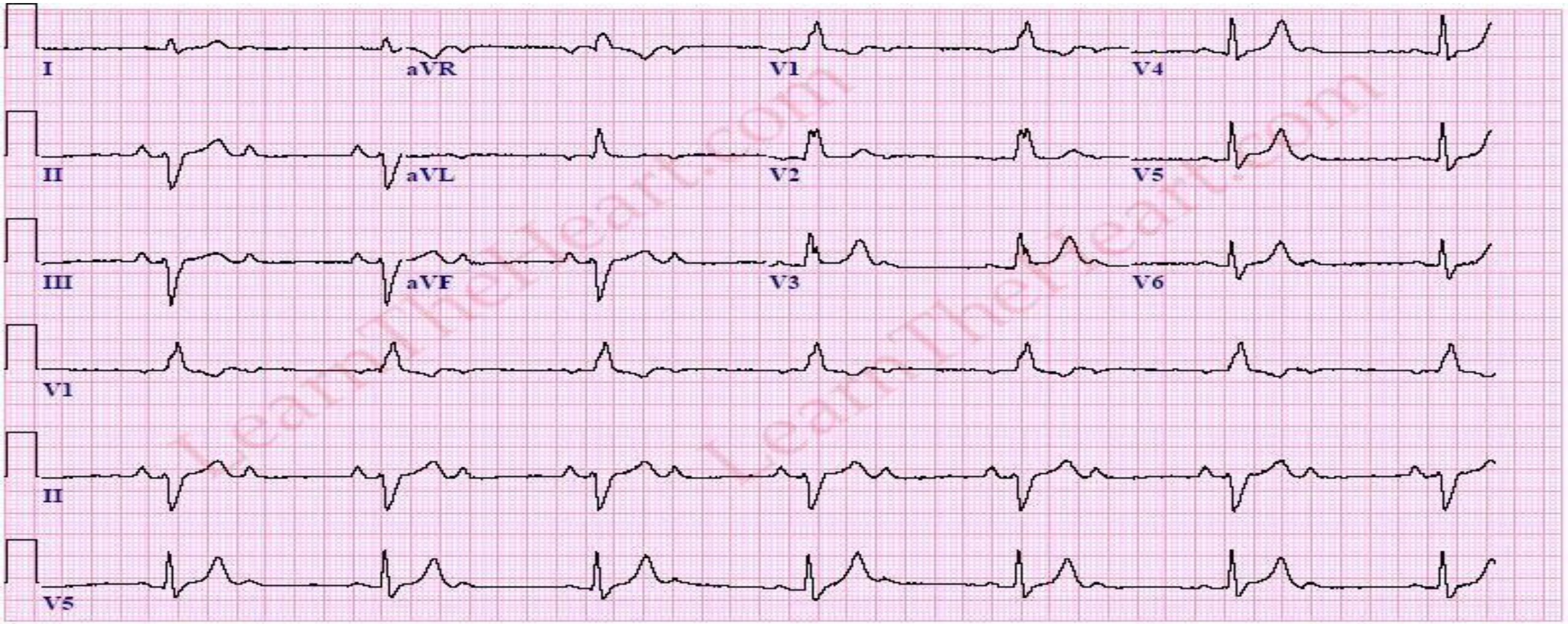
QRS wave abnormalities



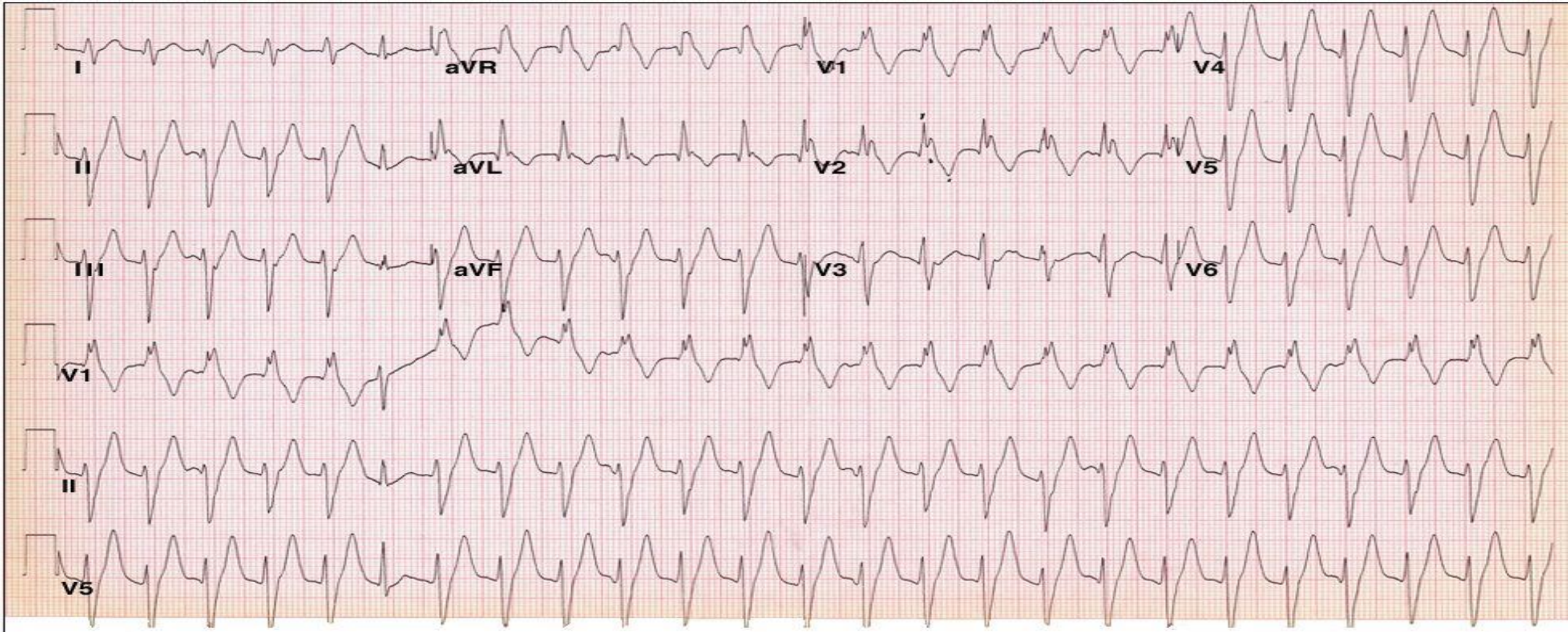
QRS wave abnormalities



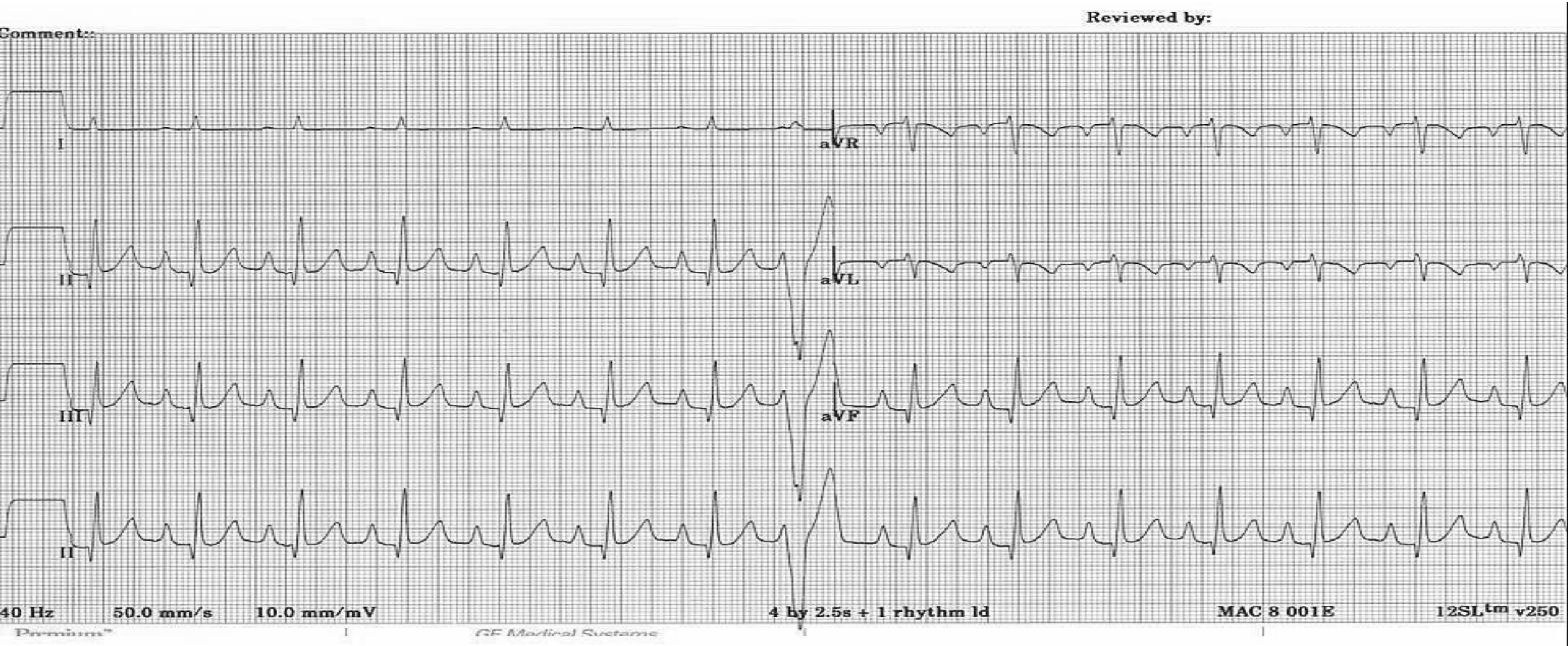
QRS wave abnormalities



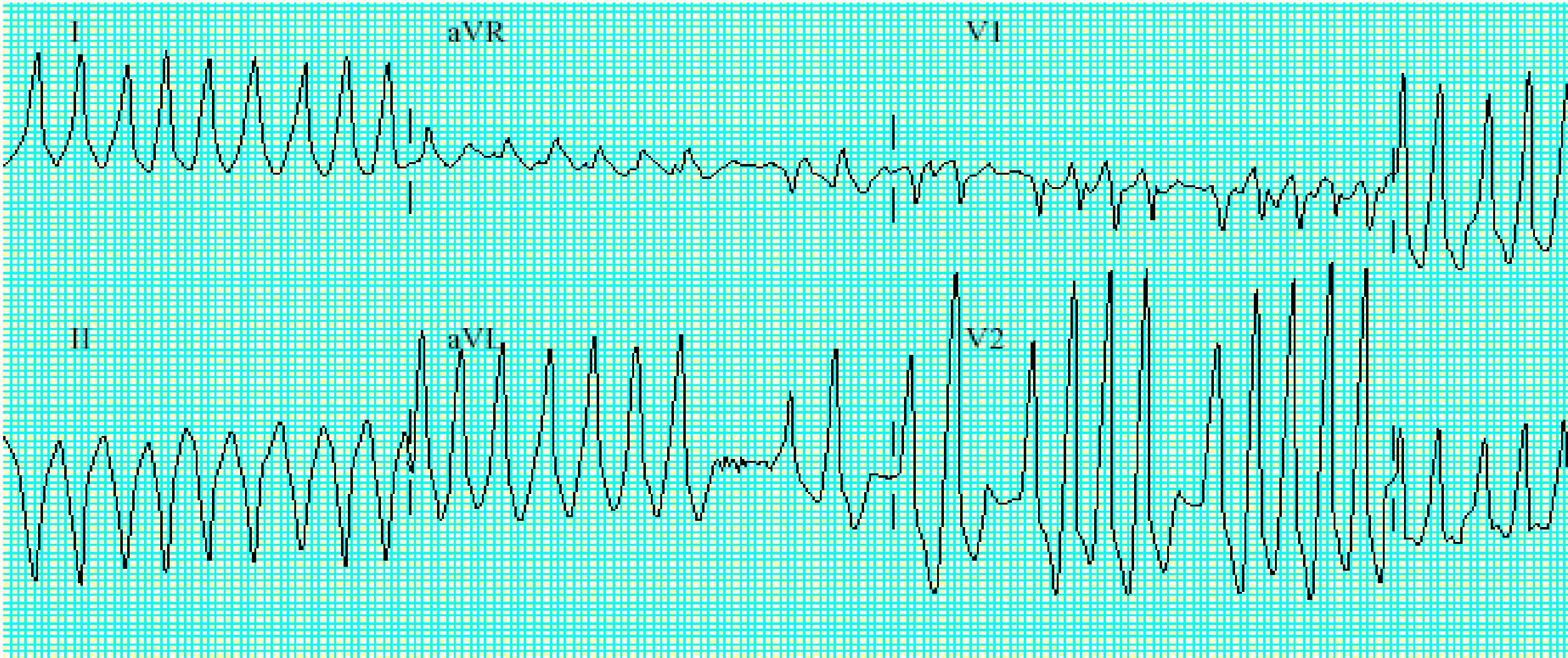
QRS wave abnormalities



QRS wave abnormalities



QRS wave abnormalities



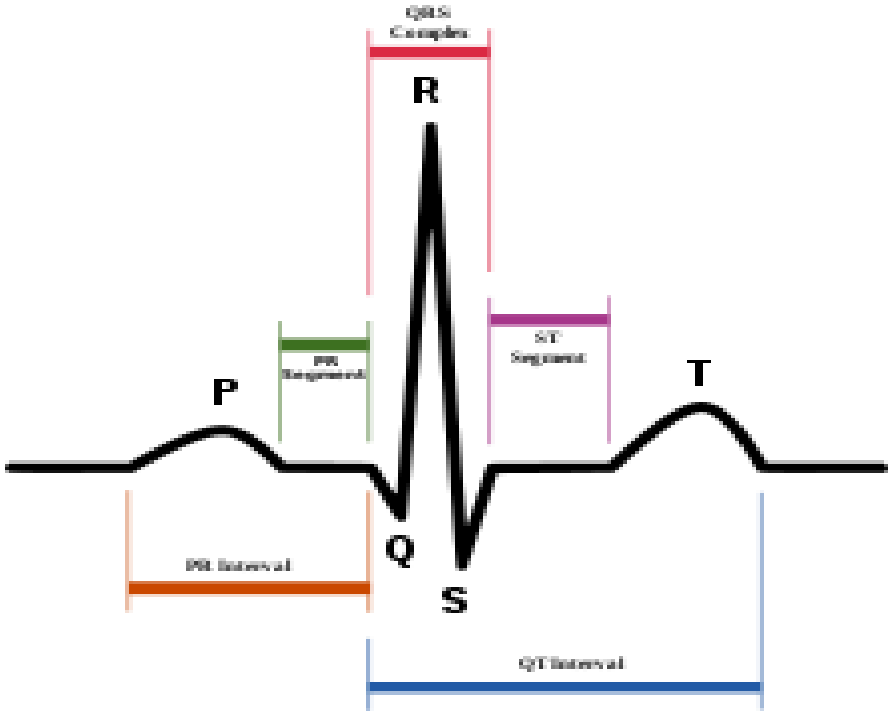
ECG comment

1. Identification
2. Standardization
3. Voltage
4. Axis
5. Rate
6. Rhythm
7. P-wave
8. PR interval
9. QRS wave
10. **ST segment**
11. T wave

ST segment

1. Def.:
2. It represents …
3. Deviation :

   Iso-electric

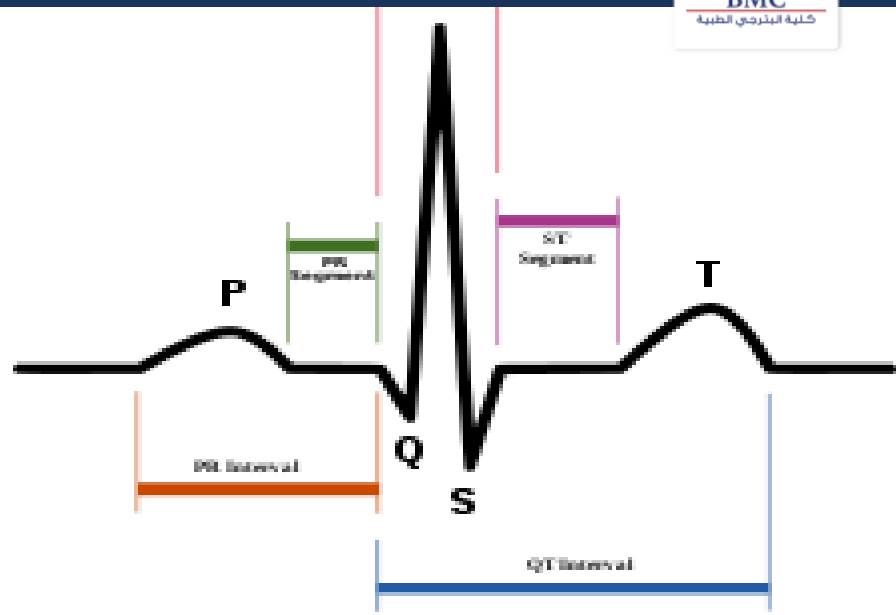


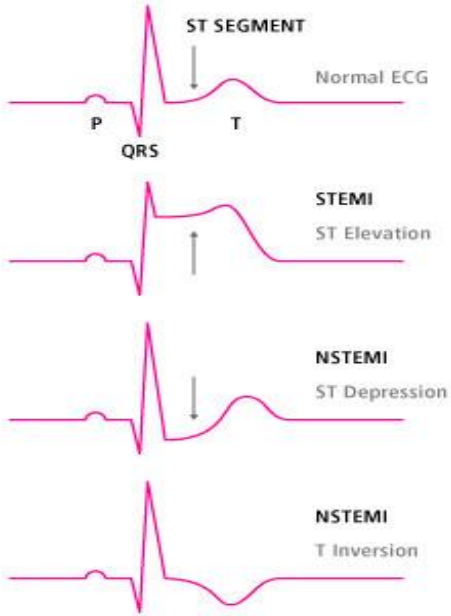
ST seg. abnormalities

1. Deviation :

   Elevated

   Depressed



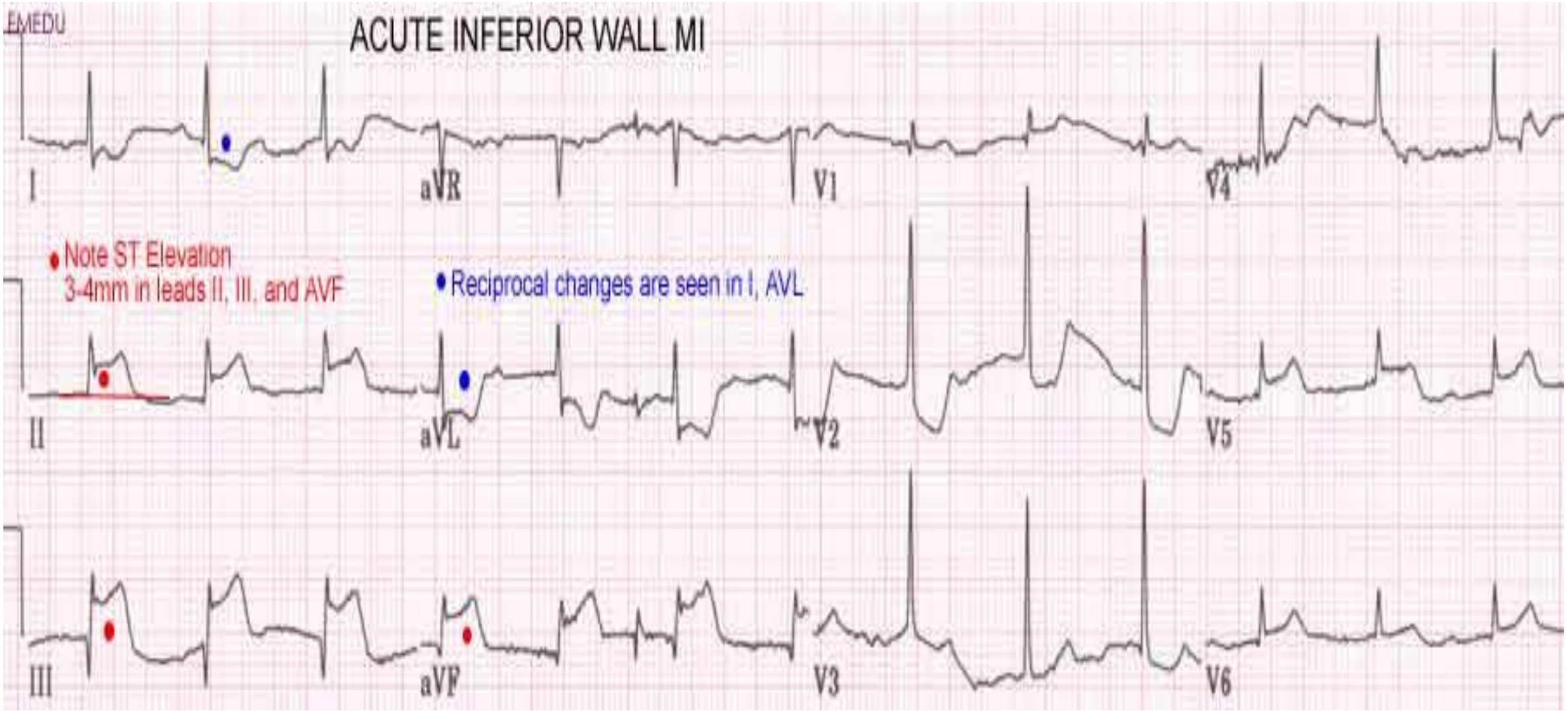


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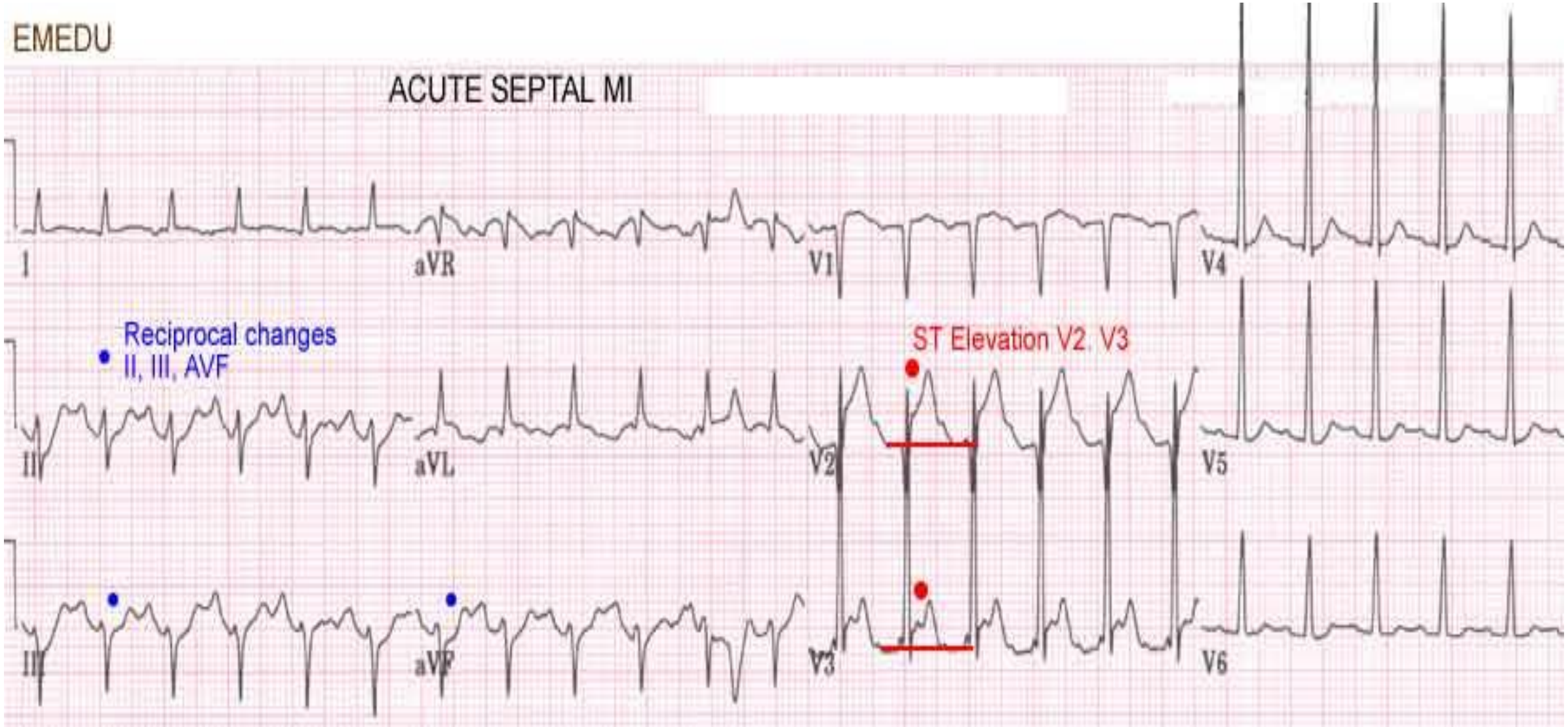
ST seg. abnormalities



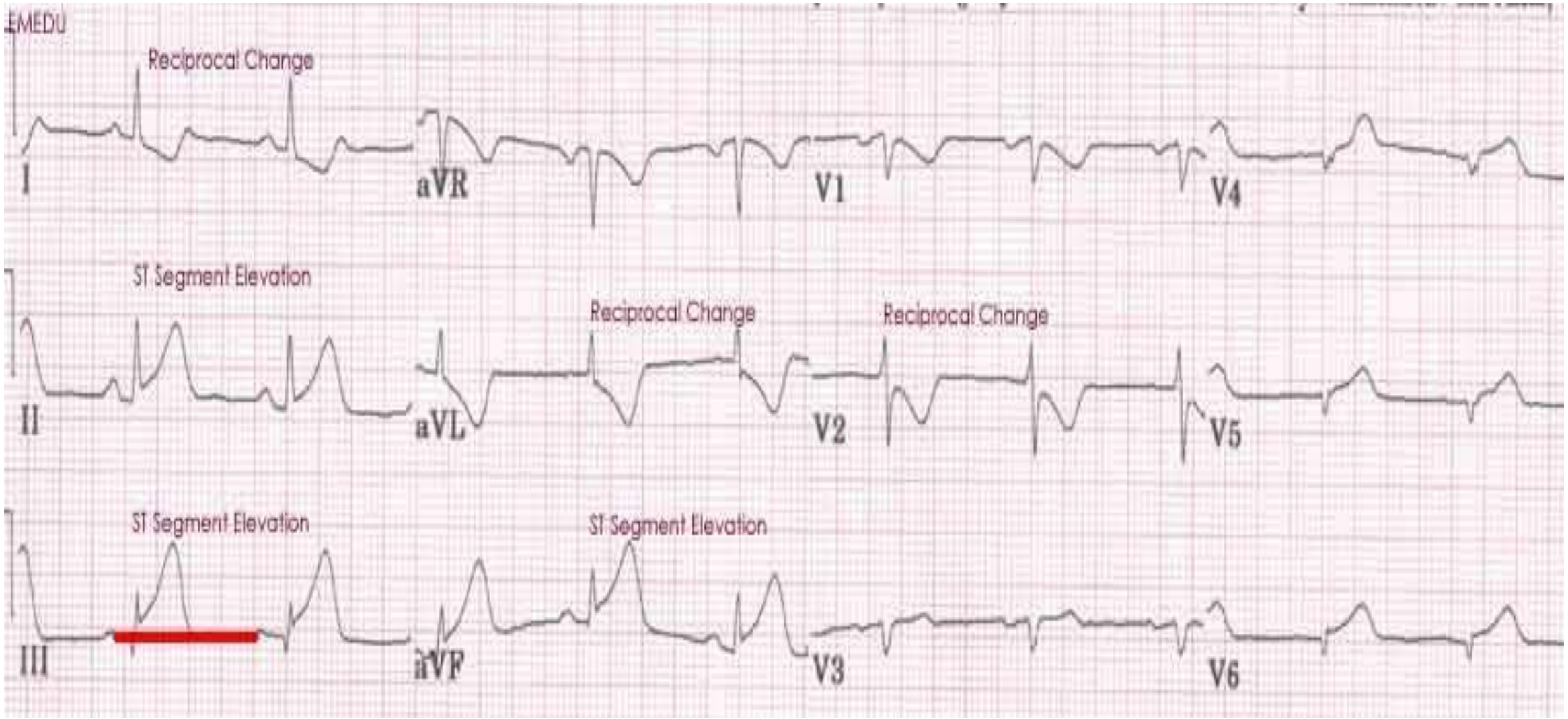
ST seg. abnormalities



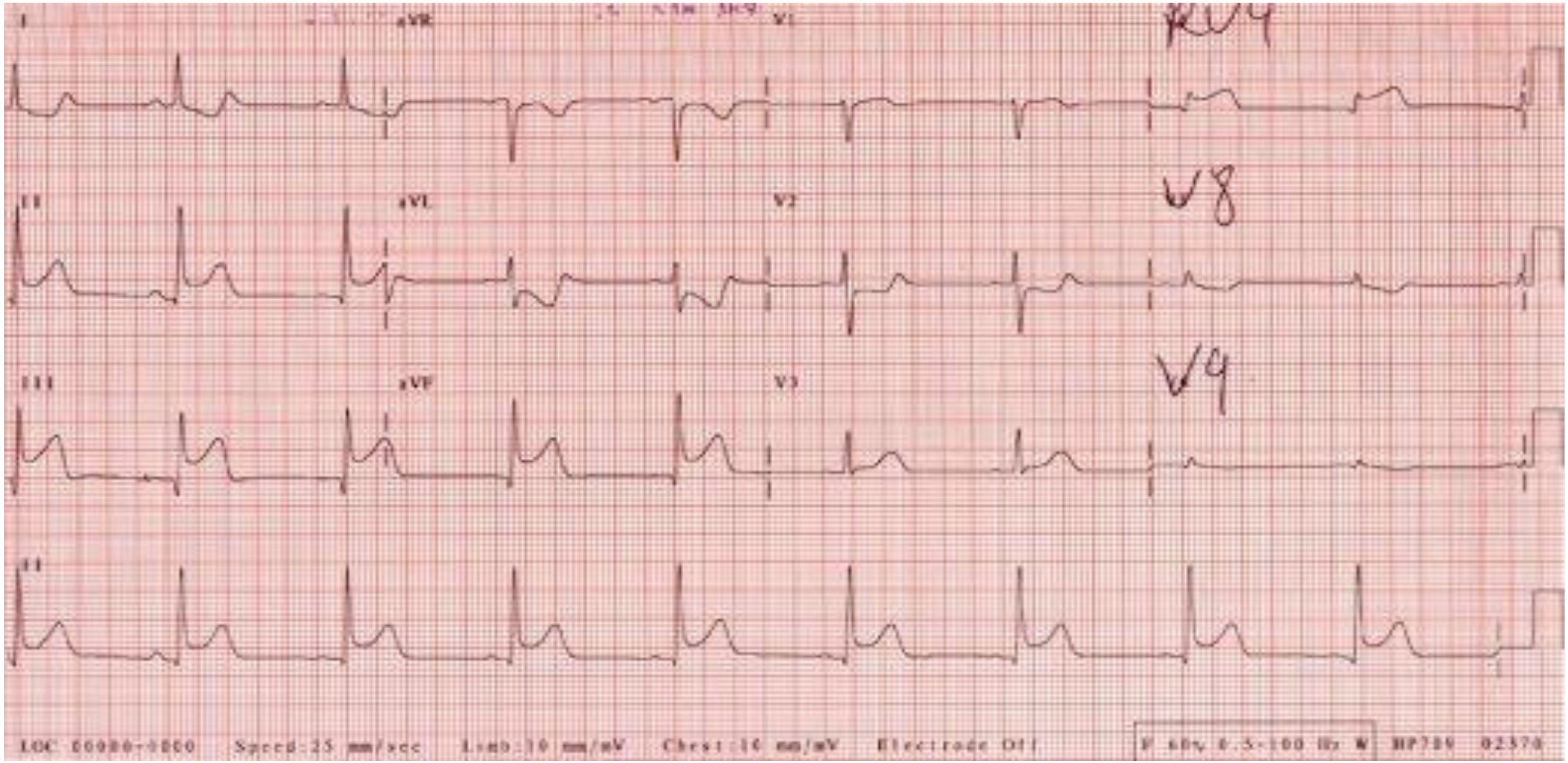
ST seg. abnormalities



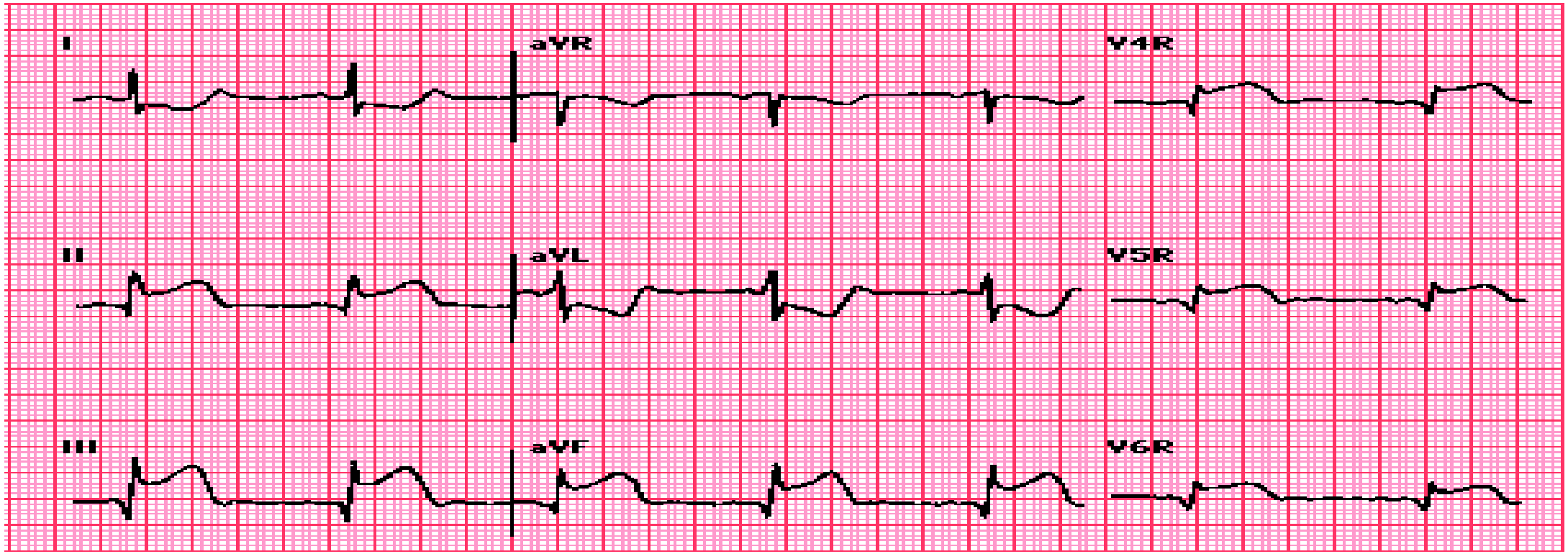
ST seg. abnormalities



ST seg. abnormalities

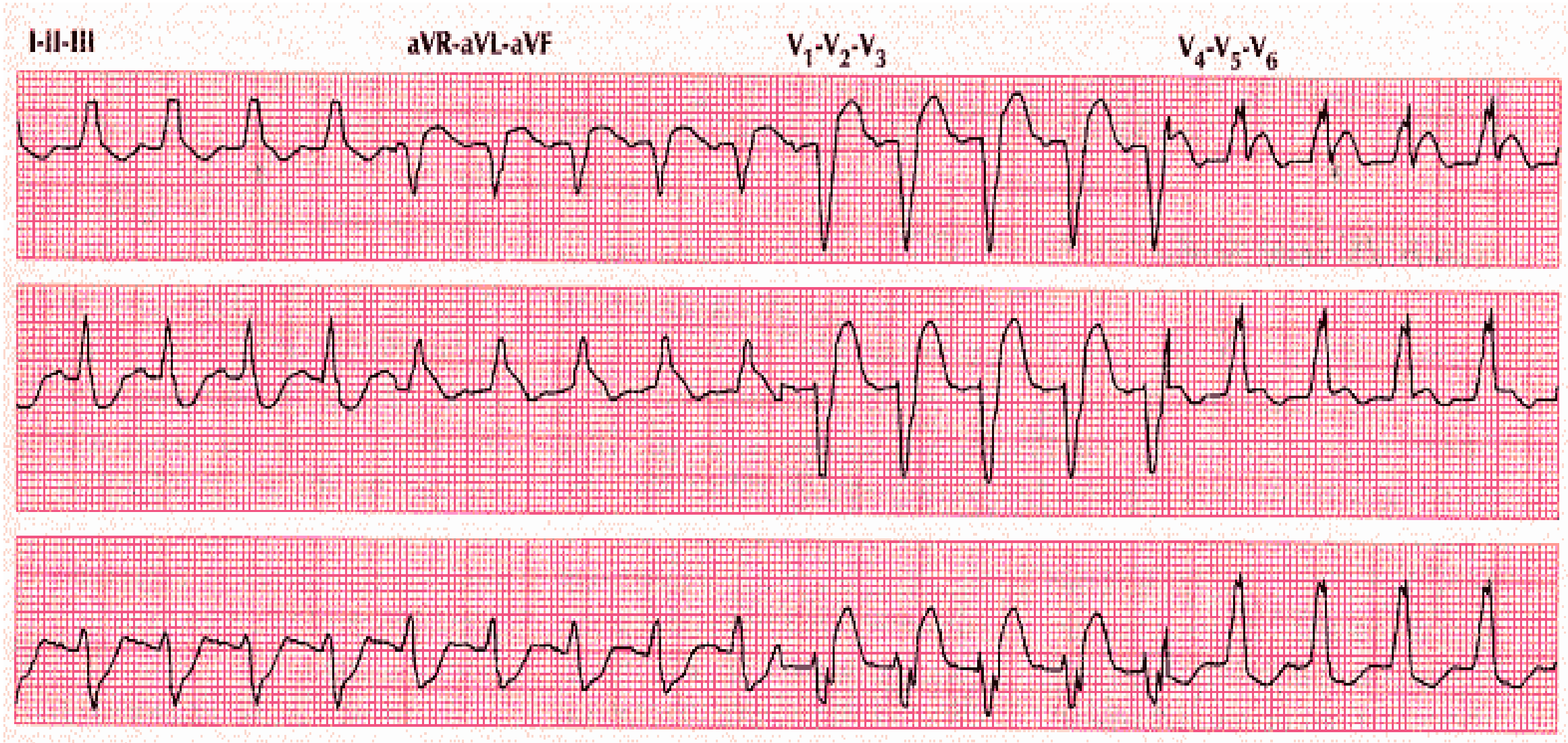


ST seg. abnormalities

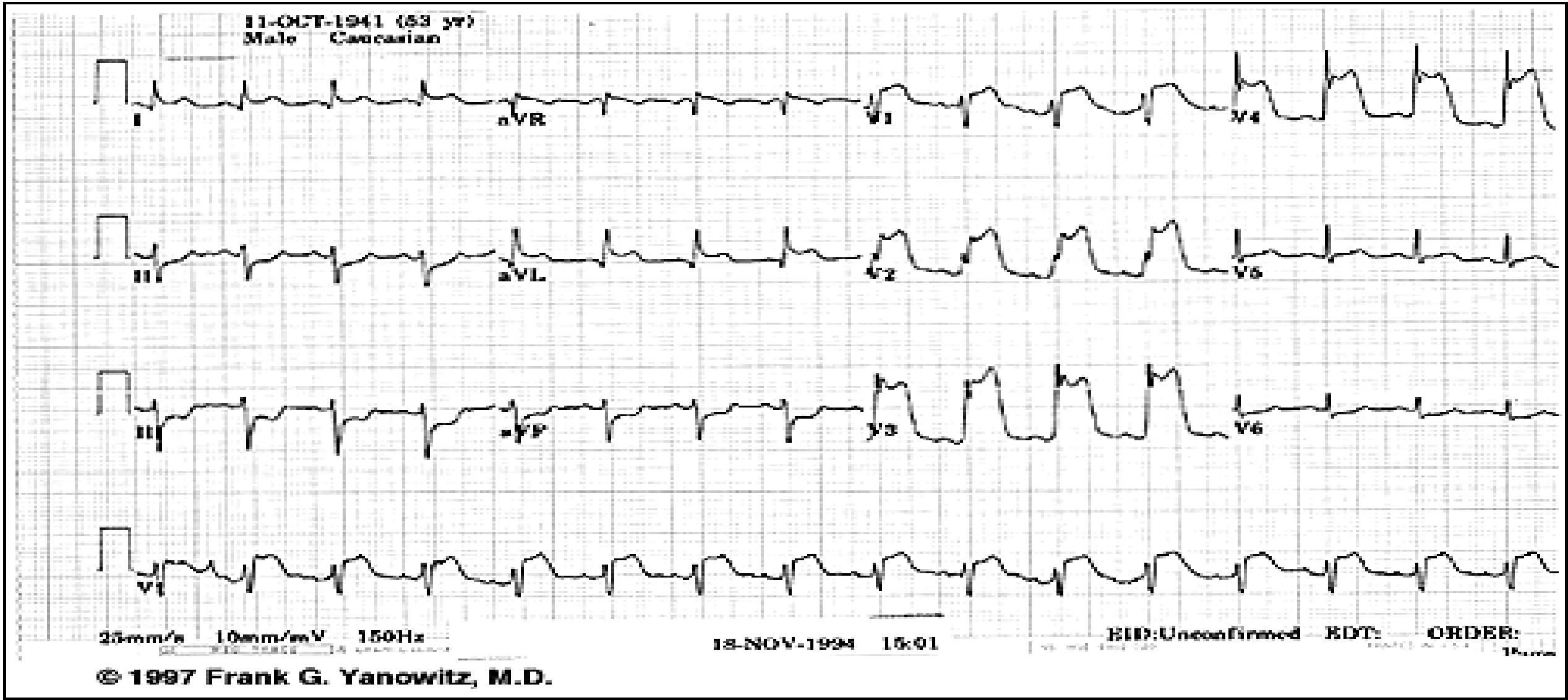


Acute inferior and right ventricular myocardial infarction
Electrocardiogram shows Q waves and prominent doming ST segment elevation in II, III, and aVF, findings which are characteristic of an acute inferior myocardial infarction. ST elevation in the right precordial leads — V4R, V5R, and V6R — indicates right ventricular involvement as well (arrows). The ST depressions in leads I and aVL represent reciprocal changes. Courtesy of Ary Goldberger, MD.

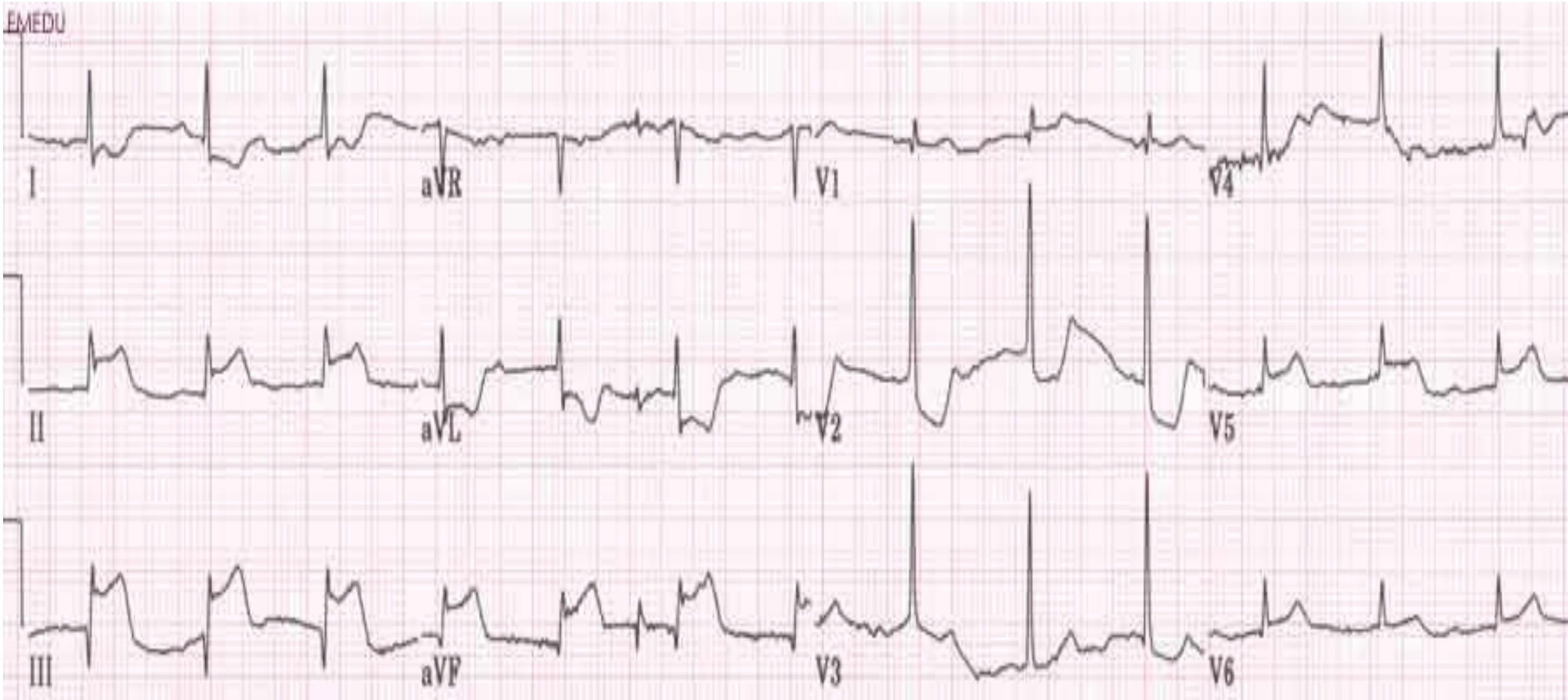
ST seg. abnormalities



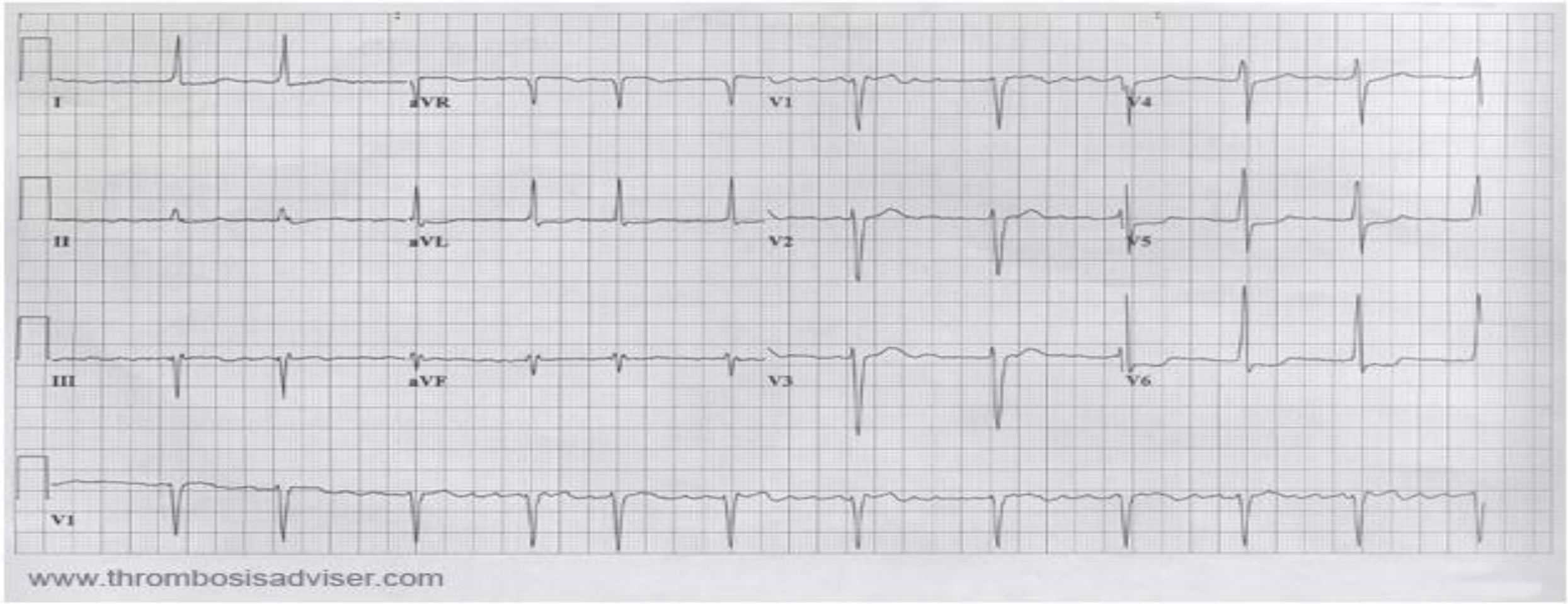
ST seg. abnormalities



ST seg. abnormalities



ST seg. abnormalities

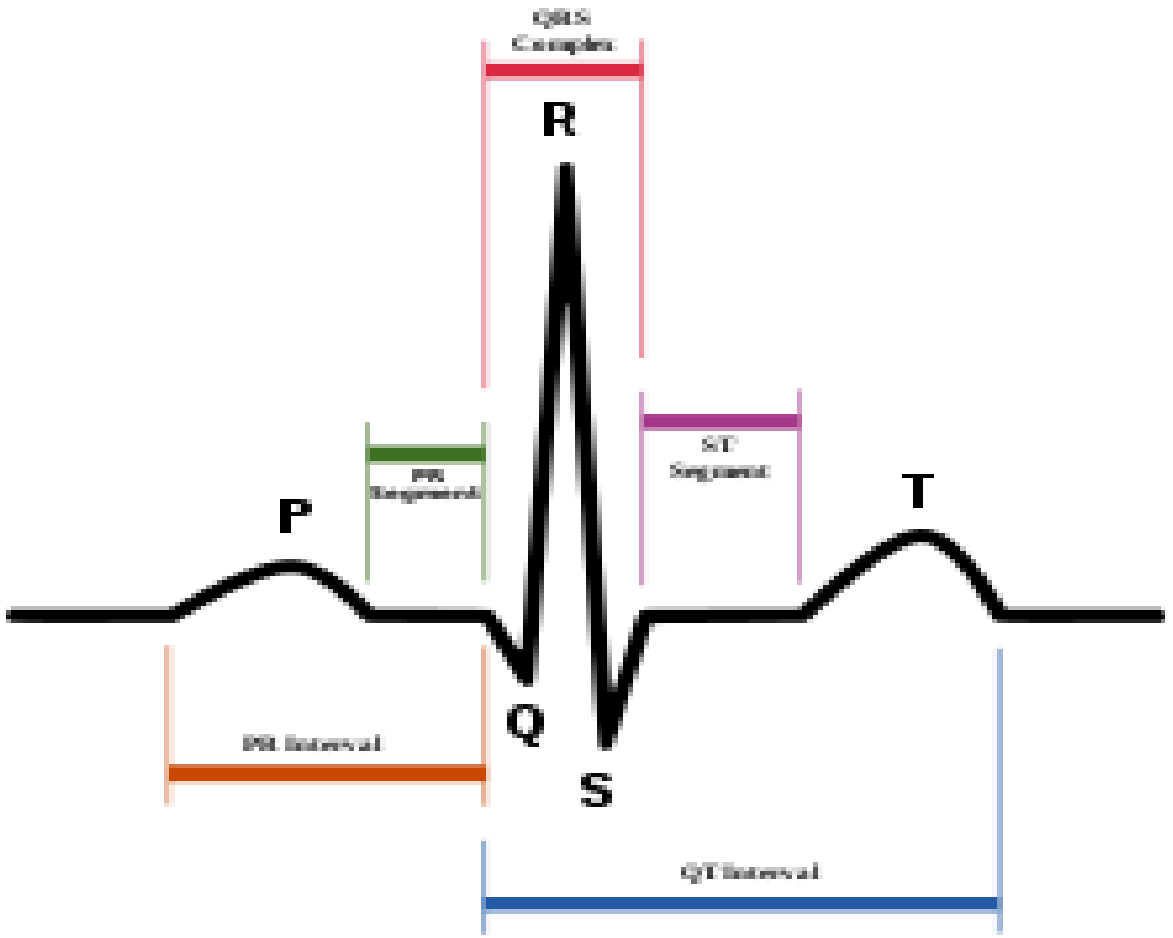


ECG comment

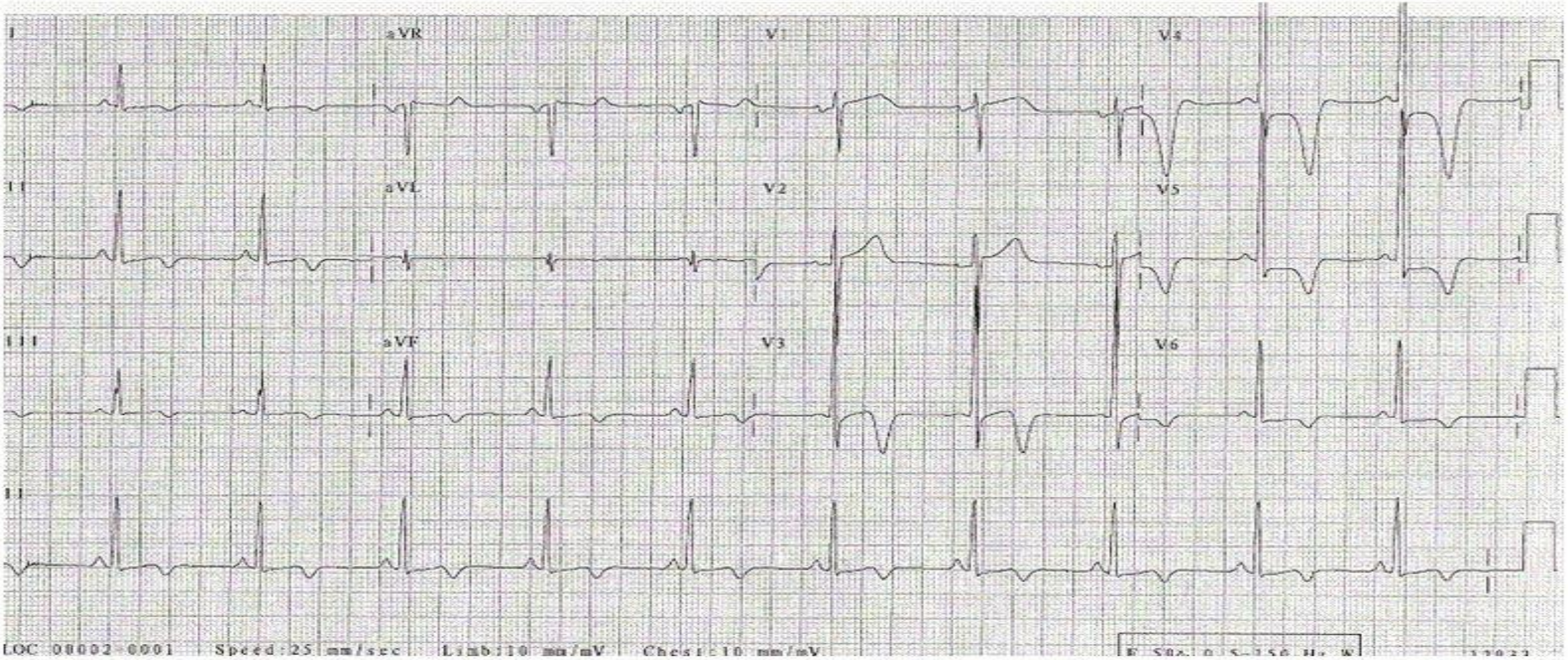
1. Identification
2. Standardisation
3. Voltage
4. Axis
5. Rate
6. Rhythm
7. P-wave
8. PR interval
9. QRS wave
10. ST segment
11. **T wave**

T-wave

1. Direction:
   aVR, V1,2 - others
2. Amplitude:
3. Duration:
4. Shape:



T-wave Abnormalities



BMC
Batterjee Medical College

one-minute paper (1st sem. 26-27)

Thank You

Questions ???